Import Price Shocks, FX-Monetary Policies, and Real Income Stabilization

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Abstract

Import-dependent economies often experience real depreciations during essential-import price spikes, as in the recent energy shock episodes. When some households are borrowing constrained, such concurrent depreciations are costly: driven by the Marshall-Lerner condition, they amplify real income volatility. Conventional wisdom nonetheless advocates a laissez-faire foreign exchange (FX) policy response that accommodates them. In a two-agent open economy, we show that optimal policy instead leans against depreciations by opening interest parity gaps and inducing stronger-than-laissez-faire capital inflows to indirectly boost household real incomes. Exchange rate management becomes a real income stabilization tool and, under nominal wage rigidity, an anti-inflation policy: stabilizing real incomes amounts to fighting imported inflation. Optimal monetary policy complements it with a tighter-than-conventional stance. Their roles are not substitutable: interest parity limits monetary policy's real income protection. Quantitatively, FX policy fights imported inflation more effectively than monetary policy.

Keywords: Exchange rate management; FX interventions; optimal monetary policy; heterogeneous agents; import price shocks; Marshall-Lerner condition; real income channel; energy shocks.

JEL Classifications: E52, F31, F32, F38, F41, D52.

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1 Introduction

Recent years have repeatedly confronted import-dependent economies with large temporary spikes in the world prices of essential imports. The 2022 energy crisis, the rare-earth disruptions of 2025 and the renewed volatility of energy prices in 2026 (IEA 2022, 2026*a*,*b*) shared the same structure: essential goods that were hard to substitute away from in the short run became abruptly more expensive. For an import-price-taking economy, this is a negative terms-of-trade shock: the country must give up more of its output to buy any given amount of imports. Because demand for essential imports is price inelastic, the import bill rises and the shock becomes a real income shock: more domestic output must be exported rather than consumed at home. In addition, import-dependent countries are often subject to the Marshall-Lerner condition: alongside inelastic import demand, they face downward-sloping, price-elastic world demand for their exports. To finance a larger import bill, the country must export more, but foreigners buy more only if domestic goods become cheaper, i.e., if the real exchange rate depreciates. Import price spikes can then be accompanied by concurrent real depreciations, which endogenously worsen the terms of trade and amplify current real income losses. The 2022 energy crisis illustrates this pattern in Japan: as world energy prices rose, the yen depreciated in real terms, and at the peak of the crisis, energy was 43 percent more expensive in domestic consumption units than the rise in its world price alone implied (Figure 1).[1]

Essential imports, such as energy, are often intermediate inputs, complementary to domestic production. When the terms of trade deteriorate, through a higher world price, a real depreciation or both, and these imports cannot be forgone, a larger share of output must be paid abroad and a smaller share remains as income for domestic factors, whose prices, real wages among them, come under downward pressure. In monetary economies where nominal wages adjust slowly, real wages must fall through a rise in the domestic price level: a terms-of-trade shock can then trigger domestic inflation. Japan's experience in 2022 fits this pattern: domestic inflation rose, nominal wages moved slowly and real wages fell.[2] The right panel of Figure 1 shows real wages falling and moving against the real exchange rate; this negative co-movement reflects an endogenous transfer of purchasing power to foreigners.[3]

A rise in world import prices is exogenous, but domestic policy can influence how much the real exchange rate depreciates during such spikes. How, then, should import-dependent economies conduct their monetary and exchange rate policies[4] in response? Should they resist the contemporaneous real depreciation or let the real exchange rate move freely, and what path of inflation should they choose while the shock lasts? The conventional wisdom on these questions comes from the representative-agent small-open-economy New Keynesian framework. In terms of exchange rate policy, this framework adopts Friedman's case for flexible exchange rates: given a terms-of-trade shock, it prescribes a laissez-faire response that leaves capital flows

[1] Other major energy importers showed the same pattern: the amplification was 27 percent in South Korea, 22 percent in the euro area and 10 percent in India (IMF 2026; BLS 2026; FRED 2026*a*; Statistics Bureau of Japan 2026; FRED 2026*b*; IMF 2025).

[2] Between January 2021 and August 2022, Japanese CPI rose by 2.9 percent and nominal wages by 1.9 percent, while real wages fell by 1.4 percent (Statistics Bureau of Japan 2026; Ministry of Health, Labour and Welfare 2023).

[3] When nominal wages are sluggish and domestic inflation rising, a real depreciation must take the form of a large nominal depreciation: between January 2021 and August 2022, the yen depreciated by 30 percent against the U.S. dollar (FRED 2026*a*).

[4] Exchange rate policy is a broad term. We define it as a policy in which the public policymaker distorts cross-border financial flows, through outright capital controls or foreign exchange interventions, to influence the path of the real exchange rate.

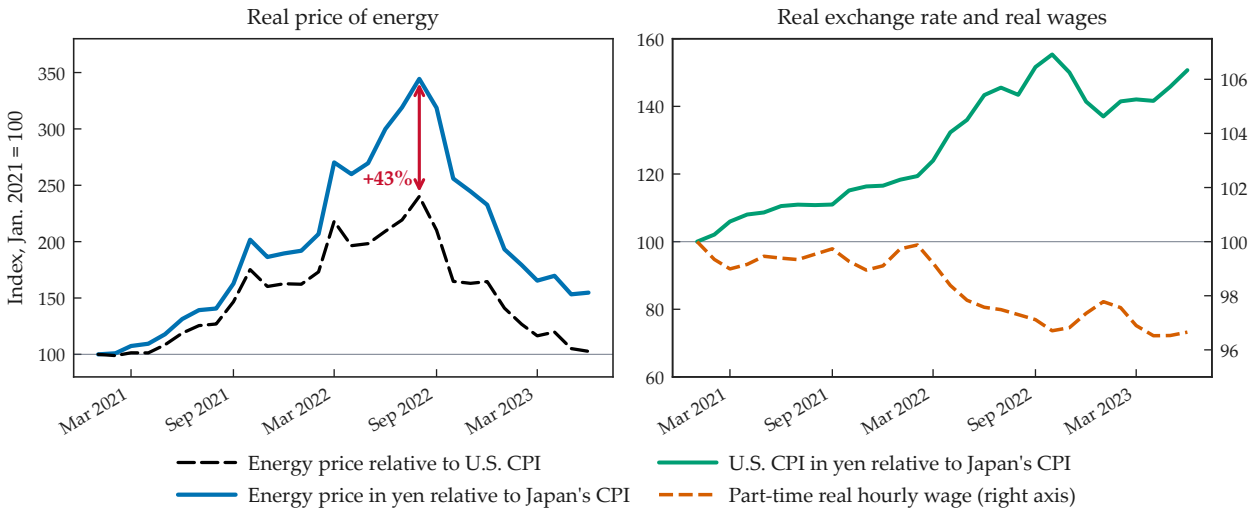


Figure 1. Japan, 2021–2023: the real price of energy, the real exchange rate and real wages

Note: Left panel: the price of energy relative to the U.S. consumption basket (the IMF energy price index in U.S. dollars deflated by the U.S. CPI, dashed) and relative to Japan's consumption basket (the same index in yen deflated by Japan's CPI, solid). Right panel: the real exchange rate (the yen-dollar rate times the U.S. CPI divided by Japan's CPI; an increase is a real depreciation) and, on the right axis, the real hourly wage of part-time workers. The solid line in the left panel equals the dashed line times the real exchange rate in the right panel: the real depreciation accounts for the gap between the two. Part-time earnings consist almost entirely of current wages, so their real wage proxies the current real income of the liquidity-constrained households in our model. Its correlation with the real exchange rate is −0.93. Sources: FRED (IMF 2026; FRED 2026*a*; BLS 2026), Statistics Bureau of Japan (2026) and Ministry of Health, Labour and Welfare (2023).

undistorted and lets the real exchange rate respond freely, acting as a shock absorber. Monetary policy is assigned to stabilize domestic inflation and real activity, depending on the nature and degree of nominal rigidities: under nominal wage rigidity, it accommodates the rise in domestic prices to let real wages fall, while stabilizing wage inflation and real activity.

This prescription is correct in these models because the relevant friction is a nominal rigidity with real costs, which gives monetary policy its role; the real income losses that a concurrent depreciation adds during an import price spike create no separate role for exchange rate policy. The laissez-faire exchange rate response rests on the assumption that agents have perfect access to international asset markets, which determine how much of the higher import bill is paid at each date and therefore how much the real exchange rate depreciates at each date and how closely it tracks import prices. By borrowing during high-price periods, households can optimally spread real income losses over time, and a transitory import price spike is then just a present-value real income loss. Exchange rate policy cannot undo this loss and has no reason to distort when households pay for it.

In line with recent heterogeneous-agent literature, we relax this assumption of perfect household access to international asset markets. We introduce a mass of liquidity-constrained households who cannot borrow against future incomes and are therefore unable to smooth volatile real incomes over time. We ask: when an open economy with heterogeneous households faces essential-import price shocks and is subject to the Marshall-Lerner condition, what is the jointly optimal response of exchange rate and monetary policy? How does heterogeneity change it, and how far can these policies protect households who cannot borrow through the shock? Direct fiscal transfers are a natural way to augment and stabilize real incomes; we

instead study environments where lump-sum transfers to constrained households are unavailable.

We argue that, under these assumptions, the conventional wisdom no longer holds. Even in a real economy without nominal rigidities, household borrowing constraints and the Marshall-Lerner condition turn the real exchange rate from a shock absorber into a shock amplifier. Given temporary spikes in essential-import prices, a laissez-faire exchange rate response becomes costly because it endogenously amplifies domestic real income volatility, thereby making the consumption path of borrowing-constrained households more volatile. The Marshall-Lerner condition is responsible for this amplification: when some households cannot borrow, the economy pays more of the higher import bill through contemporaneous exports. Foreigners buy these additional exports only at a lower price, so the real exchange rate depreciates in the same period in which the import price spikes. On the one hand, this depreciation raises export revenues; on the other, the endogenous price response finances the additional exports by further eroding households' current real incomes, with the largest burden falling on the borrowing-constrained, who consume out of current incomes. Because the spike is temporary, the later fall in world prices brings an appreciation and a recovery of real incomes. The real exchange rate and exports thus co-move strongly with world import prices, and domestic real incomes become more volatile. The Marshall-Lerner condition governs the extent of this volatility: the lower the export revenue elasticity, the stronger the co-movement and the more volatile the real incomes. In a monetary economy with nominal wage rigidity, real depreciations erode real wages through higher inflation, and domestic prices also become intertemporally more volatile.

Despite the real source of this volatility, the recent heterogeneous-agent New Keynesian (TANK/HANK) literature has largely studied these shocks through monetary policy's role in protecting real incomes, while adopting the laissez-faire exchange rate policy response from representative-agent frameworks. We instead study exchange rate management through capital flows as the primary real income stabilization tool against terms-of-trade shocks, irrespective of whether nominal rigidities are present. Our main result is that a passive, laissez-faire exchange rate response, which lets the real exchange rate depreciate freely and contemporaneously with world import prices, is costly for constrained households. An optimal policy that cares about the intertemporal consumption path of these households should instead actively lean against such depreciations to protect their real incomes and stabilize them over time.

How does exchange rate policy work? It can neither borrow on behalf of constrained households nor transfer borrowed resources to them directly. Instead, its operations in the foreign exchange market rely on the Marshall-Lerner condition to appreciate the real exchange rate and indirectly boost households' real incomes. During a spike, policy induces stronger net capital inflows than under laissez-faire. This borrowed foreign purchasing power substitutes for export revenues, and the Marshall-Lerner condition then ensures that the real exchange rate appreciates relative to laissez-faire. Once the spike has passed, additional capital outflows must be tolerated to repay the debt: the real exchange rate depreciates more than under laissez-faire, exports expand and real incomes fall. By distorting the path of capital flows, policy tilts the path of the real exchange rate, shifting real incomes from low-price to high-price periods and making the consumption of borrowing-constrained households less volatile over time. We identify this as the real-income stabilization channel of exchange rate policy, which works by driving wedges between domestic and world real interest rates. Relative to laissez-faire, optimal policy re-times when the country

pays for its real income losses in present value; it weakens the co-movement of the real exchange rate and exports with import prices and stabilizes the terms of trade, at the cost of more volatile capital flows, which distort the consumption path of unconstrained households.

This role for policy grows with the mass of borrowing-constrained households and with the essentiality of the import: the less elastic the import demand, the stronger the motive for exchange rate policy to respond. The policy's ability to deliver real income gains also depends crucially on the Marshall-Lerner export revenue elasticity: the less export revenues respond to the real exchange rate, the more strongly capital inflows appreciate it and boost real incomes. Although the channel is purely real, it matters for imported inflation in a monetary economy: under nominal wage rigidity, a real appreciation induced by capital inflows raises real wages by lowering domestic inflation. Real income stabilization then amounts to intertemporal price stabilization, and exchange rate policy, which has features of fiscal policy in its management of capital flows and public balance sheets, becomes an anti-inflation policy.

We also answer how the role for monetary policy changes once exchange rate policy is active. Exchange rate policy manages the path of capital flows, while monetary policy, under nominal wage rigidity, influences the level of real activity: employment and output. When imports and employment are strong complements, we find that optimal monetary policy is more contractionary than a typical optimal representative-agent rule. This is because when employment and imports expand, the import bill rises, which induces a real depreciation and a fall in real wages, redistributing resources away from constrained wage earners and towards unconstrained net borrowers in international asset markets, who can then buy more domestic goods with any given foreign-unit capital inflow. A public agent choosing monetary policy internalizes this pecuniary redistributive effect and therefore regards a marginal monetary expansion during an import price spike as costlier than in a representative-agent model, because such an expansion would endogenously redistribute real incomes away from constrained households.

Relative to a typical representative-agent economy, the jointly optimal policy in our framework pairs an active exchange rate policy, which leans against a real depreciation during import price spikes, alongside a restrained monetary policy. In an open economy small in asset markets, the roles of these policies are not substitutable. Monetary policy in isolation is bound by interest parity: it cannot drive a wedge between domestic and world real interest rates and moves capital flows only through employment and output, which weakens its grip on the real exchange rate and limits the real income stabilization it can deliver. Exchange rate policy drives these wedges directly, moving capital flows independently of real activity.

We formalize these arguments in an analytically tractable two-period, two-agent model of an open economy that is small in asset and import markets and large in its export market. The economy consists of a mass of hand-to-mouth households who consume their current incomes, and Ricardian, permanent-income households who trade in international asset markets. Both supply labor to firms that combine it with an essential imported input in fixed proportions to produce domestic output under nominal wage rigidity. In this environment we consider the jointly Ramsey-optimal monetary and capital control policies of a utilitarian public agent facing a temporary spike in imported input prices.

We then quantify these results in an infinite-horizon two-agent model with financial intermediation

frictions, in which exchange rate policy manages capital flows through foreign exchange interventions, while intermediation frictions make opening interest parity gaps costly. Optimal interventions trade off this cost against the real income stabilization objective that arises under import-price volatility. Calibrating the model to Japan, we study perfect-foresight impulse responses and a historical policy counterfactual on the realized 2022 path of world energy prices, comparing jointly optimal policy with conventional representative-agent prescriptions and with optimal monetary policy in isolation. When our two-agent economy follows the representative-agent-equivalent Ramsey rule, its real depreciation at the peak of the crisis is 3.5 times as large as in a representative-agent economy, illustrating how household heterogeneity and the Marshall-Lerner condition make the real exchange rate co-move with the shock when policy is set suboptimally. Compared with that policy, optimal policy makes energy 19.5 percent cheaper in domestic units at the peak, undoes 29 percent of the peak real-wage loss, and lowers average quarterly inflation through the peak from 0.60 to 0.23 percent. Exchange rate policy delivers most of these gains on its own: optimal exchange rate policy, with monetary policy set suboptimally, undoes 27 percent of the peak wage loss and lowers inflation to 0.28 percent. With monetary policy set optimally and exchange rate policy set suboptimally to close interest parity gaps, only 0.3 percent of the peak wage loss is undone and inflation falls only to 0.58 percent.

The rest of the paper proceeds as follows. Section 2 describes the related literature. Section 3 presents the analytical model and characterizes the optimal policy response. Section 4 develops a quantitative two-agent model and quantifies policy responses to energy price shocks. Section 5 concludes.

2 Related Literature

This paper relates to literature studying policy responses to terms-of-trade shocks in small open economies. Conventional wisdom, rooted in the case for flexible exchange rates (Friedman 1953), favors a free-floating exchange rate and undistorted capital flows, while the monetary policy response has received more attention. In representative-agent New Keynesian models, monetary policy targets nominal rigidities: it stabilizes inflation under sticky domestic prices (Clarida, Galí and Gertler 2002; Galí and Monacelli 2005) and wage inflation under nominal wage rigidity, accommodating price inflation (Aoki 2001; Erceg, Henderson and Levin 2000; Bodenstein, Erceg and Guerrieri 2008). With an endogenous terms-of-trade externality, optimal monetary policy departs from price stability to influence the terms of trade (Faia and Monacelli 2008; De Paoli 2009), on average restraining employment below its flexible-price level (Catão and Chang 2015). Recent work finds that optimal monetary policy runs positive output gaps after energy shocks (Lorenzoni and Werning 2023) and is expansionary after tariffs, with a depreciating exchange rate (Bianchi and Coulibaly 2025; Werning, Lorenzoni and Guerrieri 2025). Drechsel et al. (2026) find that stabilizing domestic prices remains optimal for commodity exporters but, for importers using energy in production, optimal policy keeps wage growth stable and lets energy costs pass into domestic prices. Egorov and Mukhin (2023) show that inflation targeting and a floating exchange rate remain optimal under dollar pricing. Following the case for flexible exchange rates, none of these papers need to distort capital flows. Our representative-agent benchmark is consistent with this literature: capital flows remain optimally undistorted and contractionary

monetary policy corrects the terms-of-trade externality. With borrowing-constrained households, we find that exchange rate policy distorts capital flows, opening interest parity gaps to lean against real depreciations during import price spikes; under nominal wage rigidity, price level stabilization then emerges from this real income stabilization objective even without sticky prices, and monetary policy is tighter than in the representative-agent benchmark.

Two-agent and heterogeneous-agent New Keynesian (TANK/HANK) studies of terms-of-trade shocks have so far adopted the representative-agent prescription of undistorted capital flows and studied the monetary policy response in isolation. Auclert et al. (2024*a*) find that energy price increases depress real incomes and output even without monetary tightening, and that an individual importer's tightening has limited effect on imported inflation. We likewise find that optimal monetary policy alone does little to stabilize real incomes, whereas exchange rate policy, by opening interest parity gaps, stabilizes them independently of the level of real activity and fights imported inflation more effectively. Chan, Diz and Kanngiesser (2024) find that, with hand-to-mouth households, optimal monetary policy is less contractionary than in representative-agent economies after energy shocks. Once exchange rate policy is active, we instead find optimal monetary policy tighter than in the representative-agent benchmark. Acharya, Challe and Corsetti (2026) find that optimal monetary policy allows a larger real depreciation after export contractions, whereas we find that, after essential-import price spikes, optimal policy leans against the depreciation. To the best of our knowledge, we are the first to study jointly Ramsey-optimal monetary and exchange rate policy in a two-agent monetary economy facing import price shocks.

More generally, the HANK literature has argued that monetary policy works largely through indirect income effects of changes in real activity (Kaplan, Moll and Violante 2018; Auclert 2019). Several papers find that, after adverse productivity shocks, optimal monetary policy is more accommodative than in representative-agent economies (Challe 2020; Acharya, Challe and Dogra 2023; Acharya and Challe 2025). In our framework, exchange rate policy also delivers indirect income effects, but by distorting capital flows, through the real exchange rate and independently of the level of real activity. Our import price shock is isomorphic to an adverse factor-biased productivity shock, yet we find optimal monetary policy tighter than in the representative-agent benchmark once exchange rate policy is active: it is exchange rate policy that lowers the domestic real interest rate below the world rate during the spike, which monetary policy, bound by interest parity, cannot do. Auclert et al. (2024*b*) show that a depreciation lowers real incomes by raising import prices relative to domestic output prices, and that leaning against it through rate hikes deepens the recession. We build on this real income channel: borrowing constraints and the Marshall-Lerner condition make the real exchange rate depreciate with the import price, and exchange rate policy uses the same channel in reverse, appreciating the real exchange rate without contracting activity. Finally, redistributive motives make optimal monetary policy more accommodative than in representative-agent economies (Bilbiie 2008; Bhandari et al. 2021; Dávila and Schaab 2023), whereas Bianchi and Van der Ghote (2026) find that foreign ownership of domestic equity leads the central bank to tighten. Our monetary tightening shares this direction but reflects the policy internalizing a redistributive price effect between domestic households due to the Marshall-Lerner condition.

We implement exchange rate policy through foreign exchange interventions, which act as an implicit

capital control tax, building on intermediary-based exchange rate determination (Gabaix and Maggiori 2015). Interventions have been studied as responses to capital flow shocks, exchange rate objectives at the zero lower bound and shallow foreign exchange markets (Cavallino 2019; Amador et al. 2020; Basu et al. 2020); we use them to stabilize real incomes and, in a monetary economy, as an anti-inflation policy. Itskhoki and Mukhin (2023) find that, with a representative household, monetary policy closes the output gap and interventions eliminate interest parity deviations. We share this assignment, but with borrowing-constrained households, interventions instead open costly parity gaps and redistributive motives open negative output gaps after essential-import price spikes. Capital controls also serve a macroprudential role (Bianchi 2011; Bianchi and Lorenzoni 2022; Biljanovska and Vardoulakis 2024) or manipulate the terms of trade (Costinot, Lorenzoni and Werning 2014). Our motive is neither: terms-of-trade manipulation is absent under our fixed-proportions baseline and, under general CES technology, restricts borrowing during the spike, whereas the heterogeneity-induced real income stabilization motive goes against it.

On capital control policies under heterogeneity, the paper most closely related to ours is Fanelli and Straub (2021). They show that, with non-homothetic Stone-Geary preferences and heterogeneous wealth, capital controls stabilize the relative price of tradeables and non-tradeables, and hence the purchasing power of hand-to-mouth households, by dampening volatility induced by capital flow shocks. We share with them the general principle that optimal capital controls internalize endogenous relative price volatility to smooth the consumption of hand-to-mouth households. However, we differ fundamentally in the shock, the endogenous relative price and the source of its volatility: we study a world essential-import price shock that endogenously amplifies the volatility of the economy's terms of trade. We do not rely on non-homothetic preferences or wealth differences, but volatility arises due to the Marshall-Lerner condition: a higher import bill requires greater contemporaneous exports, which foreigners buy only at a lower price, so a depreciation shifts excessive purchasing power abroad today and reverses when the shock subsides. Our capital control policy subsidizes capital inflows to substitute for export revenues, appreciating the real exchange rate today and depreciating it more in the future: it reduces real income volatility by making capital flows more volatile. In a monetary model with nominal wage rigidity, we illustrate how this real income stabilization translates into fighting imported inflation, and we incorporate monetary policy to study how its role changes under active capital controls.

3 Analytical Framework

This section develops a two-period model of a small open economy that imports an essential intermediate input and is populated by households who are heterogeneous in their ability to participate in asset markets and smooth consumption intertemporally. A fraction of households are hand-to-mouth and consume their current income each period, while the remaining households can save and borrow freely in international asset markets. Given a temporary spike in import prices that erodes current household real incomes, we characterize optimal monetary and exchange rate policies for this economy.

The framework is tractable enough to characterize optimal policies in closed form. We describe a utilitarian Ramsey planner's optimization problem with only two policy margins: the labor wedge and the intertemporal

wedge. We interpret the labor wedge as the margin that monetary policy can move, and the intertemporal wedge as the margin that exchange rate policy can influence. This section takes a simplified, reduced-form approach to policy implementation and leaves the full decentralization with nominal rigidities and financial intermediation frictions to the quantitative model of Section 4.

3.1 Environment

The economy operates over two periods, $t \in \{1, 2\}$. It produces a single final tradeable good and interacts with the rest of the world through three channels: it exports the domestic good, imports a productive input, and borrows or lends on international asset markets.

3.1.1 Goods and Relative Prices

Three tradeable goods circulate in the world economy. The first is the domestically produced good, produced only in the small open economy. It is consumed by domestic households and exported to the rest of the world. Let C_t, X_t, and Y_t denote aggregate domestic consumption, exports, and output. The resource constraint is given by:

$$C_t + X_t = Y_t \tag{1}$$

The second good is the foreign consumption basket, which domestic households do not consume and domestic firms do not use in production. Bonds traded in international asset markets are claims denominated in units of this good, and the good also serves as the world numeraire: we normalize its foreign currency price to $P_t^* = 1$. One unit of foreign currency then equals one unit of this good, and all world prices are expressed in units of this good. The world gross real interest rate is constant and denoted by R^*, i.e., one unit of the foreign good saved in period t returns R^* units in period $t + 1$. The economy is price-taking with respect to this interest rate.

The third good is an intermediate foreign good which the economy imports. The economy uses this good in the production of the domestic tradeable good. The world price of this input is denoted P_t^{T*} in units of the foreign good. The economy is also price-taking with respect to this relative price.

The domestically produced tradeable good is the only final good that the economy consumes. Let P_t denote the price of the domestic good in units of domestic currency and let e_t denote the nominal exchange rate, i.e., the price of the foreign currency in units of the domestic currency. With the world consumption good as the numeraire, e_t is also the price of the world consumption good in domestic currency units. Since the domestic economy consumes only the domestic good, P_t is also the domestic price level, and the real exchange rate is $\epsilon_t = \frac{e_t}{P_t}$. By the law of one price for tradeables, the inverse of the real exchange rate, $\frac{1}{\epsilon_t}$, is the world price of the domestic good in units of the world numeraire. The economy's terms of trade is then given by $\frac{P_t^{T*}}{\frac{1}{\epsilon_t}} = \epsilon_t P_t^{T*}$, which is the price of the imported input in units of the domestic good. Finally, let W_t denote the nominal wage and $w_t = \frac{W_t}{P_t}$ denote the real wage in units of the domestic good.

3.1.2 Production Technology

A unit mass of identical firms produces the domestic good, Y_t, using labor, N_t, and the imported intermediate input, M_t, according to a Leontief technology:

$$Y_t = A_t \cdot \min\left\{N_t, \frac{M_t}{\theta}\right\} \tag{2}$$

where $A_t > 0$ is total factor productivity and $\theta > 0$ is the input requirement per unit of labor employed. The Leontief specification captures an essential feature that motivates this paper: the imports are critical to the domestic economy, i.e., the imported and domestic inputs are strict complements in the short run, and a rise in the input price cannot be offset by substituting toward domestic factors. With no substitution effects, relative price changes create pure income effects, which is helpful in isolating how real income effects operate in the economy.

The firms' problem is to choose N_t and M_t to minimize the cost of producing Y_t subject to the production technology:

$$\min_{\{N_t, M_t\}} w_t N_t + \epsilon_t P_t^{T*} M_t \quad \text{s.t. (2)} \tag{3}$$

Given this Leontief technology, a cost-minimizing firm will always deploy the two inputs in fixed proportions to produce any given level of output:

$$M_t = \theta N_t \tag{4}$$

The output produced can be expressed as a linear function of labor input:

$$Y_t = A_t N_t \tag{5}$$

The firms' real marginal cost of production is therefore $\frac{w_t}{A_t} + \epsilon_t P_t^{T*} \frac{\theta}{A_t}$: each additional unit of output requires $\frac{1}{A_t}$ units of labor at wage w_t and $\frac{\theta}{A_t}$ units of the imported input at domestic cost $\epsilon_t P_t^{T*}$ per unit. Since firms are competitive, an equilibrium requires that firms make zero economic profits, i.e., the real marginal cost of production must equal one. This gives a linear expression for the real wage in terms of the price of the imported input and the real exchange rate:

$$w_t = A_t - \epsilon_t P_t^{T*} \theta \tag{6}$$

A_t denotes the total output produced per unit of labor employed, while $\epsilon_t P_t^{T*} \theta$ denotes the output paid by the firm to buy θ units of the imported input that, together with a unit of labor, produces A_t units of output. The residual, $A_t - \epsilon_t P_t^{T*} \theta$, is paid to labor. The real wage, therefore, has a linear one-for-one negative relationship with the price of the imported input and the real exchange rate. A rise in the price of the imported input or a real depreciation raises the marginal cost of production. At the prevailing wage, firms make losses and therefore decrease their demand for the input. Since inputs and labor are perfect complements, the demand for labor also falls. With less competition for workers, the wage falls one-for-one with the rise in input costs until zero profits are restored. A fall in the price of the imported input or a real appreciation has the opposite effect: firms make positive profits and expand, bidding up wages until zero profits are restored.

Let Y_t^N denote the labor income that accrues to domestic workers in period t:

$$Y_t^N = w_t N_t = (A_t - \epsilon_t P_t^{T*} \theta) N_t \tag{7}$$

Since Y_t^N is the share of gross final output that accrues to the primary domestic factors of production, it also represents the economy's GDP. The remainder, $\epsilon_t P_t^{T*} \theta N_t$, is the import bill, i.e., the share of domestic output that firms sell in the foreign exchange market in return for buying θN_t units of the imported input. The real exchange rate determines the split: a depreciation raises the import bill and compresses GDP; an appreciation does the reverse.

3.1.3 Export Demand

As illustrated by the resource constraint (1), the domestic good is either consumed domestically or exported, where X_t denotes the part of output consumed abroad in period t. Given the world price of the domestic good, $\frac{1}{\epsilon_t}$, we assume that the economy faces a downward-sloping, isoelastic export demand:

$$X_t = \zeta \epsilon_t^{1+\chi} \tag{8}$$

where $\zeta > 0$ is a scale parameter and $-(1+\chi)$ is the price elasticity of export demand. We assume that $0 < \chi < \infty$, i.e., the export demand is relatively price-elastic. Given this downward-sloping demand structure, a real depreciation is required to induce foreigners to buy more of the domestic good, and a real appreciation reduces the quantity sold abroad. Moreover, since the demand is relatively price elastic, the export revenue, $\frac{X_t}{\epsilon_t} = \zeta \epsilon_t^{\chi}$, in units of the foreign good is increasing in the real exchange rate. This is the Marshall-Lerner condition in this environment: a depreciation is required to increase the supply of foreign purchasing power in the domestic economy. We refer to χ as the Marshall-Lerner export revenue elasticity: it measures the elasticity of supply of foreign purchasing power (in units of the foreign good) with respect to the real exchange rate.

3.1.4 Households

The economy is populated by a unit mass of households of two types. A fraction $\lambda \in [0, 1)$ are hand-to-mouth (HtM) households and the remaining $1 - \lambda$ are Ricardian households. Both types consume only the domestic good and supply labor. Let C_t^p and C_t^r denote the consumption of HtM and Ricardian households in period t.

All households supply N_t units of labor and take this quantity as given. In the analytical model presented in this section, we assume that N_t is a policy instrument. We discuss this further in Section 3.1.5.

The period utility of household $i \in \{r, p\}$ is:

$$u(C_t^i) - v(N_t)$$

where $u : \mathbb{R}_{++} \to \mathbb{R}$ and $v : \mathbb{R}_{++} \to \mathbb{R}$ are twice continuously differentiable, with $u' > 0$, $u'' < 0$, $v' > 0$, and $v'' > 0$. They satisfy the standard Inada conditions: $\lim_{C \downarrow 0} u'(C) = \infty$, $\lim_{C \to \infty} u'(C) = 0$, $\lim_{N \downarrow 0} v'(N) = 0$, and $\lim_{N \to \infty} v'(N) = \infty$. All households share the same preferences. The common discount factor is $\beta \in (0, 1)$, and we impose $\beta R^* = 1$ throughout.

Hand-to-mouth households. HtM households cannot access asset markets. Given labor incomes, $w_t N_t$, they choose consumption, C_t^p, to maximize lifetime utility subject to period-by-period budget constraints:

$$C_t^p = w_t N_t \tag{9}$$

Since they do not participate in asset markets, the budget constraints imply that they consume their current labor income each period, and income shocks therefore pass into their consumption one for one.

Ricardian households. Ricardian households, in addition to consuming the domestic good and supplying labor, also participate in international asset markets. Let f^* denote a Ricardian household's net foreign asset position in period 1, denominated in units of the foreign good, with $f^* > 0$ denoting saving and $f^* < 0$ borrowing. Given labor income, the real exchange rate, the world interest rate, and government policy (τ, T), a Ricardian household's problem is to choose (C_t^r, f^*) to maximize lifetime utility subject to budget constraints:

$$\begin{aligned} \max_{\{C_1^r,\, C_2^r,\, f^*\}} \quad & u(C_1^r) + \beta\, u(C_2^r) \qquad \text{s.t.} \\ C_1^r + \epsilon_1 f^*(1+\tau) &= w_1 N_1 + T \\ C_2^r &= w_2 N_2 + \epsilon_2 R^* f^* \end{aligned} \tag{10}$$

where (τ, T) are policy instruments discussed in Section 3.1.5. The optimal asset choice satisfies the Euler equation:

$$u'(C_1^r)\, \epsilon_1 (1+\tau) = \beta R^*\, u'(C_2^r)\, \epsilon_2 \tag{11}$$

3.1.5 Fiscal-Monetary Policy

We assume that public policy has access to two margins: the intertemporal wedge and the labor wedge. The policy that influences the intertemporal wedge is interpreted as the exchange rate policy and is implemented through a capital control tax τ on the foreign asset positions of Ricardian households. Tax revenue is rebated lump-sum to Ricardian households as a transfer T. The public agent's budget constraint is:

$$(1-\lambda)\, \tau\, \epsilon_1 f^* = (1-\lambda)\, T \tag{12}$$

The public agent's ability to influence the labor wedge is modeled in a reduced-form manner: the public agent directly sets employment, N_t, and we interpret this policy as monetary policy. This keeps the analytical framework clean while preserving the trade-offs that monetary policy faces. We defer the full decentralization to Section 4 with nominal wage rigidities and financial intermediation frictions. There, monetary policy sets the inflation rate and exchange rate policy is implemented through FX interventions. The qualitative results from the analytical model carry over to that framework.

No direct fiscal transfers. The framework has no fiscal instrument capable of making lump-sum transfers to HtM households; the capital control revenue is rebated only to Ricardians. This modeling choice keeps the focus on how exchange rate and monetary policies reach the real incomes of HtM households through pecuniary general equilibrium effects. Direct transfers to HtM households would reach them trivially.

3.1.6 Balance of Payments

The final piece in the model is the balance-of-payments (BoP) accounting condition, which is already implicit in the specified equations. Combining the household budget constraints, the firms' zero-profit condition (6), goods market clearing (1), and the fiscal budget constraint (12) yields the balance-of-payments condition, which is interpreted as the economy's external budget constraint. Denote aggregate consumption as $C_t = \lambda C_t^p + (1-\lambda) C_t^r$ and let $F^* \equiv (1-\lambda) f^*$ be the aggregate net foreign asset position of the economy. Finally, substitute in the export demand function, (8), to express the period-by-period balance of payments in units of the foreign good:

$$\begin{aligned} P_1^{T*} \theta N_1 + F^* &= \zeta \, \epsilon_1^X \\ P_2^{T*} \theta N_2 &= \zeta \, \epsilon_2^X + R^* F^* \end{aligned} \tag{13}$$

The BoP constraints have a natural interpretation of equating demand and supply of foreign purchasing power[5] in the domestic economy's foreign exchange market. Firms are always net buyers of foreign purchasing power: they produce the domestic good and must pay for the imported input, exchanging domestic output for foreign purchasing power. Foreigners are always net buyers of the domestic good and suppliers of foreign purchasing power. Ricardian households switch sides depending on the sign of financial flows. When $F^* < 0$, Ricardians are net buyers of the domestic good in the foreign exchange market: they borrow foreign purchasing power and use it to buy domestic goods, joining foreigners on the supply side. When $F^* > 0$, they are net sellers of the domestic good in the foreign exchange market: they surrender domestic output out of their current labor income in exchange for future claims on foreign purchasing power, effectively joining firms on the demand side. The real exchange rate clears this market each period: it moves until the demand for foreign purchasing power equals the supply.

Combining the two period-by-period BoP constraints yields the intertemporal budget constraint for the economy:

$$P_1^{T*} \theta N_1 + \frac{1}{R^*} P_2^{T*} \theta N_2 = \zeta \, \epsilon_1^X + \frac{1}{R^*} \zeta \, \epsilon_2^X \tag{14}$$

The left-hand side of (14) is the present value of import expenditure in units of the foreign good, while the right-hand side is the present value of export revenue in foreign units. The economy cannot be a net borrower or lender in the long run: it cannot run a trade deficit or surplus in present value terms.

Define the trade balance in units of the foreign good:

$$NX_t \equiv \zeta \, \epsilon_t^X - P_t^{T*} \theta N_t \tag{15}$$

$NX_t > 0$ is a surplus and $NX_t < 0$ is a deficit. The trade balance is increasing in ϵ_t and decreasing in N_t: under the Marshall-Lerner condition, depreciation raises export revenue; higher employment raises the consumption of the intermediate input and hence increases the import bill. The intertemporal constraint

[5] We use the term 'foreign purchasing power' to describe the real exchange between the domestic economy and rest of the world because it does not matter whether foreigners pay in units of foreign good or the intermediate input while buying the domestic good: the foreign good converts into the intermediate input in world markets at an exogenous price P_t^{T*}. In other words, the foreign good is not a medium of exchange in the model. Alternatively, consider a setting with only home and foreign goods, where the foreign good directly enters domestic production technology with labor in fixed proportions and a factor intensity of $\frac{1}{P_t^{T*}\theta}$. An import price shock is then isomorphic to a factor-biased domestic productivity shock.

requires $NX_1 + R^{*-1}NX_2 = 0$.

3.1.7 Equilibrium Definition

Given fiscal-monetary policy, (τ, T, N_t), an equilibrium is a set of allocations and prices, $\{C_t^p, C_t^r, F^*, M_t, Y_t, X_t, w_t, \epsilon_t\}$ such that:

i. HtM households satisfy their budget constraints, (9).
ii. Ricardian households satisfy the Euler equation, (11) and the budget constraints in (10).
iii. Firms satisfy the cost minimization conditions, (4), (5); and wage determination condition, (6).
iv. Export demand satisfies (8).
v. The fiscal budget constraint, (12), holds.
vi. The domestic goods market clears, i.e., (1) holds.
vii. The labor market clears, i.e., the labor demand by the firms equals the labor supply, N_t.
viii. The balance of payments conditions, (13), hold.

3.2 Optimal Policy Problem

The equilibrium definition in Section 3.1.7 describes the set of all feasible decentralized equilibria, indexed by the public agent's choice of fiscal-monetary policy (τ, T, N_t). We now describe the optimal policy problem of a utilitarian public agent that chooses a sequence of policies, $\{N_t, \tau, T\}$, to maximize social welfare subject to the equilibrium conditions listed in Section 3.1.7.

We take the primal approach and state the optimization problem of a Ramsey planner who directly chooses $\{N_t, \epsilon_t\}$ subject to the implementability constraints implied by household and firm optimization. We then illustrate that the Ramsey planner's solution can be implemented as an equilibrium with an appropriate choice of capital control policy, $\{\tau, T\}$.

Consider the following constrained optimization problem:

$$\max_{\{N_t, \epsilon_t\}} \sum_{t=1}^{2} \beta^{t-1} \left[\lambda u\left(C_t^p(N_t, \epsilon_t)\right) + (1-\lambda) u\left(C_t^r(N_t, \epsilon_t)\right) - v(N_t)\right] \qquad \text{s.t.}$$
$$NX_1(N_1, \epsilon_1) + \frac{1}{R^*} NX_2(N_2, \epsilon_2) = 0 \tag{16}$$

where:

$$C_t^p(N_t, \epsilon_t) = Y_t^N(N_t, \epsilon_t) \tag{17}$$

$$C_t^r(N_t, \epsilon_t) = Y_t^N(N_t, \epsilon_t) - \frac{\epsilon_t NX_t(N_t, \epsilon_t)}{1-\lambda} \tag{18}$$

and $Y_t^N(N_t, \epsilon_t)$ and $NX_t(N_t, \epsilon_t)$ are given by labor income equation (7) and the trade balance equation (15), respectively. Equation (17) is obtained by combining the HtM households' budget constraint, (9), with wage determination equation (6). Equation (18) is obtained by combining the Ricardian households' budget constraints in (10) with the labor income equation, (7), and the balance of payments conditions, (13). This equation implies that Ricardian households are net borrowers in period t when $NX_t < 0$, consuming more

than their period labor income, and net savers when $NX_t > 0$, consuming less. Since Ricardians save and borrow in units of the foreign good, the domestic consumption that such saving or borrowing buys is the product of the real exchange rate, ϵ_t, and the trade balance in foreign units, NX_t. Finally, the constraint in (16) is the intertemporal budget constraint, (14): the present discounted value of net export revenue must equal zero.

We denote the solution to the optimization problem, (16), as the constrained efficient allocation. In Appendix A.1, we derive the first-order conditions for this problem and obtain the system of equations that characterize the solution. The solution in $\{N_t, \epsilon_t\}$ is characterized by an Euler-like intertemporal condition, (74), an intratemporal consumption-labor condition, (78) for $t = 1, 2$, and the intertemporal budget constraint in (16). In Appendix A.2 we discuss sufficient conditions that ensure that the solution is a unique maximum.

The constrained planning problem, (16), internalizes all equilibrium constraints except the Ricardian Euler equation, (11), and the goods-market-clearing condition, (1). The Ricardian Euler equation is not an implementability constraint for the planner because the planner can distort the intertemporal margin: the intertemporal wedge chosen by the Ramsey planner can be supported in the decentralized economy through an appropriate choice of τ. The goods-market-clearing condition is implied by Walras' identity. Lemma 1 formally shows that the solution to (16) can be supported as a decentralized equilibrium with an appropriate choice of policy, $\{\tau, T\}$. We define this decentralized equilibrium as the Ramsey equilibrium.

Lemma 1. *Let $\{N_t, \epsilon_t\}$ be a solution to the constrained optimization problem in (16). Then there exists a policy, $\{\tau, T\}$, such that the solution to (16) is implementable as an equilibrium defined in Section 3.1.7.*

Proof. See Appendix A.3. □

Before we proceed to characterize the constrained efficient allocation under different assumptions on λ and import price sequences, $\{P_t^{T*}\}$, it is useful to state a few definitions. Define the average marginal utility, $\bar{u}'_t$, as the population-weighted average of the marginal utilities of the two types of households:

$$\bar{u}'_t \equiv \lambda\, u'(C_t^p) + (1-\lambda)\, u'(C_t^r) \tag{19}$$

Define the labor wedge, τ_t^N, as the deviation of the average marginal rate of substitution between consumption and leisure from the real wage:

$$1 - \tau_t^N \equiv \frac{v'(N_t)}{w_t \bar{u}'_t} \tag{20}$$

Using the Ricardian Euler equation, (11), the intertemporal wedge corresponds to the capital control tax, τ:

$$1 + \tau \equiv \frac{\beta R^*\, u'(C_2^r)\, \epsilon_2}{u'(C_1^r)\, \epsilon_1} \tag{21}$$

In Appendix A.1, we use the first-order conditions of the Ramsey planner's problem to obtain closed-form expressions for the optimal labor and intertemporal wedges in terms of the optimal allocation, $\{N_t, \epsilon_t\}$. The optimal labor wedge is given by equation (80), and the optimal intertemporal wedge is given by equation (81).

Policy is said to be operative on a margin if it selects a non-zero wedge on that margin. On the other hand, an equilibrium is defined as laissez-faire if the policy chooses no distortion of the intertemporal and labor margins, i.e., sets the wedges to zero: $\tau = 0$ and $\tau_t^N = 0$. In what follows, we characterize the constrained efficient allocation under different assumptions on λ and import prices, P_t^{T*}, and compare this solution with laissez-faire equilibria and equilibria with sub-optimally chosen wedges.

3.3 Optimal Policy: Representative-Agent Benchmark

The representative-agent case is nested in the heterogeneous-agent problem by setting $\lambda = 0$: no household is borrowing constrained and the household's Euler equation always holds. Proposition 1 characterizes its constrained efficient allocation.

Proposition 1. *If $\lambda = 0$, then given any sequence of import prices and productivity, $\{P_t^{T*}, A_t\} > 0$, the constrained efficient allocation involves:*

- *No intertemporal distortions: $\tau = 0$.*
- *A strictly positive labor wedge that implements the optimal terms-of-trade:*

$$\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*} \theta}{w_t} > 0$$

Proof. See Appendix A.4. □

First, with a single household type that participates freely in international asset markets, there is no role for policy to distort the intertemporal margin. The proposition establishes that the undistorted Ricardian Euler equation coincides with the planner's optimality condition: the agent's borrowing and saving already smooth consumption optimally for any path of import prices and productivities, and the planner has no incentive to distort the intertemporal price. Therefore, in the decentralized economy, the optimal capital control tax is zero.

Second, given that, under the Marshall-Lerner condition, the economy faces an upward-sloping supply of foreign purchasing power in the form of export revenues, the model features a classic terms-of-trade distortion: competitive firms do not internalize the effect of their import demand on the real exchange rate. Each additional unit of import demand must depreciate the real exchange rate to induce foreigners to supply additional foreign purchasing power. This depreciation increases the firms' costs in the form of domestic output paid to foreigners: not only for the marginal unit of foreign purchasing power supplied by them but also for all inframarginal units. Noting that exports, X_t, can be written as a product of the real exchange rate, ϵ_t, and the foreign good supplied by foreigners, $\zeta \epsilon_t^\chi$, we have:

$$dX_t = \epsilon_t \chi \zeta \epsilon_t^{\chi-1} d\epsilon_t + \zeta \epsilon_t^\chi d\epsilon_t$$

where the first term is the direct quantity effect: a real depreciation increases the quantity of the foreign good supplied at the margin; and the second term is the inframarginal price effect: a real depreciation increases the price of the foreign good for all inframarginal units supplied by foreigners. Holding NX_t

constant, taking the total differential of (15) yields:

$$P_t^{T*}\theta dN_t = \chi\zeta\epsilon_t^{\chi-1}d\epsilon_t$$

Together this gives us:

$$dX_t = \epsilon_t P_t^{T*}\theta dN_t + \frac{1}{\chi}\epsilon_t P_t^{T*}\theta dN_t$$

where the first and second terms have the same interpretation as above. The second term is the terms-of-trade effect, which constitutes a resource loss for the economy that price-taking firms do not internalize. Such terms-of-trade effects have been discussed in the classic optimal trade policy literature at least since Johnson (1953), are closely related to the general-equilibrium argument for optimal tariffs in Dixit and Norman (1980), and arise in a small open economy whenever export demand has finite elasticity (Corden 1974; De Paoli 2009). A Ramsey planner internalizes such terms-of-trade effects and considers the marginal cost of expanding employment and imports to be higher than private agents do. Define $\hat{w}_t$ as the effective marginal product of labor, i.e., the net resources retained by the domestic economy when employment is increased by a marginal unit after accounting for the terms-of-trade effect:

$$\hat{w}_t \equiv A_t - \frac{dX_t}{dN_t} = w_t - \frac{1}{\chi}\epsilon_t P_t^{T*}\theta$$

The planner values the marginal unit of labor at $\hat{w}_t$, and the constrained efficient allocation therefore restricts employment and imports below the laissez-faire level, which reduces output and exports and results in a more appreciated real exchange rate in equilibrium relative to the competitive outcome[6]. The optimal labor wedge is proportional to the inframarginal price effect term, $\frac{1}{\chi}\epsilon_t P_t^{T*}\theta$, derived above.

This classic terms-of-trade distortion is neither a focus nor a requirement for the main results of the paper. It arises in our framework because the Marshall-Lerner export revenue elasticity is finite. Moreover, to isolate the impact of heterogeneity on optimal policy, it is useful to identify and separate out the effect of this terms-of-trade distortion in the overall policy response, and the representative-agent framework serves as the benchmark for that purpose.

Before proceeding to the heterogeneous-agent model, it is useful to conduct a comparative statics exercise with respect to the import price, P^{T*}. Lemma 2 characterizes the constrained efficient allocation for the case where import prices are constant over time, $P_1^{T*} = P_2^{T*} = P^{T*}$, and then discusses how an increase in the period 1 price induces households to borrow in international asset markets to smooth this income shock.

Lemma 2. *Let* $\lambda = 0$, $A_t = A > 0$ *for* $t \in \{1, 2\}$, *and* $\beta R^* = 1$. *Let all other parameters satisfy the conditions*

[6] In our framework import-export tariffs are unavailable. However, under Leontief production imports move one for one with employment, and the labor wedge therefore internalizes the terms-of-trade distortion and implements the same allocation an optimal tariff would implement. With isoelastic export demand the distortion is purely static: the labor wedge corrects it period by period independently, and the intertemporal wedge provides no additional leverage for this purpose. It follows that if labor supply is perfectly inelastic, the planner has no instrument with which to internalize the distortion, yet the optimal intertemporal wedge is still zero and the constrained efficient allocation coincides with the competitive laissez-faire allocation. Section 3.5 shows that this is special to Leontief production: under a general CES technology facing unequal import prices, neither wedge alone corrects the distortion and the planner distorts both margins. The Leontief production framework is therefore a special and analytically useful case because any intertemporal distortions chosen when $\lambda > 0$ reflect the role of household heterogeneity alone.

of Lemma A.1.

i. ***Symmetric equilibrium.*** *If* $P_1^{T*} = P_2^{T*} = P^{T*}$*, then the constrained efficient allocation satisfies*

$$N_1 = N_2, \qquad \epsilon_1 = \epsilon_2, \qquad NX_1 = NX_2 = 0$$

ii. ***Comparative statics.*** *In addition, assume that utility over consumption satisfies* $u'(C) = C^{-\sigma}$ *with* $\sigma \geqslant \frac{1}{1+\chi}$*. Fix* $P_2^{T*} = P^{T*}$ *and let* $P_1^{T*} > P^{T*}$*. Then the constrained efficient allocation satisfies*

$$NX_1 < 0 < NX_2$$

Proof. See Appendix A.5. □

When import prices are equal across periods, part (i) establishes that the constrained efficient allocation is symmetric: employment, the real exchange rate, and the trade balance are identical in the two periods, and trade is intratemporally balanced. With $\beta R^* = 1$ and no asymmetry between the two periods, the representative agent has no incentive to shift resources intertemporally, and equal consumption across periods is optimal. The labor wedge is active in each period, statically correcting the terms-of-trade distortion, but the intertemporal wedge is undistorted.

Part (ii) considers a temporary increase in the import price in period 1, with $P_1^{T*} > P^{T*}$. The high input price reduces real wages and compresses real income in period 1. The representative agent responds by borrowing freely in international asset markets at the world interest rate to smooth consumption across periods, running a trade deficit in period 1 and a surplus in period 2. The planner sets $\tau = 0$: the privately optimal current account path is also socially optimal, and there is no role for intertemporal policy. The labor wedge corrects the terms-of-trade distortion in each period independently.

The representative-agent benchmark therefore delineates the role of each policy instrument. Given the absence of import-export tariffs, the labor wedge through monetary policy internalizes the terms-of-trade distortion period by period, and the optimal capital control tax is zero for any sequence of import prices. This outcome is a close analog of conventional policy prescriptions in representative-agent small open economy models: since there are no frictions in asset markets, real income shocks do not call for interventions in foreign exchange markets. In response to import price spikes, the exchange rate moves freely as households optimally smooth consumption. However, when a fraction of households cannot borrow against future income to smooth consumption, asset markets need not be left undistorted. The following subsections discuss optimal policies in a heterogeneous-agent economy with a non-zero mass of hand-to-mouth households.

3.4 Heterogeneous Agents ($\lambda \in (0, 1)$)

We now turn to the case $\lambda \in (0, 1)$. A fraction λ of households cannot borrow against future income. When import prices are temporarily high, Ricardian households respond by borrowing in international asset markets, insulating their consumption from the income shock. HtM households have no such recourse: they consume their current labor income each period and bear the full income loss contemporaneously.

This asymmetry in asset market access creates a possible role for policy.

To isolate the intertemporal dimension of optimal policy, we begin with the case of inelastic labor supply, $N_t = \bar{N}$. With employment fixed exogenously, the only instrument available to the public agent is the capital control tax, τ, and the Ramsey planner's problem reduces to choosing a point on the feasible frontier defined by the intertemporal budget constraint. The labor wedge and its interaction with the intertemporal wedge are taken up in Section 3.4.2.

3.4.1 Inelastic Labor Supply ($N_t = \bar{N}$)

Equal import prices. We begin with the case $P_1^{T*} = P_2^{T*} = P^{T*}$. With $\beta R^* = 1$ and no asymmetry between the two periods, neither household type has an incentive to shift resources intertemporally. The privately optimal asset position for Ricardian households is $f^* = 0$, implying balanced trade in each period. With $NX_1 = NX_2 = 0$, both household types consume their current labor income in each period. Since the income path is identical across periods, consumption is equal over time for both types, and since trade is balanced in each period, Ricardian and HtM households also consume equally within each period: $C_t^p = C_t^r$ for $t \in \{1, 2\}$.

This allocation is optimal for both household types. HtM households would choose it even if they had access to asset markets, and in this special case their inability to smooth consumption is inconsequential. The planner therefore has no incentive to distort the intertemporal margin, and the constrained efficient allocation coincides with laissez-faire: $\tau = 0$ and the exchange rate path is undistorted. This result is the direct counterpart of part (i) of Lemma 2: with equal prices and $\beta R^* = 1$, the intertemporal wedge is zero both in the representative-agent economy and in the heterogeneous-agent economy. Lemma 3 formally summarizes this result.

Lemma 3. *Assume inelastic labor supply:* $N_t = \bar{N} > 0$. *Let* $\lambda \in (0, 1)$, $A_t = A > 0$ *for* $t \in \{1, 2\}$, *and* $\beta R^* = 1$. *Let* $P_1^{T*} = P_2^{T*} = P^{T*} > 0$. *Then the constrained efficient allocation involves no intertemporal distortions:* $\tau = 0$, *i.e., it coincides with the laissez-faire equilibrium, and satisfies:*

$$\epsilon_1 = \epsilon_2, \qquad NX_1 = NX_2 = 0, \qquad C_1^r = C_1^p = C_2^r = C_2^p$$

Proof. Follows as a Corollary to Proposition 5. □

The laissez-faire equilibrium under $P_1^{T*} > P_2^{T*}$. Now consider the case where $P_1^{T*} > P_2^{T*} = P^{T*}$. Before turning to optimal policy we characterize the laissez-faire equilibrium, which provides the reference point against which optimal policy is measured. The high input price in period 1 compresses real labor incomes, and Ricardian households borrow in international asset markets to smooth consumption over their lifetime. As in the representative-agent case of Lemma 2, the economy runs a trade deficit in period 1 and a surplus in period 2: $NX_1 < 0 < NX_2$.

The heterogeneity in asset market access, however, introduces a dimension that is absent in the representative-agent economy. Ricardian households supplement their compressed period-1 labor income with the resources

borrowed from abroad, consuming strictly more than their current incomes. HtM households, by contrast, consume their current labor income each period. In period 1, Ricardians consume strictly more than HtM households, $C_1^r > C_1^p$; in period 2 the pattern reverses, $C_2^r < C_2^p$, as Ricardians service their debt while HtM households enjoy the full recovery of labor incomes. The laissez-faire equilibrium therefore features a consumption differential between the two household types that varies over the income cycle. Lemma 4 formally establishes these properties.

Lemma 4. *Assume inelastic labor supply:* $N_t = \bar{N} > 0$. *Let* $\lambda \in (0,1)$, $A_t = A > 0$, *for* $t \in \{1,2\}$, *and* $\beta R^* = 1$. *Let* $P_2^{T*} = P^{T*} > 0$ *and* $P_1^{T*} > P^{T*}$. *Consider the laissez-faire equilibrium, which imposes* $\tau = 0$. *The equilibrium satisfies:*

$$NX_1 < 0 < NX_2, \qquad C_1^r > C_1^p, \qquad C_2^r < C_2^p$$

Proof. See Appendix A.6. □

A perturbation along the feasible frontier. Before characterizing optimal policy, it is useful to examine how a capital control tax affects equilibrium outcomes. The intertemporal budget constraint, $NX_1 + R^{*-1}NX_2 = 0$, defines the set of feasible decentralized equilibria that can be reached for some choice of τ. We take the laissez-faire equilibrium characterized in Lemma 4 as the reference point and consider the impact of an increase in the capital control tax[7]. By distorting Ricardian households' saving decisions, this policy affects the intertemporal path of the real exchange rate, and through it both the sharing of resources between the domestic economy and the rest of the world and the distribution of domestically retained resources between the two household types.

A higher capital control tax, $d\tau > 0$, raises the private cost of Ricardian foreign saving and equivalently lowers the cost of foreign borrowing, inducing additional period-1 borrowing. Let $B \equiv -NX_1 > 0$ denote period-1 borrowing; the tax generates an additional $dB > 0$. By the intertemporal budget constraint, this additional borrowing must be repaid: $dNX_2 = R^* dB > 0$. In what follows, we trace the equilibrium consequences of a perturbation, $dB > 0$.

The period-1 balance of payments condition can be written as:

$$P_1^{T*}\theta N = \zeta \epsilon_1^X + B$$

Written this way, the BoP condition is the market-clearing condition for foreign purchasing power in the domestic economy's foreign exchange market. The left-hand side is firms' demand for foreign purchasing power to pay for the imported input, which is perfectly inelastic in equilibrium because labor supply is fixed. The right-hand side is the supply of foreign purchasing power: foreigners supply $\zeta \epsilon_1^X$ by purchasing the domestic good, and Ricardians supply B through their borrowing in international asset markets. By the Marshall-Lerner condition, the purchasing power supplied by foreigners is increasing in ϵ_1: a real depreciation induces them to buy more of the domestic good and consequently supply more in return, and

[7] The economic intuition discussed in this subsection holds for any point on the intertemporal budget constraint satisfying $NX_1 < 0 < NX_2$, given any import price sequence $\{P_1^{T*}, P_2^{T*}\}$; the laissez-faire allocation is a convenient starting point.

a real appreciation does the reverse. Since the net demand by firms is perfectly inelastic, a rise in B crowds out export revenues one-for-one. In other words, Ricardians bring additional foreign purchasing power and seek to exchange it for the domestic good, creating an excess supply that they clear by bidding down the real exchange rate. Taking the total differential of the BoP condition shows that the equilibrium requires a real appreciation:

$$d\epsilon_1 = -\frac{dB}{\chi\zeta\epsilon_1^{\chi-1}} < 0 \tag{22}$$

The lower the Marshall-Lerner export revenue elasticity, the larger the appreciation that a given increase in borrowing produces. By inducing Ricardians to borrow more, the public agent substitutes borrowed funds for export revenues as the source of foreign purchasing power for firms, reducing the need to sell domestic output to foreigners in the current period. To trace how this changes the flow of domestic output in the foreign exchange market, we multiply the period-1 BoP condition by ϵ_1 to express it in domestic units:

$$\epsilon_1 P_1^{T*}\theta N = \zeta\epsilon_1^{1+\chi} + \epsilon_1 B$$

The left-hand side is the share of domestic output sold by firms in the foreign exchange market in return for buying the foreign input, which represents a share of firms' production costs. The first term on the right-hand side represents exports, X_1, and the second term, $\epsilon_1 B$, is the domestic output purchased by Ricardians with borrowed foreign resources. Taking the total differential gives:

$$P_1^{T*}\theta N\, d\epsilon_1 = (1+\chi)\zeta\epsilon_1^{\chi}\, d\epsilon_1 + (\epsilon_1 dB + B d\epsilon_1) \tag{23}$$

Since $d\epsilon_1 < 0$, the left-hand side of (23) measures the total cost savings for firms. As Ricardians bid down the real exchange rate, the domestic-good cost of the imported input, $\epsilon_1 P_1^{T*}$, and therefore firms' marginal cost of production, falls. At the prevailing real wage, hiring a unit of labor and buying θ units of input to produce A units of output now creates a positive profit opportunity. Firms attempt to expand production and demand more inputs, and since inputs and labor are perfect complements, the demand for labor also expands. With the labor endowment fixed at N in equilibrium, competition for workers instead bids up the real wage until positive profit opportunities have been eliminated. The firms' zero-profit condition, (6), implies that the real wage must rise one-for-one with a real appreciation. In other words, cost savings by firms are passed through to workers in the form of higher real labor incomes. The increase in labor income is therefore:

$$dY_1^N = -P_1^{T*}\theta N\, d\epsilon_1 > 0 \tag{24}$$

Workers gain these resources, and the right-hand side of (23) shows who gives them up. The net cost reduction for firms can be decomposed into two components. The first expression is $dX_1 = (1+\chi)\zeta\epsilon_1^{\chi}\, d\epsilon_1$, the decrease in exports. Using (22) we have:

$$dX_1 = -\left(\epsilon_1 dB + \frac{1}{\chi}\epsilon_1 dB\right) < 0 \tag{25}$$

The decrease in exports itself decomposes into a quantity effect, $\epsilon_1 dB$, and an inframarginal price effect, $\frac{1}{\chi}\epsilon_1 dB$. The quantity effect is the domestic output no longer sold to foreigners because Ricardian borrowing replaces foreign-supplied purchasing power at the margin. The price effect arises because the Marshall-

Lerner export revenue elasticity is finite: a real appreciation decreases the domestic output paid to foreigners at the margin and against all inframarginal units of foreign purchasing power supplied by them.

The second expression on the right-hand side of (23) represents the net change in the quantity of the domestic good bought by Ricardians with the borrowed foreign good. The quantity effect, $\epsilon_1 dB$, is positive: an increase in borrowing allows Ricardians to buy more of the domestic good at any given price. The price effect, $B d\epsilon_1$, is negative: a real appreciation lowers the domestic-good value of all foreign resources exchanged by Ricardians, not only the marginal unit. The sign of the net change depends on whether the quantity effect dominates the price effect or vice versa.

Substituting (24) and (25) into (23), the net increase in labor income can be expressed as the sum of two inframarginal price effects associated with a real exchange rate appreciation:

$$dY_1^N = \left(\frac{\epsilon_1}{\chi} - B \frac{d\epsilon_1}{dB} \right) dB > 0 \tag{26}$$

The first term is the inframarginal price effect on foreigners: an appreciation lowers the domestic output paid against all inframarginal units of foreign purchasing power supplied through exports. The second term is the inframarginal price effect on Ricardians: an appreciation lowers the domestic-good value of all inframarginal foreign resources borrowed by Ricardians[8]. Both household types therefore gain through higher labor income, financed by foreigners and Ricardians giving up their claims on domestic resources.

Thus, a subsidy to Ricardian borrowing induces a real exchange rate appreciation, which, through cost reduction for firms, results in an endogenous transfer of real resources toward labor incomes and away from foreign exchange market transactions. Since labor incomes are gained by both Ricardians and HtM households, this policy redistributes through pecuniary general equilibrium effects. Exports fall, while HtM consumption rises one-for-one with labor income. The change in Ricardian consumption is ambiguous: the labor-income gain and the positive quantity effect of additional borrowing are offset partly or fully by the inframarginal price effect:

$$dC_1^p = dY_1^N > 0$$

$$dC_1^r = dY_1^N + \frac{1}{1-\lambda} \left[\epsilon_1 dB + B \, d\epsilon_1 \right]$$

However, in period 2 Ricardians face a larger debt burden. The intertemporal resource constraint requires that an appreciation in period 1 must be followed by a depreciation in period 2. The economy must export more in period 2 to pay back the larger period-1 borrowing. A real depreciation in period 2 does an analogous endogenous resource transfer but in the opposite direction: it reduces labor incomes and increases the share of output sold to foreigners. Using the period-2 BoP condition with $dNX_2 = R^* dB > 0$:

$$d\epsilon_2 = \frac{R^* dB}{\chi \zeta \epsilon_2^{\chi - 1}} > 0 \tag{27}$$

A larger repayment obligation depreciates the real exchange rate in period 2. This raises the domestic-good price of the imported input and hence firms' marginal cost, inducing them to reduce production. At the

[8]The positive quantity effect on Ricardians cancels against the negative quantity effect on foreigners. The net gain in labor income is the sum of the two inframarginal price effects.

prevailing wage, firms make losses. Labor demand falls, and the real wage declines until zero profits are restored. Hence labor income falls:

$$dY_2^N = -P_2^{T*}\theta N \, d\epsilon_2 < 0 \tag{28}$$

Since HtM households consume current labor income, their consumption falls in period 2. Ricardians are worse off as well: labor income falls and they face a higher repayment burden in domestic units, and both the quantity and price effects are unambiguously negative:

$$dC_2^p = dY_2^N < 0$$

$$dC_2^r = dY_2^N - \frac{1}{1-\lambda}\left[\epsilon_2 R^* dB + R^* B \, d\epsilon_2\right] < 0$$

The net fall in labor income can be decomposed in a way that parallels the period-1 expression:

$$dY_2^N = -\left[\frac{\epsilon_2}{\chi} - B\frac{d\epsilon_2}{dB}\right] R^* dB \tag{29}$$

The first term is the inframarginal export-price effect: the depreciation increases domestic output sold to foreigners against all inframarginal units of foreign purchasing power supplied by them. The second is the inframarginal price effect on Ricardian repayment obligations: a depreciation increases the domestic-good value of their existing debt burden. Since $B > 0$, this effect places part of the burden directly on Ricardians and dampens the fall in labor income.

Thus, a marginal increase in period-1 borrowing redistributes resources toward labor income in period 1 and away from labor income in period 2. Exports fall in period 1 and rise in period 2: HtM households, who live off current labor income, gain in the first period and lose in the second, while Ricardians face an ambiguous period-1 effect and an unambiguous period-2 loss. This intertemporal endogenous redistribution of resources is driven by the path of the real exchange rate, which policy can choose by distorting the intertemporal margin.

The planner's first-order condition. To characterize optimal policy, we return to the primal problem, (16), where a utilitarian Ramsey planner directly chooses the optimal exchange rate path by internalizing the pecuniary general equilibrium effects of exchange rate movements. In Appendix A we derive the Ramsey planner's first-order condition with respect to ϵ_t, which we restate below to make explicit the net marginal benefits of an appreciation at time t:

$$(\beta R^*)^{t-1}\left[\bar{u}_t'\left(\frac{\epsilon_t}{\chi} - NX_t\frac{d\epsilon_t}{dNX_t}\right) + u'(C_t^r)\epsilon_t + u'(C_t^r)NX_t\frac{d\epsilon_t}{dNX_t}\right] = \mu \tag{30}$$

where $\bar{u}_t'$ is average marginal utility, defined in (19), and μ is the Lagrange multiplier on the intertemporal budget constraint. The left-hand side captures the net benefit of a marginal appreciation in period t. The first term is the labor-income gain valued at average marginal utility. The term in parenthesis is identical to the net change in labor income decomposed into two inframarginal price effects, as illustrated in equations (26) and (29) above. The second term is the direct quantity effect on the Ricardian foreign asset position. The third is the price effect: an appreciation lowers the domestic-good value of all inframarginal foreign

resources exchanged by Ricardians. The first and second terms have positive signs. The third term's sign depends on the sign of NX_t: if the Ricardians are net borrowers ($NX_t < 0$), the price effect is negative, and if they are net savers ($NX_t > 0$), the price effect is positive.

Equation (30) can be rearranged to group appreciation-induced resource transfers among agents into two net transfers. First, a net transfer between foreigners and the domestic economy, and, second, a net transfer between the two household types within the domestic economy:

$$(\beta R^*)^{t-1}\left[u'(C_t^r)\epsilon_t\left(1+\frac{1}{\chi}\right)+\lambda\big(u'(C_t^p)-u'(C_t^r)\big)\left(\frac{\epsilon_t}{\chi}-NX_t\frac{d\epsilon_t}{dNX_t}\right)\right]=\mu \tag{31}$$

The first term is interpreted as a net resource transfer from foreigners to domestic agents valued at Ricardian marginal utility: an appreciation reduces exports at the margin and lowers the domestic output paid against inframarginal foreign purchasing power. The second term captures the net resource transfer between domestic households. It represents the pecuniary general equilibrium effect of a real appreciation on HtM households through their labor incomes. Since a share λ of the overall increase in labor income accrues to HtM households, it is effectively equivalent to a net resource transfer from Ricardians to HtMs. The planner values this transfer at the marginal utility difference between the two types: how much more HtM households value these resources relative to Ricardians.

The Lagrange multiplier μ captures the marginal cost of a real exchange rate appreciation in period t through a tightening of the intertemporal budget constraint. Feasibility dictates that an appreciation in period t must be followed by a depreciation in period $t+1$. A marginal depreciation at time $t+1$ relaxes the intertemporal budget constraint, the marginal utility benefit of which is also μ. An optimizing planner equates the benefits of inducing a marginal appreciation at time t to the costs of inducing a marginal depreciation at time $t+1$. This yields the planner's intertemporal optimality condition:

$$\epsilon_1\,u'(C_1^r)+\Omega_1=\beta R^*\Big[\epsilon_2\,u'(C_2^r)+\Omega_2\Big] \tag{32}$$

where

$$\Omega_t\equiv\frac{\lambda\big[u'(C_t^p)-u'(C_t^r)\big]\left(\frac{\epsilon_t}{\chi}-NX_t\frac{d\epsilon_t}{dNX_t}\right)}{\frac{1+\chi}{\chi}} \tag{33}$$

Ω_t is the redistributive wedge that captures the social value of a net resource transfer between domestic households per unit of the resource transfer between foreigners and the domestic economy induced by a marginal real exchange rate change at time t.

Optimal policy under $P_1^{T*} > P_2^{T*}$. The intertemporal optimality condition (32) and the intertemporal budget constraint characterize the constrained efficient allocation for any import price sequence. From Lemma 4, the laissez-faire allocation satisfies $NX_1^{LF} < 0 < NX_2^{LF}$, which implies $C_1^{r,LF} > C_1^{p,LF}$ and $C_2^{r,LF} < C_2^{p,LF}$. HtM households therefore have strictly higher marginal utility than Ricardians in period 1 and strictly lower marginal utility in period 2, giving $\Omega_1^{LF} > 0$ and $\Omega_2^{LF} < 0$. Moreover, the laissez-faire allocation satisfies the Euler equation: $\epsilon_1^{LF}u'(C_1^{r,LF}) = \beta R^*\epsilon_2^{LF}u'(C_2^{r,LF})$. Evaluating the left- and

right-hand sides of (32) at the laissez-faire allocation, we get:

$$\epsilon_1^{LF}\, u'(C_1^{r,LF}) + \Omega_1^{LF} \;>\; \epsilon_2^{LF}\, u'(C_2^{r,LF}) + \Omega_2^{LF}$$

Thus the marginal benefit of a real exchange rate appreciation in period 1 exceeds the marginal cost of the required depreciation in period 2 at the laissez-faire allocation whenever $P_1^{T*} > P_2^{T*}$. The inability of HtM households to intertemporally smooth consumption in response to a transitory fall in real incomes makes their income and consumption excessively volatile from the perspective of a utilitarian planner. By inducing a real exchange rate appreciation in period 1, and a depreciation in period 2, the planner can tilt the economy's access to resources toward the high-input price period and away from the low-input price period. This motive is absent when all households participate in asset markets; with a mass of constrained households, the same tilt induces an endogenous resource transfer toward HtM households in the high-input price period and away from them in the low-input price period.

The period-1 appreciation reduces the domestic-good cost of the imported input, raises firms' residual profits at the prevailing wage, and through the zero-profit condition (6) bids wages up until zero profits are restored. Labor income rises in period 1, and since HtM households consume current labor income one-for-one, their consumption rises. In period 2, the depreciation raises input costs, compresses labor incomes, and reduces HtM consumption. By tilting the exchange rate path, the planner makes HtM consumption intertemporally smoother than under laissez-faire: the compression in period 1 is partially offset, and the recovery in period 2 is partially dampened. For Ricardian households, the change in consumption in period 1 is ambiguous: the labor income gain and the positive quantity effect are partially or fully offset by the appreciation-induced reduction in the domestic-good value of their foreign borrowing. In period 2 their consumption unambiguously falls, as they face both lower labor income and a higher domestic-good cost of repayment. The net effect on social welfare is strictly positive, however: at utilitarian Pareto weights, a marginal appreciation at $t = 1$ with the required depreciation at $t = 2$ always raises welfare starting from the laissez-faire allocation.

Therefore, optimal policy involves a more appreciated real exchange rate in period 1 and a more depreciated real exchange rate in period 2 relative to laissez-faire: $\epsilon_1^O < \epsilon_1^{LF}$ and $\epsilon_2^O > \epsilon_2^{LF}$. The real exchange rate consequently co-moves less with the import price. Moreover, the trade deficit must be larger in period 1 and the trade surplus must be larger in period 2 relative to laissez-faire. At the same time, this appreciation partially offsets the exogenous spike in the world input price by compressing its domestic-good equivalent in the high-price period at the cost of a higher domestic equivalent in the low-price period, thereby stabilizing the country's terms of trade. Since real labor income, Y_t^N, i.e., the country's GDP, moves one-for-one against the terms of trade, it is also relatively stabilized under the conditions of Lemma A.2: higher in the high-input price period and lower in the low-input price period under the constrained efficient allocation. Proposition 2 summarizes these results.

Finally, the constrained efficient allocation involves a strictly positive intertemporal wedge, i.e., in a decentralized equilibrium, this allocation is implementable through a strictly positive capital control tax on Ricardian asset holdings, or equivalently a subsidy on Ricardian borrowing. The tax on Ricardian saving raises their private cost of accumulating foreign assets, inducing them to borrow more in period 1 than

they would choose under laissez-faire. The public agent in effect works through the Ricardian households' asset market participation and the pecuniary general equilibrium effects of real exchange rate changes. In Proposition 2 we establish that whenever $P_1^{T*} > P_2^{T*}$, the constrained efficient allocation is decentralized with a strictly positive capital control tax at $t = 1$, i.e., $\tau > 0$:

$$\tau = \frac{\Omega_1^O - \beta R^* \Omega_2^O}{\epsilon_1^O \, u'(C_1^{r,O})} > 0 \tag{34}$$

Proposition 2. *Assume that labor supply is inelastic,* $N_t = \bar{N} > 0$. *Let* $\lambda \in (0,1)$, $A_t = A > 0$ *for* $t \in \{1,2\}$, $\beta R^* = 1$, $0 < \chi \leqslant 1$, $P_2^{T*} = P^{T*} > 0$, *and* $P_1^{T*} > P^{T*}$. *Let* $(\epsilon_t^O, C_t^{r,O}, C_t^{p,O}, NX_t^O, Y_t^{N,O})$ *denote the constrained efficient allocation and* $(\epsilon_t^{LF}, C_t^{r,LF}, C_t^{p,LF}, NX_t^{LF}, Y_t^{N,LF})$ *denote the laissez-faire allocation. The constrained efficient allocation has the following properties:*

i. The intertemporal wedge is strictly positive: $\tau > 0$.

ii. The real exchange rate co-moves less with the import price than under laissez-faire. It is strictly more appreciated in period 1 *and strictly more depreciated in period* 2*:*

$$\epsilon_1^O < \epsilon_1^{LF}, \qquad \epsilon_2^O > \epsilon_2^{LF}$$

iii. The trade deficit is strictly larger in period 1*, and the trade surplus is strictly larger in period* 2*:*

$$NX_1^O < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^O$$

iv. The input price in domestic tradeable units is strictly lower in period 1 *and strictly higher in period* 2*:*

$$\epsilon_1^O P_1^{T*} < \epsilon_1^{LF} P_1^{T*}, \qquad \epsilon_2^O P_2^{T*} > \epsilon_2^{LF} P_2^{T*}$$

v. Real labor income is strictly higher in period 1 *and strictly lower in period* 2*:*

$$Y_1^{N,O} > Y_1^{N,LF}, \qquad Y_2^{N,O} < Y_2^{N,LF}$$

vi. Under the conditions of Lemma A.2, the input price and real labor income are more stable across periods than under laissez-faire:

$$|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}|$$
$$\left|Y_2^{N,O} - Y_1^{N,O}\right| < \left|Y_2^{N,LF} - Y_1^{N,LF}\right|$$

Proof. See Appendix A.7. □

In the high-input price period, exchange rate policy suppresses exports below their laissez-faire level and retains a larger share of domestic output within the economy. Due to inframarginal price effects on foreigners and Ricardians, the resources thus retained flow disproportionately toward labor incomes, and since HtM households consume their current labor income without any recourse to asset markets, this channel raises their consumption in states where import prices are high. In future states when the shock has subsided, the exchange rate depreciates to service the accumulated external obligation, reversing the resource transfer: the export share rises, the domestic share contracts, and real wages fall. The net effect is a partial smoothing of the income and consumption path of HtM households relative to laissez-faire: exchange

rate policy provides a degree of implicit insurance against transitory import price shocks even in the absence of direct fiscal transfers or explicit insurance markets. Proposition 3 summarizes the corresponding changes in output shares.

Proposition 3. *Assume that conditions of Proposition 2 hold. Relative to the laissez-faire equilibrium, the Ramsey equilibrium involves:*

- *a lower export share of output, a higher domestic share, a higher HtM share of output, and an ambiguous change in the Ricardian share of output in the high input price period, i.e., at* $\mathtt{t} = 1$.
- *a higher export share of output and a lower domestic share, a lower HtM and Ricardian share of output in the low input price period, i.e., at* $\mathtt{t} = 2$.

Proof. See Appendix A.8. □

We interpret the intertemporal policy characterized in this subsection as the real-income stabilization channel of exchange rate policy, and we refer to the public agent's reason for distorting the intertemporal margin as the real-income stabilization motive. By distorting the intertemporal margin, the public agent changes the net supply of foreign purchasing power in the domestic economy's foreign exchange market, which in turn changes the real exchange rate and imported-input costs. Firms pass these cost changes into wages, thereby affecting the real incomes of constrained households. The resources arrive indirectly: these pecuniary general equilibrium effects originate in the actions of Ricardians in the foreign exchange market and reach households who do not themselves respond to changes in intertemporal prices.

The economics of our framework naturally shares its flavor with the real income channel of monetary policy in closed-economy HANK models, delivered there through monetary policy's influence on the intertemporal margin. In both cases, policy has no direct way of reaching borrowing-constrained households and must influence their real incomes and consumption through indirect pecuniary general equilibrium effects. There, monetary policy moves the domestic real interest rate, unconstrained households move spending across time, and the resulting change in aggregate demand influences employment and with it the real incomes of borrowing-constrained households. Our open economy framework differs in several ways. First, general equilibrium effects reach constrained households through changes in relative prices rather than through the level of employment or production. By altering the intertemporal price faced by Ricardians, policy influences their participation in the foreign exchange market, and with it the real exchange rate, the imported-input costs faced by firms, and therefore the real wages received by all workers through the zero-profit condition. These effects are operative, and in fact strongest, when employment is perfectly inelastic, as we illustrate in Section 3.5. Second, the source of the indirect general equilibrium effect on constrained households in our framework has no counterpart in closed-economy HANK models: the Marshall-Lerner condition, i.e., an upward-sloping supply of foreign purchasing power through export revenue. When Ricardians borrow more, they crowd out foreigners and appreciate the real exchange rate, which lowers firms' marginal cost and, through the zero-profit condition, feeds into higher real wages. Policy can therefore divert resources away from exports and toward domestic households by inducing stronger Ricardian participation in asset markets. The Marshall-Lerner export revenue elasticity determines how far foreigners are crowded out

and how much the real exchange rate responds: the more elastic the supply, the smaller the exchange rate response to Ricardian participation and the weaker the general equilibrium effect on real wages. Third, in an economy small with respect to global asset markets, the intertemporal price is disciplined by those markets and intertemporal distortions lie beyond the limits of monetary policy. Delivering the real income gains therefore necessitates a separate policy instrument that directly distorts the intertemporal margin, which in our analytical model is a capital control tax and, as we illustrate in Section 4, is equivalently implementable through foreign exchange interventions. Our framework illustrates that the real income influence of policies distorting the intertemporal margin extends beyond the monetary policy setting in which HANK models study it.

Monetary policy in a small open economy still plays a role in distorting the labor wedge and therefore influences real outcomes. In the next subsection, we relax the inelastic labor supply assumption to isolate this role, and show that household heterogeneity modifies the optimal labor wedge beyond the static terms-of-trade correction and that employment-import complementarity requires monetary and exchange rate policies to be set in coordination.

3.4.2 Elastic Labor Supply

The preceding subsection fixes employment at $\bar{N}$ and characterizes the optimal intertemporal wedge in isolation. In this subsection we endogenize the level of employment and input utilization and characterize the two margins jointly in our heterogeneous-agent economy. We interpret the labor wedge as implementable through monetary policy in a decentralized economy with nominal rigidities, while our analytical framework maintains a reduced-form specification where the public agent directly chooses employment. We illustrate that the interaction between household heterogeneity and employment-import complementarity creates a novel role for policy to distort the labor wedge.

The representative-agent framework of Section 3.3 establishes that monetary policy already has a role: the labor wedge corrects for the terms-of-trade distortion that arises because competitive firms do not internalize the effect of their employment and import decisions on the real exchange rate. Facing an upward-sloping supply of foreign purchasing power, an increase in their collective import demand depreciates the real exchange rate and the economy must give up more domestic output to foreigners against all inframarginal units of foreign purchasing power supplied by them. The public agent internalizes this and restricts the economy's employment and import levels below laissez-faire with a strictly positive labor wedge, $\tau_t^N > 0$. There we showed that a Ramsey planner values a marginal unit of employment at the effective marginal product, i.e., the net output retained by the economy for a marginal increase in employment, defined as $\hat{w}$:

$$\hat{w}_t = w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi}$$

where the second term is the terms-of-trade correction. The planner's first-order condition with respect to employment in the representative-agent economy is given by $v'(N_t) = \hat{w}_t u'(C_t)$. To isolate any distortion to the labor wedge beyond the static terms-of-trade correction, we define the terms-of-trade-adjusted labor

wedge, using the average marginal utility $\bar{u}'_t$, as:

$$1 - \hat{\tau}^N_t \equiv \frac{v'(N_t)}{\hat{w}_t \bar{u}'_t} \tag{35}$$

Proposition 1 establishes that the representative-agent constrained efficient allocation satisfies $\hat{\tau}^N_t = 0$ and $\tau = 0$ for any sequence of import prices.

Define the RA-equivalent policy as the one that imposes these same restrictions in the heterogeneous-agent economy: $\hat{\tau}^N_t = 0$ for $t \in \{1, 2\}$ and $\tau = 0$. In other words, the RA-equivalent policy satisfies the undistorted Ricardian Euler equation and implements a labor wedge that just corrects for the static terms-of-trade distortion. Together with the intertemporal budget constraint, it satisfies:

$$\epsilon_1 u'(C^r_1) = \beta R^* \epsilon_2 u'(C^r_2) \tag{36}$$

$$v'(N_t) = \hat{w}_t \bar{u}'_t \qquad t \in \{1, 2\} \tag{37}$$

In what follows, we derive the jointly optimal intertemporal and labor wedges in the heterogeneous-agent economy and compare them to the RA-equivalent policy wedges. The optimal intertemporal wedge carries the same sign as in the previous subsection, and the optimal terms-of-trade-adjusted labor wedge is strictly positive, $\hat{\tau}^N_t > 0$, whenever the trade balance is nonzero, as it is when $P^{T*}_1 \neq P^{T*}_2$.

A perturbation to N_t. Before characterizing the planner's optimality condition with respect to employment, it is useful to consider the partial effect of a perturbation to N_t, $dN_t > 0$, on equilibrium outcomes, holding the trade balance, NX_t, fixed at a given level. Since the trade balance is fixed, the perturbation is purely intratemporal.

Through labor-market clearing, an increase in labor supply, which we interpret as a marginal monetary expansion, requires that firms hire all additional labor and produce more. Since labor and imported inputs are perfect complements, the labor market and the foreign exchange market must clear simultaneously. An increased labor supply creates downward pressure on the real wage, lowering marginal costs and opening up positive profit opportunities that draw firms into demanding more labor and more imported inputs.

Holding NX_t fixed, the foreign purchasing power needed to pay for the additional imports must come from higher export revenues. Because the supply of foreign purchasing power is increasing in the real exchange rate, financing the larger import bill requires a real depreciation. Taking the total differential of the trade balance, $NX_t = \zeta \epsilon^{\chi}_t - P^{T*}_t \theta N_t$ and imposing $dNX_t = 0$ gives:

$$d\epsilon_t = \frac{P^{T*}_t \theta}{\chi \zeta \epsilon^{\chi-1}_t} dN_t > 0 \tag{38}$$

Firms' import bill in domestic units goes up, not only because imports have increased at the margin, but also because the real exchange rate depreciates and increases the cost of all inframarginal imports. Exports expand with the depreciation, and Ricardian households face a pecuniary general equilibrium effect: the depreciation changes the domestic-good value of their net financial flows, NX_t:

$$\epsilon_t P^{T*}_t \theta dN_t + d\epsilon_t P^{T*}_t \theta N_t + d\epsilon_t NX_t = dX_t \tag{39}$$

In equilibrium, zero profits must be restored: as the real exchange rate depreciates, the real wage must fall one-for-one until positive profit opportunities are eliminated:

$$dw_t = -P_t^{T*}\theta \, d\epsilon_t < 0 \tag{40}$$

In other words, the inframarginal increase in firms' costs is paid for entirely by workers, who give up real wages. The net effect on labor income is therefore ambiguous. The direct effect of higher employment raises labor income, and the indirect general-equilibrium effect through the real exchange rate and real wage lowers it:

$$dY_t^N = w_t \, dN_t + N_t \, dw_t = (A_t - \epsilon_t P_t^{T*}\theta)dN_t - d\epsilon_t P_t^{T*}\theta N_t \gtreqless 0 \tag{41}$$

where the first term is the direct effect at the prevailing wage and the second the general-equilibrium effect through the real exchange rate. Combining with (39) we get:

$$A_t dN_t - dX_t = dY_t^N - d\epsilon_t NX_t \tag{42}$$

where the left-hand side, $A_t dN_t - dX_t = \hat{w}_t dN_t$, is the net gain to the domestic economy: marginal output retained domestically after accounting for increased exports, and the right-hand side shows how this additional output is distributed between labor income and this price effect on Ricardians.

Since HtM households consume their current labor income, $dC_t^p = dY_t^N$. Ricardians face the same change in labor income and, in addition, face the price effect on their net financial flows:

$$dC_t^r = dY_t^N - \frac{1}{1-\lambda} NX_t d\epsilon_t$$

where the sign of the price effect depends on NX_t: a depreciation raises the domestic-good value of the foreign resources Ricardians bring into the economy when they are net borrowers, and raises the domestic-good cost of the resources they send abroad when they are net savers. Combining household budgets with Equation (42), the net changes in consumption of the two household types can be expressed as:

$$dC_t^p = \hat{w}_t dN_t + NX_t d\epsilon_t \tag{43}$$

$$dC_t^r = \hat{w}_t dN_t - \frac{\lambda}{1-\lambda} NX_t d\epsilon_t \tag{44}$$

The above equations imply that the price effect on Ricardians is equivalent to a net transfer of resources between the two household types, and we refer to it as the redistributive price effect. For a marginal increase in employment, the domestic economy gains $\hat{w}_t$ units of output, but the distribution of these resources between the two household types is determined endogenously through the real exchange rate and the sign of the country's trade balance. If the country is a net importer/net borrower ($NX_t < 0$), a real depreciation implies an endogenous transfer from HtM households to Ricardians, and if the country is a net exporter/net saver ($NX_t > 0$), the direction of the transfer is reversed.

To summarize, a marginal increase in employment, holding the trade balance fixed, induces a real depreciation and a fall in real wages. This depreciation raises exports and allows firms to import and produce more, and the domestic economy potentially gains resources from this increased production. However, the real depreciation creates a redistributive price effect on Ricardian financial flows, due to which such gains are unequally distributed between the two household types. In other words, by increasing employment a planner

can potentially generate more resources for the domestic economy, but the distribution of these resources is asymmetric and determined endogenously when households differ in their asset market participation.

The planner's first-order condition. A utilitarian Ramsey planner internalizes both the resources a marginal unit of employment creates for the economy and how they are distributed. In Appendix A we derive the planner's first-order condition with respect to employment, which we restate below:

$$v'(N_t) = \hat{w}_t \bar{u}'_t + \lambda \left[u'(C^p_t) - u'(C^r_t) \right] NX_t \left. \frac{d\epsilon_t}{dN_t} \right|_{NX_t} \tag{45}$$

The left-hand side is the social marginal disutility of labor. The first term on the right-hand side is the social value of the resources a marginal unit of employment creates: after accounting for the terms-of-trade distortion, the economy retains $\hat{w}_t = A_t - \frac{1+\chi}{\chi} \epsilon_t P^{T*}_t \theta$ units of output, valued at the average marginal utility, i.e., as they would be valued if every household received an equal share of them. The second term corrects for the fact that they do not, since the depreciation that pays for the additional imports affects Ricardians through the redistributive price effect on their financial flows. As discussed previously, this transfer runs from HtM households to Ricardians when Ricardians are net borrowers ($NX_t < 0$) and in the opposite direction when they are net savers ($NX_t > 0$). The planner values it at the marginal utility difference between the two types. This redistributive price effect disappears in a representative-agent economy, where $\lambda = 0$; when trade is intratemporally balanced, $NX_t = 0$; or when the two household types consume the same amount of the good, $C^p_t = C^r_t$.

A strictly positive terms-of-trade-adjusted labor wedge under $NX_t \neq 0$. Given any sequence of import prices and productivity, $\{P^{T*}_t, A_t\}$, the constrained efficient allocation satisfies the intertemporal budget constraint, $NX_1 + R^{*-1} NX_2 = 0$, together with the optimal intertemporal condition, (32), derived in the previous subsection, and the intratemporal optimality conditions, (45), for each t. The optimal terms-of-trade-adjusted labor wedge is the redistributive price effect term measured relative to the social value of the resources a marginal unit of employment creates:

$$\hat{\tau}^N_t = - \frac{\lambda \left[u'(C^p_t) - u'(C^r_t) \right] NX_t \left. \frac{d\epsilon_t}{dN_t} \right|_{NX_t}}{\hat{w}_t \bar{u}'_t} \tag{46}$$

If the solution satisfies $NX_t = 0$, employment choice just accounts for the terms-of-trade distortion and the wedge is zero. If it involves $NX_t \neq 0$, then the wedge is strictly positive. This is because, with strictly concave utility, $u(.)$, the sign of $[u'(C^p_t) - u'(C^r_t)] NX_t$ is always negative whenever $NX_t \neq 0$. This follows from $C^r_t - C^p_t = -\epsilon_t NX_t / (1 - \lambda)$. If Ricardians are net borrowers in a period ($NX_t < 0$), then they consume more than HtM households, and consequently have a lower marginal utility. Similarly, if Ricardians are net savers in a period ($NX_t > 0$), then they consume less than HtM households, and have a higher marginal utility. The Ramsey planner internalizes the redistributive price effect on Ricardian financial flows, which arises due to employment-import complementarity and endogenously shifts resources away from the higher marginal utility household type. For a utilitarian planner, this endogenous redistribution imposes an additional welfare cost of increasing employment at the margin beyond the labor disutility

cost. Consequently, whenever $NX_t \neq 0$, optimal policy involves a strictly positive labor wedge correction beyond the terms-of-trade correction. Proposition 4 formally summarizes this result.

Proposition 4. *Given any* $\{P_t^{T*}, A_t\}$, *and* $\lambda \in (0,1)$, *consider the constrained efficient allocation that solves* (16). *Assume that it satisfies* $\hat{w}_t > 0$ *for* $t \in \{1,2\}$. *Then this allocation involves a strictly positive terms-of-trade-adjusted labor wedge,* $\hat{\tau}_t^N = 1 - \frac{v'(N_t)}{\hat{w}_t \bar{u}_t'} > 0$, *whenever* $NX_t \neq 0$.

Proof. See Appendix A.9. □

Optimal policy under $P_1^{T*} = P_2^{T*}$. The preceding discussion allows us to characterize optimal policy in this case almost immediately because it is analogous to the representative-agent case discussed in Proposition 1 and Lemma 2 of Section 3.3. With $A_t = A$, $P_1^{T*} = P_2^{T*}$, and $\beta R^* = 1$, the planner's per-period objectives, augmented by the multiplier on the intertemporal budget constraint, differ only by the positive factor β (Appendix A.2). Under the conditions of Lemma A.1, they therefore have the same static maximizer, and the constrained efficient allocation is symmetric: $N_1 = N_2$ and $\epsilon_1 = \epsilon_2$. Therefore, the two periods must then have the same trade balance; the intertemporal budget constraint implies that $NX_1 = NX_2 = 0$. Ricardians neither borrow nor save, and both household types consume their current labor income in each period and $C_t^p = C_t^r$ for $t \in \{1,2\}$.

With balanced trade and equal consumption across household types, the redistributive price effect term in the planner's first-order condition with respect to employment, (45), vanishes. The employment optimality condition reduces to $v'(N_t) = \hat{w}_t \bar{u}_t'$, which is identical to the RA-equivalent policy condition (37), i.e., the solution involves a strictly positive labor wedge that accounts for the static terms-of-trade distortion: $\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P^{T*} \theta}{w_t} > 0$, and the terms-of-trade-adjusted labor wedge is zero: $\hat{\tau}_t^N = 0$. Since the planner chooses a symmetric allocation, it satisfies the undistorted Ricardian Euler equation, and $\tau = 0$. The constrained efficient allocation therefore coincides with the RA-equivalent policy in this case. Proposition 5 formally summarizes optimal policy with symmetric primitives.

Proposition 5. *Let* $\lambda \in (0,1)$, $A_t = A > 0$, $P_t^{T*} = P^{T*} > 0$ *for* $t \in \{1,2\}$, *and* $\beta R^* = 1$. *Assume the conditions of Lemma A.1. Consider the constrained efficient allocation that solves* (16). *This solution involves:*

- *No intertemporal distortion:* $\tau = 0$.
- *A strictly positive labor wedge implementing the optimal terms-of-trade:* $\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P^{T*} \theta}{w_t} > 0$, *i.e., the terms-of-trade-adjusted labor wedge is zero:* $\hat{\tau}_t^N = 0$.
- *Symmetric equilibrium across periods:* $N_1 = N_2$, $\epsilon_1 = \epsilon_2$, $NX_1 = NX_2 = 0$, *and* $C_1^p = C_1^r = C_2^p = C_2^r$.

Proof. See Appendix A.10. □

Optimal policy under $P_1^{T*} > P_2^{T*}$. When the import price is temporarily higher in period 1, the RA-equivalent policy no longer implements the constrained efficient allocation. With borrowing-constrained households, the utilitarian planner instead chooses both a positive intertemporal wedge and a positive terms-of-trade-adjusted labor wedge. The intertemporal wedge moves the real exchange rate path, redistributing

resources toward labor incomes and away from exports in the high-price period and reversing this pattern in the low-price period. The labor wedge moves employment and imported-input demand within each period, accounting for terms-of-trade effects and for the redistributive price effect on Ricardian financial flows, which shifts resources between the two household types. Proposition 6 formally summarizes that implementing the constrained efficient allocation requires distorting both margins. The optimal policy prescription therefore differs from the representative-agent benchmark: it requires a combination of exchange rate policy that distorts the intertemporal margin and monetary policy that distorts the consumption-leisure margin, both performing distinct roles in allocating resources across time and between household types.

Proposition 6. *Let* $\lambda \in (0,1)$, $A_t = A > 0$, $P_1^{T*} > P_2^{T*} = P^{T*} > 0$, *and* $\beta R^* = 1$. *Assume the conditions of Lemma A.1. Consider the constrained efficient allocation that solves* (16). *Assume that this solution satisfies* $\hat{w}_t > 0$ *for* $t \in \{1,2\}$. *Assume further that* $0 < \chi \leqslant 1$ *and preferences satisfy* $u'(C) = C^{-\sigma}$ *with* $\sigma > 1/(1+\chi)$. *Then the constrained efficient allocation involves:*

- *A strictly positive intertemporal wedge,* $\tau > 0$.
- *A strictly positive terms-of-trade-adjusted labor wedge,* $\hat{\tau}_t^N > 0$ *for* $t \in \{1,2\}$.

Proof. See Appendix A.11. □

Proposition 6 only allows comparison between optimal policy and the RA-equivalent policy in proportionate terms: signs and magnitudes of the two wedges. These wedge comparisons do not imply an unambiguous ranking of employment levels across policy regimes. The redistributive price effect term in (45) implies that monetary policy tends to restrain employment in period 1. At the same time, a positive intertemporal wedge implies that exchange rate policy tends to appreciate the period-1 real exchange rate relative to the RA-equivalent allocation, which raises $\hat{w}_1$ and tends to push employment in the opposite direction. The ranking of N_1 and similarly N_2 across regimes is therefore ambiguous. While not globally provable, using numerical simulations in the next section, we show that under reasonable parameterizations, the statements of Propositions 2 and 3 hold true even under elastic labor supply: optimal policy appreciates the real exchange rate and worsens the trade balance in the high-price period, then depreciates the exchange rate and improves the trade balance in the low-price period, reducing the real exchange rate's co-movement with import prices relative to the RA-equivalent allocation. At the same time, optimal policy stabilizes the domestic-units price of the imported input, i.e., the terms of trade, $\epsilon_t P_t^{T*}$, and consequently stabilizes domestic labor income, Y_t^N, making them less volatile across periods relative to the RA-equivalent allocation. Finally, optimal policy lowers the export share of output and raises the HtM share of output in the high-price period, and reverses this pattern in the low-price period, relative to the RA-equivalent allocation.

Employment-import complementarity together with household heterogeneity requires the two policies to be set in coordination. This is because both policies are linked through the same equilibrium object, the real exchange rate. In the high-price period the intertemporal distortion induces Ricardians to borrow more and to supply additional foreign purchasing power to the domestic foreign exchange market. This borrowed purchasing power attempts to substitute for export revenues and appreciate the real exchange rate. However, as Ricardians attempt to bid down the real exchange rate, firms see positive profit opportunities

and attempt to expand their labor and import demand. In equilibrium, how much the real exchange rate appreciates and consequently how much the real wage rises depends on how much monetary policy allows labor supply to expand in response to higher demand. The more monetary policy allows employment and imports to expand, the weaker is the appreciation that exchange rate policy is able to deliver and the smaller is the gain in real wages of workers. Workers gain real incomes but by working longer hours, while a relatively depreciated real exchange rate disproportionately benefits Ricardian borrowers, whose foreign borrowing becomes more valuable in domestic units through the redistributive price effect of a monetary-policy-induced relative depreciation. By contrast, if monetary policy restricts labor supply, expanded firm demand instead bids up real wages and the real exchange rate appreciates more. Workers gain real incomes in the form of higher earnings per hour while Ricardian borrowing loses value in domestic units. At the optimum, exchange rate policy reaches the intertemporal margin directly and sets the path of borrowing and repayment, leaving monetary policy no intertemporal task. The imports an additional hour requires can be paid for by current exports, through a depreciation within the period, or by borrowing repaid later, and with the borrowing path set optimally the public agent is indifferent between the two at the margin. Optimal monetary policy therefore values the hour within the period, balancing the marginal costs of labor disutility and the redistributive price effect against the marginal benefit of higher net output. Both policies together determine the path of employment and the real exchange rate.

We have established a heterogeneity-induced real income role for monetary policy that fits in the class of the real income channel of monetary policy emphasized in HANK models. By influencing the level of inflation, monetary policy distorts the labor wedge and directly affects employment levels and therefore the real incomes of households. In typical HANK models, monetary expansions after adverse productivity shocks can raise employment and current earnings of constrained households. The implication is that monetary policy is relatively more expansionary than in a representative-agent world. In our import-dependent open economy studied here, employment is complementary to imported inputs. In states with high import prices, higher employment raises the import bill; with the Marshall-Lerner condition holding, the required increase in imports depreciates the real exchange rate. The depreciation creates the standard terms-of-trade effect, lowering real wages, and under heterogeneity it also has an asymmetric pecuniary effect, the redistributive price effect: it raises the domestic-good value of Ricardian net inflows, which distributes the aggregate real income gains disproportionately toward borrowers, i.e., Ricardian households, and away from HtM households. A utilitarian planner internalizes this endogenous redistribution and considers employment expansion in high-price states to be costly.

So far we have assumed a specific production and import structure in our framework, which has helped with analytical tractability. In the next subsection we consider alternative assumptions and present a comparative statics exercise.

3.5 Comparative Statics

The previous subsections characterized how heterogeneity in asset market access shapes the optimal intertemporal and labor wedges under utilitarian welfare when the economy faces a transitory import price spike, in a framework tractable enough to characterize the constrained efficient allocation analytically. This

subsection relaxes the two assumptions that carry that tractability. First, we replace the Leontief technology with a CES technology in labor and imported inputs that nests it, and vary the elasticity of substitution and the inverse Frisch elasticity of labor supply. Second, we consider an economy in which imports are final goods instead of production inputs. Both exercises trace how the optimal wedges vary with the degree of essentiality of imports, and both show that the intertemporal wedge attributable to heterogeneity is largest when imports are least substitutable.

3.5.1 CES Production Technology

As an alternative to the Leontief technology described in Section 3.1.2, we consider a CES production technology that combines labor and imported inputs to produce the domestic output:

$$Y_t = A_t \left[(1-\alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}} \tag{47}$$

where $\xi > 0$ is the elasticity of substitution between labor and imported inputs; the Leontief technology of the previous subsections is nested as the limit $\xi \to 0$. For the numerical simulations, households' utility is $u(C) = C^{1-\sigma}/(1-\sigma)$ and labor disutility is $v(N) = \eta N^{1+\phi}/(1+\phi)$, where ϕ is the inverse Frisch elasticity of labor supply and $\phi \to \infty$ is the perfectly inelastic case considered previously.

Appendix A.12 states the utilitarian Ramsey planner's primal problem under this technology and shows that ξ and ϕ jointly govern the degree of essentiality of imports: as $\xi \to 0$ and $\phi \to \infty$, the price elasticity of import demand approaches zero. We vary these two parameters and solve for the optimal wedges under an illustrative calibration described in the appendix.

Figures A.1, A.2, and A.3 in Appendix A.12 report the optimal intertemporal and labor wedges over (ξ, ϕ) for representative-agent $(\lambda = 0)$ and heterogeneous $(\lambda = 0.35)$ economies under equal $(P_1^{T*} = P_2^{T*})$ and unequal $(P_1^{T*} > P_2^{T*})$ import price paths. First, when import prices are equal across periods, the optimal intertemporal wedge is zero for every pair (ξ, ϕ), for $\lambda = 0$ and $\lambda > 0$ alike: the no-intertemporal-distortion result holds across the entire parameter space. With import and export tariffs absent, the planner corrects the static terms-of-trade distortion through the labor wedge alone; the distortion arises because additional export revenue is earned only at a lower export price, and this correction is positive throughout and increases as the elasticity of substitution approaches zero. Second, when import prices are unequal, the no-intertemporal-distortion result does not extend to general CES production: neither wedge alone can fully correct the terms-of-trade distortion, and the planner distorts both margins simultaneously. The intertemporal motive is the capital-control motive of Costinot, Lorenzoni and Werning (2014): the economy takes the world interest rate as given yet holds market power in its own exports, and intertemporal distortions are generated by the time variation in the incentive to exercise that market power, which the import price spike creates and equal prices remove. Leontief production is the special case in which this motive disappears, as the analytical framework established and the exercise confirms. Imports then move one for one with employment, and the period-by-period labor wedge implements the entire terms-of-trade correction, leaving no motive on the intertemporal margin; with perfectly inelastic labor supply the labor wedge is unavailable, yet the planner leaves the intertemporal margin undistorted despite the time-varying

incentive, and the constrained efficient allocation coincides with laissez-faire. Proposition 1 establishes this result analytically; any intertemporal distortion that arises with $\lambda > 0$ under Leontief production therefore reflects the role of heterogeneity alone.

Away from the Leontief limit, both wedges are nonzero even with $\lambda = 0$: the level of a wedge in the heterogeneous-agent economy mixes the corrections a representative-agent planner would already choose with the part contributed by heterogeneity. Differencing the wedges in the two economies facing the same import price path therefore helps isolate the role of heterogeneity. Let $\Delta\tau$ and $\Delta\tau_1^N$ denote the differences between the optimal intertemporal and period-1 labor wedges of the heterogeneous economy and those of the representative-agent economy. Figure 2 plots both differences against the elasticity of substitution for a range of labor supply elasticities. Both differences are positive throughout and largest as $\xi \to 0$ and $\phi \to \infty$ jointly, the limits in which imports become essential to domestic activity. The two motives trade off on the intertemporal margin: with unequal import prices, the representative-agent terms-of-trade motive sets a negative intertemporal wedge, restricting borrowing in the high-price period (Figure A.1, panel (c)), whereas heterogeneity raises the wedge toward expanding borrowing, weakening the terms-of-trade distortion and reversing it as imports become essential (panel (d)). Intertemporal distortions, in the form of either capital controls or foreign exchange interventions, are therefore not equally desirable across imported goods. For imports with close substitutes or a price-elastic demand the additional wedges are small, and under Cobb-Douglas production negligible: the case for intertemporal distortions rests strongly on the necessity character of imports. Similarly, the labor wedge difference is increasing in the degree of essentiality, and the complementarity between the policies that distort these margins strengthens with it: the intertemporal policy appreciates the real exchange rate, and the more tightly imports move with employment, the larger the wedge monetary policy must set to keep an employment and import expansion from undoing the appreciation, and the more of the cost saving reaches constrained households through real wages. Heterogeneity thus widens the scope of both instruments selectively: the more essential the import, the larger the distortion each of them is called on to deliver.

In Appendix B.8 we obtain the infinite-horizon counterpart. The no-shock intertemporal wedge remains zero with and without constrained households, while an import-price spike produces a dynamic terms-of-trade motive away from Leontief that the public agent weighs against the carry cost of an interest parity gap.

3.5.2 Imports as Final Goods

The discussion so far has treated imports as intermediate inputs. In this subsection we depart from that assumption and instead consider imports as final goods. We illustrate that the real-income stabilization motive of exchange rate policy, operating through the intertemporal margin, holds when imports are final goods, while the role of monetary policy becomes much more limited. Production of domestic output is linear in labor, and households consume both domestic and imported goods, with Stone-Geary preferences over the two:

$$u(C, M) = \frac{\left[C^{1-\omega}(M - \underline{M})^{\omega}\right]^{1-\sigma}}{1-\sigma}, \qquad M \geqslant \underline{M}$$

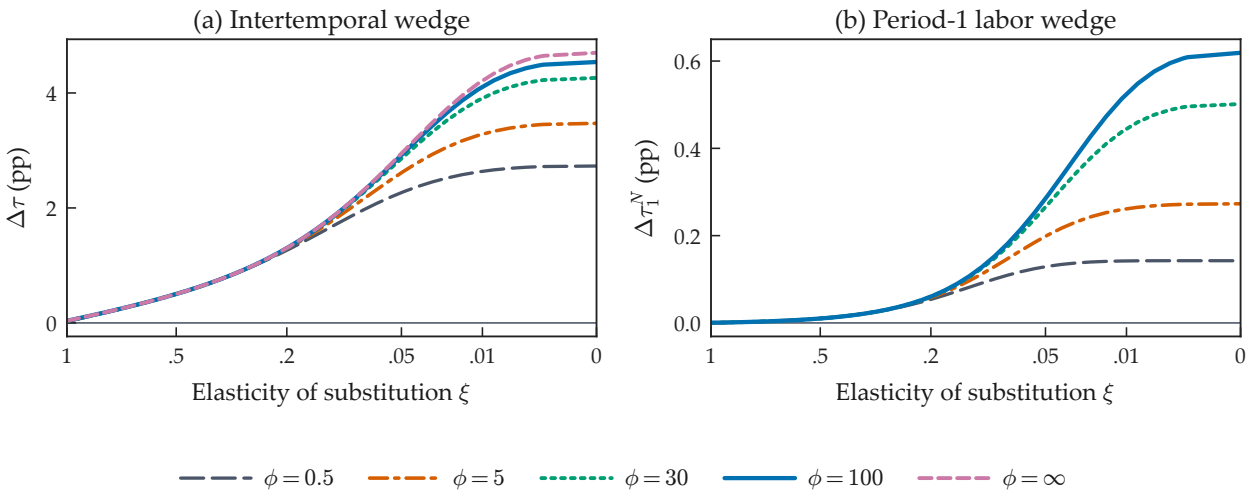


Figure 2. Optimal wedges: isolating the role of heterogeneity

Note: The horizontal axis uses a cube-root scale. The $\phi = \infty$ series appears only in the left panel because the labor margin is absent.

In Appendix A.13 we describe the environment with these assumptions, state the constrained problem, and present the comparative statics results. We conduct a comparative statics exercise to illustrate how the optimal intertemporal and labor wedges vary with the degree of essentiality of imports measured by the Marshallian price elasticity of demand for imported goods. We show that this elasticity is governed by the subsistence level $\underline{M}$ and the share parameter ω: with $\underline{M} = 0$ preferences are Cobb-Douglas and the elasticity is one; with $\underline{M} > 0$, as $\omega \to 0$ the elasticity falls toward zero and demand becomes perfectly inelastic at the subsistence level. We illustrate that the optimal intertemporal wedge attributable to heterogeneity rises with the degree of essentiality and is largest in the limit where demand is perfectly price inelastic. Through the intertemporal margin, exchange rate policy supports the real incomes of constrained households by cutting exports and retaining more output for domestic consumption while import prices are high, and reverses the pattern once they fall. The role for stabilization is increasing in the degree of essentiality of imports, supporting a stronger case for intertemporal distortions when imports are necessities. The optimal labor wedge attributable to heterogeneity, however, is small in this environment and smallest when import demand is perfectly inelastic. The role for monetary policy beyond correcting nominal rigidities and the static terms-of-trade distortion is limited in this case. The complementarity between the two policies that arises when imports are necessary inputs in production does not extend to the case where imports are final goods.

For the remainder of the paper we return to the original specification, in which imports are production inputs under the Leontief technology. The next section embeds the analytical framework in a two-agent, infinite-horizon quantitative model with nominal wage rigidities and financial intermediation frictions. In this extended framework, FX interventions decentralize the intertemporal policy, while monetary policy sets the inflation rate and controls the labor wedge. The quantitative model carries over the economic intuition and compares optimal policy with restricted counterfactual policies under energy price shocks.

4 Quantitative Model

The analytical framework characterizes the two policy margins in closed form under a reduced-form policy structure, in which the public agent sets employment directly and distorts the intertemporal margin through a capital control tax. In this section we develop an infinite-horizon two-agent economy in which those margins are carried by the instruments that monetary and fiscal authorities operate. Monetary policy sets a nominal interest rate and moves the labor wedge through nominal wage rigidity, and exchange rate policy takes the form of foreign exchange intervention through the public balance sheet, where the intertemporal wedge appears as a deviation from interest parity. Holding that deviation open costs the economy a carry loss, and the size of the distortion that policy should choose therefore becomes a quantitative question.

We solve the Ramsey problem of this economy, establish the conditions under which its solution is implemented by the two instruments, calibrate the model to Japan as an energy importer, and compare Ramsey optimal policy with restricted optimal counterfactual policies along a transitory energy price spike and the realized 2021-2023 path of world energy prices. We then solve the economy without lump-sum transfers.

4.1 Environment

Time is discrete and indexed by $t \geqslant 1$. The economy is deterministic and faces an exogenously specified path of imported input prices and total factor productivity, $\{P_t^{T*}, A_t\}_{t\geqslant 1}$ where $P_t^{T*} > 0$ and $A_t > 0$ for all t and bounded. The preferences, household types, final domestic good production technology, export demand, and relative-price definitions are as in Section 3. The intertemporal policy is implemented through foreign exchange interventions in the presence of financial intermediation frictions, while the labor wedge is implemented through monetary policy in the presence of nominal wage rigidities.

4.1.1 Households

As before, a fraction $(1-\lambda)$ of households are Ricardian and a fraction λ are HtM. Ricardian households trade one-period domestic nominal bonds. Let $B_{t+1}^{M,r}$ denote nominal bond holdings chosen in period t, and let R_{t+1}^{M} denote the gross nominal return paid on bonds carried into period $t+1$. Ricardian households solve

$$\begin{aligned} \max_{\{C_t^r, B_{t+1}^{M,r}\}_{t\geqslant 1}} \sum_{t=1}^{\infty} \beta^{t-1}\left[u(C_t^r) - v(N_t)\right] \quad \text{s.t.} \\ P_t C_t^r + B_{t+1}^{M,r} = W_t N_t + R_t^M B_t^{M,r} + T_t \end{aligned} \tag{48}$$

Let $\Pi_{t+1} \equiv P_{t+1}/P_t$ and $R_{t+1} \equiv R_{t+1}^M/\Pi_{t+1}$ denote gross inflation and the gross real interest rate in domestic units. The solution to the Ricardian household problem is characterized by an Euler equation:

$$\frac{u'(C_t^r)}{\beta u'(C_{t+1}^r)} = R_{t+1} \tag{49}$$

and a no-Ponzi-transversality condition:

$$\lim_{T\to\infty} \frac{B_{T+1}^r}{\prod_{s=2}^{T} R_s} = 0 \tag{50}$$

where $B^r_{t+1} \equiv B^{M,r}_{t+1}/P_t$ denotes the Ricardian household's bond holdings in real domestic units. Ricardian households participate in domestic asset markets but not directly in international asset markets. HtM households remain excluded from all asset markets and direct fiscal transfers. They consume their current labor incomes each period:

$$C^p_t = w_t N_t \tag{51}$$

As before, both household types work the same number of hours in equilibrium, N_t, and do not choose those hours themselves. The labor aggregator and the unions described next determine them.

4.1.2 Labor Aggregator

The final good production technology is inherited from the analytical model, with firms combining a final labor service and imported inputs to produce the final good. The final labor service, N_t, is produced by a competitive labor packer who aggregates over $k \in [0,1]$ tasks performed by task-specific labor, $N_{k,t}$, using a CES technology:

$$N_t = \left(\int_0^1 N_{k,t}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} \, dk \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}}, \qquad \varepsilon_w > 1$$

Let $W_{k,t}$ denote the nominal wage for task k. Cost minimization gives task demand and the aggregate wage index:

$$N_{k,t} = \left(\frac{W_{k,t}}{W_t} \right)^{-\varepsilon_w} N_t, \qquad W_t = \left(\int_0^1 W_{k,t}^{1-\varepsilon_w} \, dk \right)^{\frac{1}{1-\varepsilon_w}} \tag{52}$$

4.1.3 Unions

We adapt the nominal wage setting and rigidity structure from Erceg, Henderson and Levin (2000) and Auclert, Rognlie and Straub (2024). For each task, $k \in [0,1]$, workers are represented by a union. Taking the aggregator's labor demand function for the task k as given, the union sets the nominal wage, $W_{k,t}$, and calls on all members to work the same number of hours, $N_{k,t}$. For each task, k, workers are representative of the population: a mass λ are HtM and $(1-\lambda)$ are Ricardian. The union seeks to maximize the population-weighted utility of its members. The union also faces private quadratic utility costs for changing nominal wages. The union's optimization problem is to maximize:

$$\sum_{s \geqslant 0} \beta^s \left(\lambda u\big(C^p_{t+s}\big) + (1-\lambda) u\big(C^r_{t+s}\big) - v\left(N_{t+s}\right) - \frac{\psi_{nr}}{2} \left(\frac{W_{k,t+s}}{W_{k,t+s-1}} - 1 \right)^2 \right) \tag{53}$$

where $\psi_{nr} > 0$ parametrizes the degree of nominal wage rigidity. Each union is infinitesimal and only takes into account its marginal effect on every household's consumption and labor supply. By symmetry, in equilibrium $W_{k,t} = W_t$ and $N_{k,t} = N_t$ for all k. In Appendix B.1 we derive the unions' optimal wage and labor supply decisions that yield the aggregate wage Phillips curve of the form:

$$v'(N_t) = (1 - \tau^N_t) w_t \bar{u}'_t \tag{54}$$

where $(1-\tau_t^N)$ represents the labor wedge and is given by:

$$1-\tau_t^N = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}}\Pi_t - 1\right)\frac{w_t}{w_{t-1}}\Pi_t - \beta\left(\frac{w_{t+1}}{w_t}\Pi_{t+1} - 1\right)\frac{w_{t+1}}{w_t}\Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{nr}} w_t N_t \bar{u}_t'} \tag{55}$$

where $\mu_w \equiv \varepsilon_w/(\varepsilon_w - 1)$ is the wage markup and $\bar{u}_t'$ is the average marginal utility as defined in equation (19).

4.1.4 Financial Intermediaries

The financial intermediation structure in our model is adapted from Gabaix and Maggiori (2015). The world is populated by a unit mass of identical financial intermediaries who intermediate borrowing and lending transactions between domestic and world financial markets. These intermediaries are assumed to be owned outside the small open economy and take offsetting asset positions in domestic and world markets:

$$\frac{F_{t+1}}{e_t} + F_{t+1}^* = 0 \tag{56}$$

where F_{t+1} is the domestic asset position in units of domestic currency and F_{t+1}^* is the asset position in the international asset markets, in units of the world numeraire, i.e., the foreign good. A positive position denotes a saving and a negative position denotes a borrowing. The intermediaries' gross returns in foreign good units, v_{t+1}, are given by:

$$v_{t+1} = R_{t+1}^M \frac{F_{t+1}}{e_{t+1}} + R^* F_{t+1}^* \tag{57}$$

There is a limited repayment commitment on the part of intermediaries: each period, immediately after taking their asset positions, an intermediary can divert a portion $\Gamma\left|F_{t+1}^*\right|$, with $\Gamma > 0$, of the position it intermediates: $\left|F_{t+1}^*\right|$. If the intermediary diverts the funds, the position is unwound and the lenders recover a portion $\left(1-\Gamma\left|F_{t+1}^*\right|\right)$ of their claims $\left|F_{t+1}^*\right|$. Since the lenders anticipate the incentives of the intermediary to divert funds, the intermediary is subject to a credit constraint such that discounted returns from intermediation are weakly higher than the returns from diverting the funds:

$$\frac{1}{R^*} v_{t+1} \geqslant \Gamma\left|F_{t+1}^*\right| \times \left|F_{t+1}^*\right| \tag{58}$$

The intermediary's constrained optimization problem is then given by:

$$\max_{\{F_{t+1}^*, F_{t+1}\}} \frac{1}{R^*} v_{t+1} \quad \text{s.t. (56), (58)}$$

The solution to this optimization problem yields the intermediary's asset demand rule in the world asset markets:

$$F_{t+1}^* = \frac{1}{\Gamma}\left[1 - \frac{R_{t+1}^M}{R^*}\frac{e_t}{e_{t+1}}\right] = \frac{1}{\Gamma}\left[1 - \frac{R_{t+1}}{R^*}\frac{\epsilon_t}{\epsilon_{t+1}}\right] \tag{59}$$

Equation (59) shows that the intermediary's asset demand is linearly related to the interest parity gap $R_{t+1}\frac{\epsilon_t}{\epsilon_{t+1}} - R^*$. Define the domestic real interest rate in foreign units as $\tilde{R}_{t+1}$:

$$\tilde{R}_{t+1} \equiv R_{t+1}\frac{\epsilon_t}{\epsilon_{t+1}} \tag{60}$$

If $\tilde{R}_{t+1} = R^*$, intermediary asset demand is zero. When $\tilde{R}_{t+1} < R^*$, the intermediary saves in foreign markets and borrows in domestic markets; when $\tilde{R}_{t+1} > R^*$, the intermediary borrows in foreign markets and saves in domestic markets.

4.1.5 Monetary Policy

Monetary policy sets the path of domestic nominal interest rates, $\{R^M_{t+1}\}_{t\geqslant 1}$ where the nominal rate satisfies the Fisher equation:

$$R^M_{t+1} = R_{t+1}\Pi_{t+1} \tag{61}$$

4.1.6 Fiscal Policy

The public agent can directly participate in domestic as well as foreign asset markets, bypassing any private intermediaries. Let $B^{M,G}_{t+1}$ denote government domestic bond holdings in domestic currency units and let B^{G*}_{t+1} denote government's foreign asset position in units of the foreign good. The period-by-period government budget constraint in domestic currency units is given by:

$$B^{M,G}_{t+1} + e_t B^{G*}_{t+1} + (1-\lambda)T_t = R^M_t B^{M,G}_t + e_t R^* B^{G*}_t$$

where T_t denotes lump-sum taxes or transfers to Ricardian households. Define $B^G_{t+1} \equiv \frac{B^{M,G}_{t+1}}{P_t}$ as the government's domestic bond holdings in real domestic units. The government budget constraint in real domestic units is then given by:

$$B^G_{t+1} + \epsilon_t B^{G*}_{t+1} + (1-\lambda)\frac{T_t}{P_t} = R_t B^G_t + \epsilon_t R^* B^{G*}_t \tag{62}$$

The government is subject to no-Ponzi conditions in both markets:

$$\lim_{T\to\infty} \frac{B^G_{T+1}}{\prod_{s=2}^T R_s} = 0, \qquad \lim_{T\to\infty} \frac{B^{G*}_{T+1}}{{R^*}^{T-1}} = 0 \tag{63}$$

together with a consistency condition on interest rates which ensures that government asset positions satisfy no-Ponzi conditions in domestic as well as foreign units:

$$0 < \inf_{T\geqslant 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} \leqslant \sup_{T\geqslant 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} < \infty \tag{64}$$

4.1.7 Domestic Bonds Market Clearing

Domestic bonds market clearing requires that the asset positions by Ricardians, government, and financial intermediaries must sum to zero: $(1-\lambda)B^{M,r}_{t+1} + B^{M,G}_{t+1} + F_{t+1} = 0$. Define the aggregate Ricardian position in real domestic units as $B_{t+1} \equiv (1-\lambda)B^r_{t+1}$. Using the intermediary balance sheet condition, (56), and expressing the bonds market clearing condition in real domestic units, we have:

$$B_{t+1} + B^G_{t+1} = \epsilon_t F^*_{t+1} \tag{65}$$

4.1.8 Balance of Payments and Intertemporal Budget Constraint

The balance of payments condition is implied by the household, firm and government budget constraints together with goods and domestic bonds market clearing. In Appendix B.2, we derive the period-by-period balance of payments condition which is given by:

$$P_t^{T*}\theta N_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^X + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* + R^* B_t^{G*} \tag{66}$$

The left-hand side represents the demand for foreign purchasing power by firms, private intermediaries, and the government at time t. The right-hand side represents the supply of foreign purchasing power from export revenues, which rise with a real depreciation under the Marshall-Lerner condition, and from inherited foreign positions. The BoP conditions represent the effective external budget constraint of the economy. The period-by-period constraints can be combined to obtain the intertemporal budget constraint:

$$\sum_{t=1}^{\infty} \frac{\frac{1}{\Gamma}\left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}} + \sum_{t=1}^{\infty} \frac{P_t^{T*}\theta N_t}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^X}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1} \tag{67}$$

See Appendix B.2 for the derivation of this condition. The first term on the left-hand side represents the present discounted value of the carry cost associated with allowing interest parity deviations, i.e., allowing intermediaries to take non-zero asset positions in the domestic bonds market. This cost arises due to the assumption that intermediaries are owned outside the small open economy and must be compensated over and above the world interest rate for participating in the domestic bonds market. The second term represents the present discounted value of the import bill, and the first term on the right-hand side represents the present discounted value of export revenues. The second and third terms on the right-hand side represent inherited government and intermediary claims at $t = 1$.

4.1.9 Decentralized Equilibrium

Given the initial states: $\{R_1^M, B_1, B_1^{G*}, \epsilon_0, F_1^*, w_0\}$, exogenous sequence of import prices and productivity, $\{P_t^{T*}, A_t\}_{t=1}^{\infty}$, and fiscal-monetary policy: $\{B_{t+1}^G, B_{t+1}^{G*}, R_{t+1}^M, \frac{T_t}{P_t}\}_{t=1}^{\infty}, R_1$, a decentralized equilibrium consists of an allocation: $\{C_t^r, C_t^p, B_{t+1}, N_t, M_t, Y_t, F_{t+1}^*, X_t\}_{t=1}^{\infty}$ and prices $\{\epsilon_t, w_t, \Pi_t, R_{t+1}\}_{t=1}^{\infty}$ that satisfy all optimality and market clearing conditions. We define the decentralized equilibrium and list all equations that characterize the equilibrium in Appendix B.3.

4.2 Optimal Policy Problem

Analogous to the approach in the analytical section, we consider the primal problem of a utilitarian Ramsey planner who directly chooses $\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t=1}^{\infty}$ to maximize welfare subject to implementability constraints implied by the decentralized equilibrium. The general Ramsey problem features a time-inconsistency problem which arises from the financial intermediation structure in the model: at $t = 1$, the claim due to the intermediaries in foreign units is given by $\frac{R_1^M}{\Pi_1}\frac{\epsilon_0 F_1^*}{\epsilon_1}$. Since the intermediaries lent in domestic currency units, Π_1 and ϵ_1 can be chosen to influence the ex-post foreign-units value of the inherited claim. For the quantitative exercise, we initialize the economy at the Ramsey steady state, in

which A_t and P_t^{T*} are constant over time. In that steady state, $R = R^*$ and hence $F_1^* = 0$. This initialization removes the planner's incentive at $t = 1$ to revalue an inherited intermediary position. However, it does not eliminate time inconsistency more generally: following a shock, the commitment allocation can generate nonzero intermediary positions that a future planner would inherit. We therefore interpret the solution to (112), stated in Appendix B.4, as a full-commitment Ramsey allocation initialized at the Ramsey steady state[9].

Lemmas B.1, B.2, B.3 stated in Appendix B.4 together establish conditions under which the constrained efficient allocation, i.e., the solution to (112), is implementable as a decentralized equilibrium. Under these restrictions, the planner can choose N_t directly, and the wage Phillips curve specified by equations (54), (55) enters only in the decentralization. As in the analytical framework, choosing N_t directly chooses the optimal labor wedge. Inflation and the nominal interest rate path, $\{\Pi_t, R_{t+1}^M\}$ can be constructed ex-post to implement the constrained efficient allocation in a decentralized equilibrium. The cost of changing nominal wages is a private cost to unions, which enters their objective and is excluded from the welfare objective: inflation affects welfare only through the real allocation it implements. The planner chooses the intertemporal wedge through $\tilde{R}_{t+1}$, and the public agent implements it through foreign exchange interventions, choosing the domestic and foreign asset positions $\{B_{t+1}^G, B_{t+1}^{G*}\}$; in the analytical framework the same wedge is a direct capital control tax. Intertemporal distortions are costly here: the carry cost in (67) is paid by the economy for departing from interest parity, a cost the capital control tax of the analytical framework does not carry. In the limit as $\Gamma \to 0$, private intermediary demand becomes infinitely elastic and public balance-sheet shifts cannot sustain an interest parity gap. If the wedge is instead implemented by the capital control tax, the limiting planner's problem coincides with the infinite-horizon version of the analytical Ramsey problem.

4.3 Quantitative Application: Energy Price Shocks and Optimal Policy

We have studied optimal policy in response to essential import price shocks in general terms; its leading application is fossil fuel price spikes in an energy-importing economy. Fossil fuels fit the theory's definition of an essential import. They are necessary to household and business energy needs, substitution away from them is limited in the short run, and their prices are set in U.S. dollars in global markets that no single importer can influence; a price spike therefore raises an import-dependent country's energy expenditure almost in proportion to its energy intensity. The 2022 energy crisis brought Brent crude and European natural gas to multiples of their pre-pandemic levels (IEA 2022; EIA 2026; ESMA 2023), and the 2026 disruption to crude and LNG flows through the Strait of Hormuz reproduced the pattern within months (IEA 2026*b*).

We calibrate the model to Japan. Japan is among the world's largest energy importers: it imports nearly 90 percent of its primary energy, was the largest importer of liquefied natural gas in 2022, and relies almost

[9] Given $F_1^* = 0$ and $\beta R^* = 1$, if $A_t = A > 0$ and $P_t^{T*} = P^{T*} > 0$ for all t, the solution to (112) is a steady state: $C_t^r = C^r$, $C_t^p = C^p$, $N_t = N$, $\epsilon_t = \epsilon$, and $\tilde{R}_{t+1} = R^*$. This property holds for any λ for which the no-shock Ramsey steady state exists. The allocation has $F_{t+1}^* = 0$ and $B_{t+1}^{G*} = B_1^{G*}$ in every period, and reoptimization therefore reproduces it under constant fundamentals. Household consumption levels and marginal utilities can remain unequal, while the steady-state intertemporal wedge remains zero.

entirely on imports for its crude oil (Agency for Natural Resources and Energy 2024). In 2022, fossil fuel import bills rose to roughly 4 percent of GDP and contributed to the largest annual trade deficit on record at 20 trillion yen (Nippon.com 2023). Japan also fits the particular sense in which the model economy is small and open. It is small with respect to world asset markets and world energy markets: it takes world interest rates and dollar energy prices as given. It is not small in the market for its own exports: Japanese output is differentiated, foreign demand for it is downward sloping and price elastic, and under this Marshall-Lerner condition earning additional foreign purchasing power within a period requires selling more cheaply.

Japan's experience during the 2022 spike, shown in Figure 1 of the introduction, illustrates the amplification the theory studies. The dashed line in the left panel plots an index of world energy prices in U.S. dollars, which peaked 172 percent above its January 2021 level. Through October 2022 the yen depreciated by 42 percent against the dollar, and Japan experienced unusually high inflation, with the CPI peaking 5.4 percent above its January 2021 level. The right panel combines the two: the CPI-deflated yen-dollar exchange rate, a measure of the real exchange rate, peaked 55 percent above its initial level. The solid line in the left panel plots the price of imported energy in units of the Japanese consumption basket: at the peak of the world price, imported energy cost 1.43 times what it would have cost had only the world price risen with no real depreciation.[10]

We use the calibrated economy to examine the assignment of the two instruments: what exchange rate policy should do and through which margin, what monetary policy should do conditional on the exchange rate policy, and whether either instrument can reproduce the other's role in its absence. We answer these questions with two imported-energy price paths. The first is an MIT shock, a perfect-foresight impulse response to a doubling of the energy price that decays at its estimated AR(1) persistence, which isolates the two roles. The second is a historical simulation along the realized path of world energy prices through the 2022 crisis, a deterministic hump-shaped sequence over 2021–2023 that peaks 157 percent above the pre-shock level, which gauges their quantitative importance. The exercise also asks whether restricted asset market participation materially amplifies the domestic consequences of an energy price shock. Appendix B.5 describes the calibration. We compare four policy regimes.

i. *RA-equivalent policy.* The heterogeneity-specific terms in the planner's optimality conditions remain at their no-shock steady-state values. Under Leontief production, the intertemporal wedge is zero, while the employment condition retains the static terms-of-trade distortion and the redistributive price effect associated with employment-import complementarity at the steady-state asset distribution.[11] Away from Leontief, the representative-agent terms-of-trade motive can also enter the intertemporal wedge, as Appendix B.8 shows.
ii. *Interest-parity policy.* Interest parity is imposed, while the employment condition responds to the shock.

[10]The figure is descriptive, and we do not attribute the real depreciation, in whole or in part, to the energy price spike. Under the Marshall-Lerner condition, however, the theory predicts a real depreciation accompanying an essential import price spike.

[11]The no-shock steady state remains heterogeneous. Ricardian households hold financial assets and receive interest income, while HtM households consume their current labor income. Their consumption levels and marginal utilities consequently differ in general. Evaluating the employment term at its steady-state value therefore does not set the terms-of-trade-adjusted labor wedge to zero; the planner retains the steady-state redistributive price effect.

iii. *FX-only policy.* The intertemporal condition responds to the shock, while the heterogeneity-specific term in the employment condition remains at its steady-state value.
iv. *Optimal policy.* Both the intertemporal and employment conditions respond to the shock.

Comparing interest-parity policy with RA-equivalent policy isolates the employment response due to heterogeneity under Leontief production. Comparing optimal policy with FX-only isolates that employment response when the intertemporal condition is chosen optimally, while comparing optimal policy with interest-parity policy identifies the contribution of the intertemporal instrument when the employment condition responds optimally. All four regimes have the same no-shock steady state, and their differences arise along the shock path.[12]

The representative-agent economy of Appendix B.6, in which no household is borrowing constrained and every household is on its Euler equation, provides the limiting case. Figures B.1 and B.2 report its allocation and implementation paths under the MIT shock. The policy comparisons in the main text instead hold the heterogeneous-agent economy and its initial distribution fixed. In the representative-agent economy the two instruments separate along the lines of Itskhoki and Mukhin (2023): exchange rate policy is assigned to the financial friction, and monetary policy to nominal rigidities. The public agent's foreign asset position takes on the financial flows that intermediaries would otherwise price: interest parity holds, no carry losses are incurred, and the intertemporal margin is left undistorted; monetary policy targets nominal rigidities and, in this economy, also corrects the static terms-of-trade distortion. With a mass of constrained households, optimal policy no longer uses the public balance sheet to undo the intermediation friction: it deliberately opens interest parity gaps and pays the carry losses the friction attaches to them, because the gap tilts the path of the real exchange rate and, through firms' costs and the zero-profit condition, the real wages and current incomes of constrained households.

The calibrated economy reproduces the amplification that the analytical framework establishes in closed form. Under RA-equivalent policy the real exchange rate depreciates as the energy price rises, and the domestic real price of energy rises 1.46 percent for every percent rise in the world price (Figure 3). Real wages, employment, and labor income fall. Since HtM households consume their current labor income, the income loss passes directly into their consumption. Constrained households cannot borrow against future income, and the economy therefore relies more heavily on contemporaneous export revenue to finance the import bill. Under the Marshall-Lerner condition, earning more export revenue requires a real depreciation, which further raises the domestic energy price and compresses real wages. Ricardian consumption instead rises on impact as Ricardian households borrow against the temporary loss, and then falls as the resulting foreign obligation is repaid.

The interest-parity policy isolates what monetary policy achieves on its own. Imposing interest parity leaves the Ricardian Euler equation undistorted, and it enters the Ramsey planner's problem as an additional

[12]The comparison begins with RA-equivalent policy because laissez-faire intermediation creates interest parity gaps and carry losses even without household heterogeneity. The public balance sheet can eliminate them at no cost, which is why optimal policy in the representative-agent economy restores interest parity. Itskhoki and Mukhin (2023) characterize optimal exchange rate and monetary policy in a representative-agent economy with this class of financial frictions. Our comparison therefore studies how heterogeneity changes policy beyond that prescription.

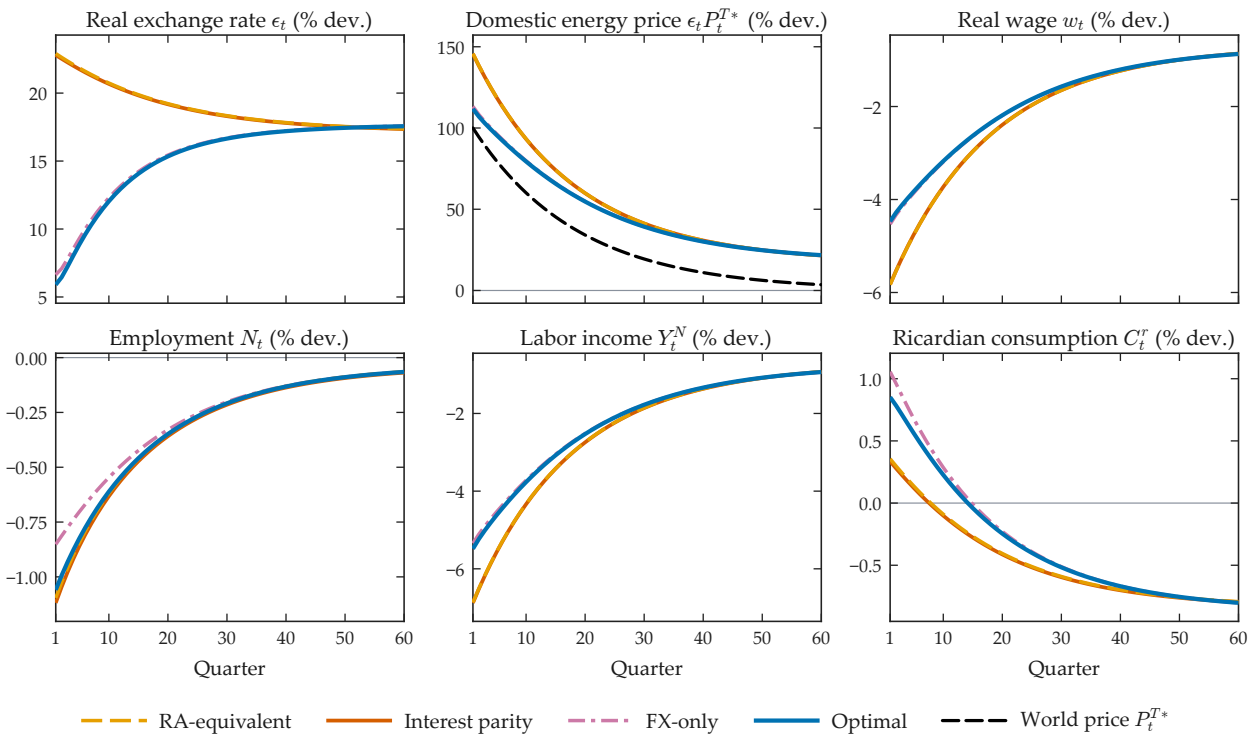


Figure 3. MIT shock: allocations and relative prices

Note: Percent deviations from the initial steady state. The solid black line is the world energy price.

constraint: the timing of the economy's foreign borrowing is then the one private agents choose at the world interest rate. The planner's employment condition responds to the shock, and because the trade balance is nonzero while the energy price is high, the redistributive price effect behind Proposition 4 is active. The public agent sets a terms-of-trade-adjusted labor wedge above its RA-equivalent level, restraining employment because an expansion would raise imported-input demand, produce a real depreciation, and shift purchasing power toward Ricardian households through the redistributive price effect on their financial flows. That restraint acts within the period, and it brings foreign purchasing power forward from the dates at which the essential import is cheap only through the borrowing Ricardian households choose in response, and that re-timing cuts the excess co-movement of the real exchange rate with the shock. Relative to RA-equivalent policy, which also imposes interest parity, this policy is more contractionary: the labor wedge is higher and employment and inflation are lower. Along the historical price path it undoes only 0.3 percent of the peak real-wage loss and lowers average quarterly inflation through the peak from 0.60 to 0.58 percent.

Exchange rate policy manages the path of capital flows by driving interest parity wedges. The FX-only policy borrows abroad in the high-price quarters, and the borrowed foreign purchasing power replaces current export revenue; the real exchange rate appreciates relative to RA-equivalent policy, the domestic cost of the imported input falls, and the real wage and labor income rise. This delivers most of the contemporaneous income stabilization and disinflation achieved by optimal policy. Under RA-equivalent policy, the depreciation lowers the real wage through firms' costs, and with nominal wages costly for unions

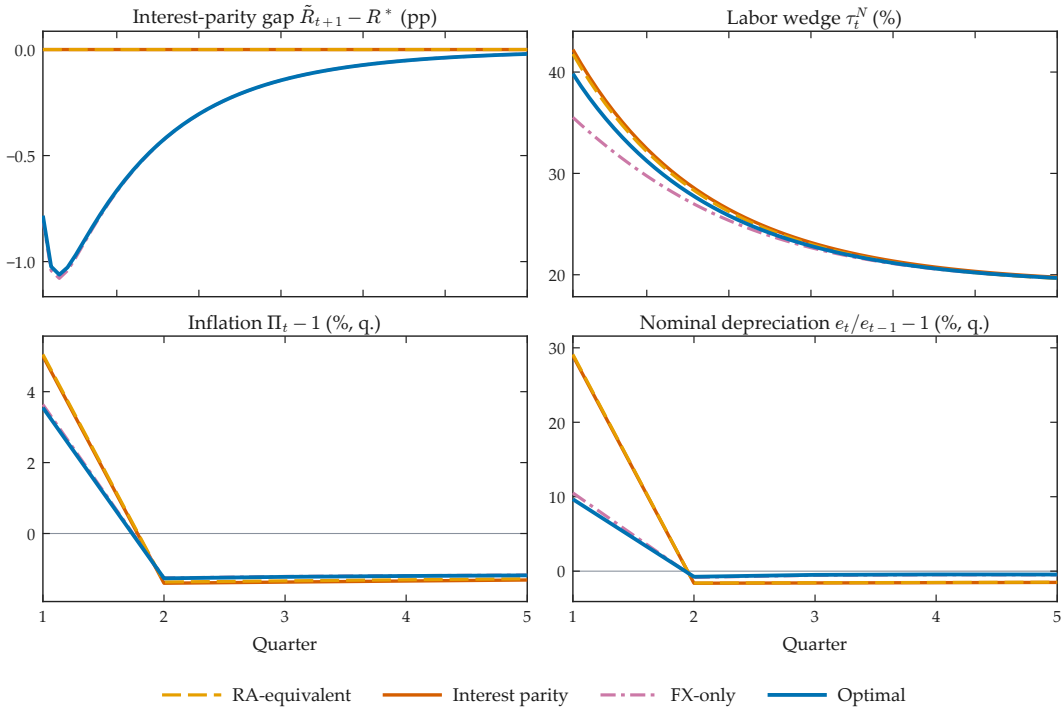


Figure 4. MIT shock: policy and implementation

to move, the required fall arrives through a rising price level: the depreciation is imported as inflation. By holding the real exchange rate stronger, the intertemporal policy reduces this imported component at its source. However, exchange rate policy alone does not determine employment or the division of the gain between wages and hours. FX-only chooses employment according to the RA-equivalent condition, which does not respond to the shock-induced redistributive price effect created when higher employment raises import demand and depreciates the real exchange rate. The cheaper imported input induces firms to expand: employment and imports rise, financing the additional imports requires export revenue, and earning it gives back part of the appreciation. The FX-only policy therefore delivers slightly higher labor income than optimal policy, through additional hours at a lower real wage. The resulting real exchange rate is weaker than under optimal policy; through the redistributive price effect, this depreciation raises what the borrowed foreign resources buy at home, in favor of Ricardian households, and their consumption in Figure 3 is highest in this regime. These responses reverse as the foreign position is repaid. Exchange rate policy therefore changes the timing and incidence of the imported-resource loss without removing it: along the historical path, the present discounted value of aggregate consumption differs by only 0.0005 percent between optimal and RA-equivalent policy.

We define monetary policy's role conditional on the intertemporal policy: alongside it, the labor wedge determines how much of the appreciation survives and in what form its gains arrive. The appreciation lowers firms' marginal cost, and an accommodating monetary stance would let employment and imports expand in response; financing the additional imports requires export revenue, and earning it would give back part of the appreciation the intertemporal policy has bought. Through the redistributive price effect, the expansion's depreciation would also raise what the borrowed positions buy at home, in favor of Ricardian

Table 1: Analytical predictions under the MIT shock

Prediction	RA-equivalent	Optimal
Intertemporal response (Prop. 2)		
Real exchange rate, high-price quarter (% dev.)	22.91	5.90
Real exchange rate, terminal steady state (% dev.)	17.16	17.69
Trade balance, high-price quarter (% of s.s. export revenue)	-106.5	-110.4
Trade balance, terminal steady state (% of s.s. export revenue)	4.05	4.17
Domestic price and labor income (Prop. 2)		
Volatility of domestic energy price (%)	109.8	79.9
Domestic energy price, terminal steady state (% dev.)	17.19	17.72
Volatility of labor income (%)	6.19	4.78
Labor income, terminal steady state (% dev.)	-0.72	-0.74
Volatility of the real wage (%)	5.18	3.79
Output shares (Prop. 3)		
Export share, high-price quarter (%)	4.40	3.65
Export share, terminal steady state (%)	4.10	4.13
HtM consumption share, high-price quarter (%)	31.69	32.15
HtM consumption share, terminal steady state (%)	33.422	33.415
Ricardian consumption share, terminal steady state (%)	62.47	62.46

households, in the quarters in which constrained households' marginal utility is highest. Optimal monetary policy therefore implements a lower inflation path than the FX-only policy: through the wage Phillips curve, lower inflation holds nominal wage growth and hours down, and employment stays below the FX-only path. The Ramsey objective contains no direct inflation loss and the resource constraint contains no wage-setting cost. Price stability nonetheless arises endogenously because, with nominal wages costly to move, stabilizing the real wage protects the current consumption of constrained households. Through the zero-profit condition, $w_t = A_t - \epsilon_t P_t^{T*}\theta$, the lower domestic imported-input cost then appears as a higher real wage rather than primarily as additional hours. The intertemporal policy thus determines the size of the cost saving, while the monetary stance determines its composition and its recipients: wages rather than hours, and labor income for all households rather than gains to the unconstrained through the redistributive price effect. Figure 4 displays the two policy margins. The optimal policy opens a negative interest parity gap, the decentralized form of the positive intertemporal wedge, and implements a labor wedge of 39.84 percent on impact, against 35.53 percent under FX-only with the same intertemporal rule. The larger labor wedge, lower inflation, and lower employment under the optimal policy establish that monetary policy is more contractionary than the RA-equivalent monetary rule used under FX-only.

Figure B.5 in Appendix B.7 reports the output reallocation under the MIT shock. Relative to RA-equivalent policy, borrowed foreign purchasing power replaces export revenue in the high-price quarters, the export share falls, and more output remains for domestic consumption. The appreciation lowers the import-bill share, while the higher real wage raises the HtM consumption share. The Ricardian share also rises slightly, a response that Proposition 3 leaves ambiguous in the high-price period. The rise reflects the net inflow: the additional foreign resources that exchange rate policy brings in accrue only to Ricardian households and tilt their consumption path toward the high-price quarters. As the foreign position is repaid, the export and import-bill shares rise relative to RA-equivalent policy and both household consumption shares fall.

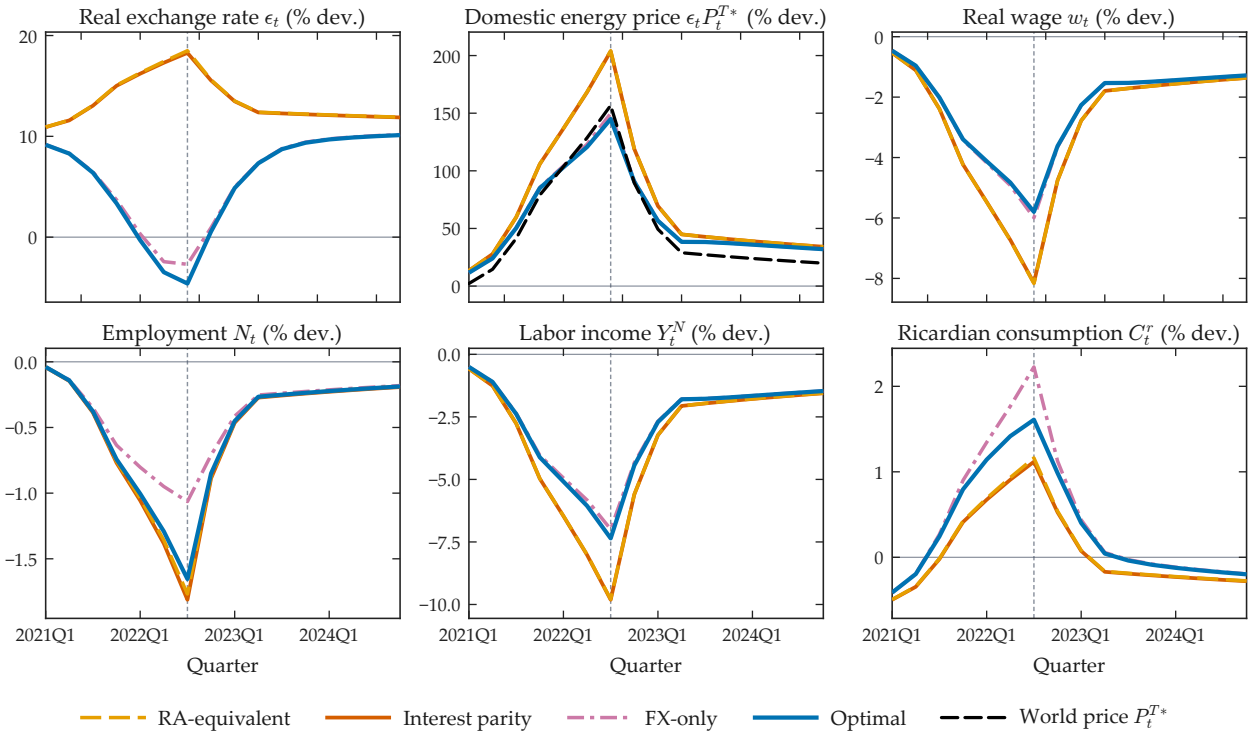


Figure 5. 2021–2023 shock: allocations and relative prices

Note: Percent deviations from the initial steady state. The solid black line is the world energy price; the vertical line marks its peak.

In Table 1 we compare the analytical predictions with the calibrated responses under the MIT shock; each prediction holds. Volatility is measured as the maximum absolute deviation from the terminal steady state, which keeps the permanent component that all regimes share out of the comparison. Under this transition-path measure, optimal policy makes the real exchange rate more volatile and reduces its co-movement with the import-price shock, while the domestic energy price, the real wage, and labor income all become less volatile: policy stabilizes them by moving the real exchange rate more. The high-price responses in Figure 3 and the terminal rows of the table show the same pattern in levels: the real exchange rate is stronger, the domestic energy price lower, and labor income higher while the import price is high, with each ranking reversed after the shock. The trade balance likewise moves from a larger deficit in the high-price quarter to a larger surplus later. The export and HtM shares move in the directions stated in Proposition 3, including the lower HtM and Ricardian shares after the shock. Finally, the negative interest parity gap in Figure 4 and the positive terms-of-trade-adjusted labor wedge implied by the optimal allocation verify the policy signs in Propositions 2, 4, and 6; the figure reports the corresponding labor wedge used in the decentralization.

We run the same comparison along the realized 2021 to 2023 path in Figure 5. Under RA-equivalent policy, every percent rise in the world price of energy raises its domestic real price by 1.30 percent at the peak. Relative to the representative-agent economy, restricted asset market participation accounts for 71 percent of the RA-equivalent real depreciation at the peak. Under optimal policy the domestic-energy amplification factor falls to 0.92, and the domestic real price of energy is 19.5 percent lower than under RA-equivalent policy at the peak. The prescribed path appreciates the real exchange rate through the run-up to its strongest

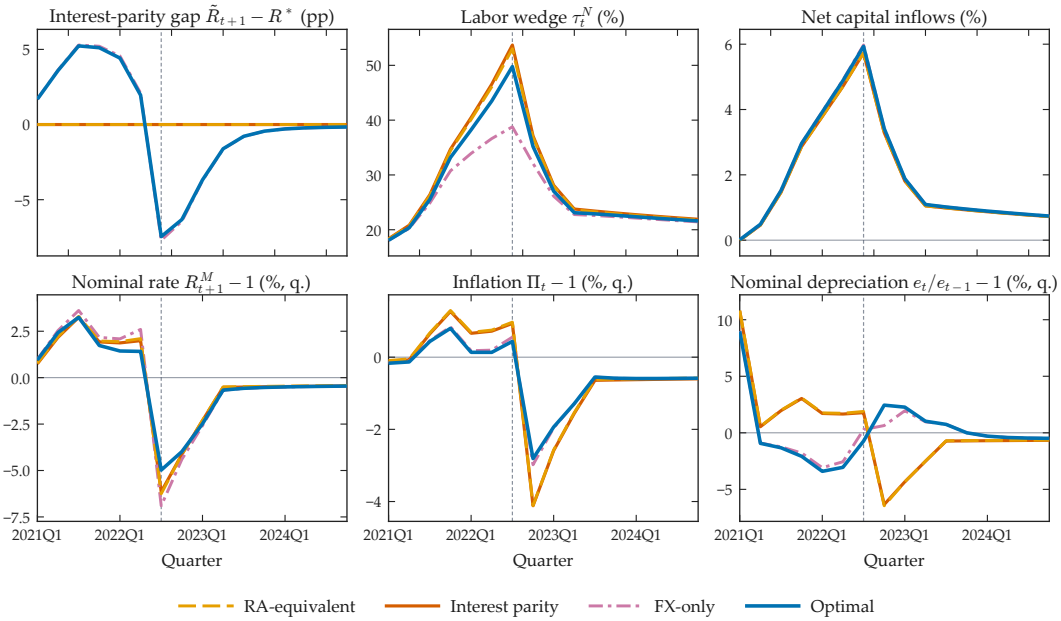


Figure 6. 2021–2023 shock: policy and implementation

Note: Net capital inflows are shown as a percent of initial steady-state gross output. The vertical line marks the energy-price peak.

value at the peak of the world price, stronger than before the shock, and depreciates it afterward as the foreign position is unwound. Optimal policy undoes 29 percent of the RA-equivalent peak real-wage loss and 25 percent of its peak labor-income loss. FX-only already undoes 27 percent of the real-wage loss and lowers average quarterly inflation through the peak from 0.60 to 0.28 percent; optimal policy lowers it further to 0.23 percent. Under optimal policy, the inflation path is also less volatile than under RA-equivalent policy both before and after the peak. Employment remains below the FX-only path, and the income protection therefore comes through a higher real wage rather than additional hours. Ricardian consumption rises as foreign purchasing power is brought into the high-price quarters and falls as the position is repaid.

We connect the two policy margins to their observable implementation in Figure 6. Net capital inflows build as the energy price rises and recede after its peak: foreign purchasing power replaces export revenue when the import bill is high and the position is unwound afterward. The interest parity gap is positive early in the run-up and turns negative as the price approaches its peak, tilting the FX interventions toward the quarters in which foreign purchasing power is most valuable. The labor wedge rises with the import price and restrains employment and imported-input demand in the quarters when exchange rate policy holds the real exchange rate strong. In absolute terms, monetary policy is contractionary during the initial run-up: under optimal policy the quarterly nominal rate rises from its no-shock steady-state value of 0.30 percent to 3.25 percent in quarter 3. We evaluate the relative stance by comparing regimes with the same intertemporal rule. Under interest parity, allowing the employment condition to respond raises the peak labor wedge from 53.03 percent under RA-equivalent policy to 53.71 percent. When the intertemporal condition is chosen optimally, moving from the RA-equivalent employment condition under FX-only to optimal policy raises the peak labor wedge from 38.78 percent to 49.78 percent. The larger response in the second comparison reflects the complementarity of the instruments: monetary policy restrains the employment and import

expansion that would otherwise offset part of the appreciation. Employment is 0.59 percentage points lower under optimal policy than under FX-only, and average quarterly inflation through the peak is 0.05 percentage points lower. With nominal wages rigid, the real wage falls through a rising price level, and a real depreciation therefore takes the form of a nominal depreciation that outpaces domestic inflation, as in Japan in 2022 (Figure 1). Following a large initial depreciation, the nominal depreciation rate turns negative through the early run-up and positive again around the peak as the foreign position starts to unwind.

Figure B.6 of Appendix B.7 reports the corresponding output shares, which move as under the MIT shock: relative to RA-equivalent policy, the export and import-bill shares fall and the HtM and Ricardian consumption shares rise during the high-price quarters, and these differences recede as the energy price falls and the foreign position is unwound.

In Table 2 we collect the effectiveness measures and policy wedges across regimes and both price paths. Under the MIT shock, optimal policy lowers the amplification factor to 1.12 and undoes 23 percent of the RA-equivalent impact real-wage loss and 20 percent of its impact labor-income loss. The domestic energy price is 13.8 percent cheaper, and quarterly inflation falls from 5.05 percent under RA-equivalent policy to 3.55 percent. Along the realized 2021 to 2023 path, these prescriptions are a full-commitment counterfactual: exchange rate policy holds the real exchange rate stronger and monetary policy restrains activity while the shock builds, after which the foreign position is unwound as the price falls.

In the calibrated economy, we find that exchange rate policy delivers most of the real-income protection and disinflation, whereas monetary policy in isolation, bound by interest parity, delivers little. Monetary policy supports exchange rate policy by restraining employment and imported-input demand, which preserves the appreciation and directs more of the gain toward real wages than hours. The signs of both wedges, the direction of the exchange rate path, and the division of the income gain accord with Section 3.

Extension without lump-sum transfers. In Appendix C.1 we withdraw the baseline model's lump-sum transfers to, and taxes on, Ricardian households and make domestic bonds real claims. When the shock

Table 2: Policy effectiveness and implementation

	RA-equivalent	Interest parity	FX-only	Optimal
Panel A. MIT shock; impact quarter				
Domestic energy-price amplification factor	1.46	1.46	1.13	1.12
Domestic energy cheaper than RA-equivalent (%)	0.0	0.1	13.2	13.8
Real-wage loss undone relative to RA-equivalent (%)	0.0	0.2	22.3	23.3
Average quarterly inflation through reference quarter (%, q.)	5.05	5.00	3.64	3.55
Interest parity gap (pp)	0.00	0.00	-0.80	-0.78
Labor wedge τ^N (%)	41.79	42.22	35.53	39.84
Panel B. 2021-2023 energy-price path; peak quarter				
Domestic energy-price amplification factor	1.30	1.30	0.96	0.92
Domestic energy cheaper than RA-equivalent (%)	0.0	0.2	17.9	19.5
Real-wage loss undone relative to RA-equivalent (%)	0.0	0.3	26.6	29.0
Average quarterly inflation through reference quarter (%, q.)	0.60	0.58	0.28	0.23
Interest parity gap (pp)	0.00	0.00	-7.70	-7.45
Labor wedge τ^N (%)	53.03	53.71	38.78	49.78

depreciates the real exchange rate, the public agent's foreign-unit reserves exceed in value its domestic-goods liabilities to Ricardian households, a surplus it can now pay out only through the return on their claims, which moves even the representative-agent economy away from interest parity. Relative to that economy, restricted asset market access reproduces the baseline policy assignment: the real exchange rate is more appreciated while import prices are high, and monetary policy sets a larger labor wedge and delivers lower inflation. The real-income stabilization channel therefore operates when all interventions must be financed through the public balance sheet at market prices.

5 Conclusion

We studied how import-dependent economies should conduct monetary and exchange rate policies when essential-import prices spike, some households are borrowing constrained and lump-sum transfers to them are unavailable. Conventional representative-agent prescriptions rest on perfect access to international asset markets and prescribe letting the real exchange rate depreciate freely as a shock absorber. We showed that borrowing constraints and the Marshall-Lerner condition turn the real exchange rate into a shock amplifier: a contemporaneous depreciation amplifies constrained households' current real income losses. A public agent who cares about constrained households' welfare finds it desirable to depart from laissez-faire and lean against this depreciation. Exchange rate policy induces stronger net capital inflows during the spike and, through the Marshall-Lerner condition, appreciates the real exchange rate relative to laissez-faire. The gains reach constrained households through indirect income effects of this appreciation, which we identify as the real-income stabilization channel of exchange rate policy. Policy re-times the real income loss: it weakens the real exchange rate's co-movement with import prices to stabilize real incomes over time. Under nominal wage rigidity, this real income protection arrives as lower inflation, and exchange rate policy therefore becomes an anti-inflation policy.

With exchange rate policy active, monetary policy's role changes. Exchange rate policy drives interest parity wedges, while monetary policy distorts the labor wedge to influence employment and output. When imports and employment are strong complements, an expansion during a spike induces a depreciation that redistributes resources from constrained wage earners to unconstrained net borrowers. Optimal monetary policy internalizes this redistributive price effect and takes a tighter-than-conventional stance that restrains employment and import demand while exchange rate policy seeks an appreciation through capital inflows.

In a quantitative two-agent model calibrated to Japan and the 2022 path of world energy prices, we found that exchange rate policy, through foreign exchange interventions, stabilizes household real incomes while accepting costly interest parity gaps. On its own, it delivers most of the real income protection and disinflation of jointly optimal policy, whereas monetary policy alone, set under interest parity, does little.

Three excluded margins mark natural directions for future research: partial or delayed fiscal transfers; policy set without commitment, where the promised unwinding of inflows may not be credible; and idiosyncratic risk under incomplete markets.

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Appendix

A Analytical Model Appendix

A.1 Optimality Conditions

Consider the two-period constrained planning problem, (16). Form the Lagrangian:

$$\mathcal{L} = \sum_{t=1}^{2} \beta^{t-1}\Big[\lambda\, u(C_t^p) + (1-\lambda)\, u(C_t^r) - v(N_t)\Big] + \mu\left[NX_1 + \frac{1}{R^*}\, NX_2\right] \tag{68}$$

where μ is the Lagrange multiplier on the intertemporal budget constraint.

FOC with respect to ϵ_t. Differentiating (68) with respect to ϵ_t, using $C_t^p = Y_t^N$ and $C_t^r = Y_t^N - \frac{\epsilon_t}{1-\lambda} NX_t$:

$$\beta^{t-1}\left[\bar{u}_t' \frac{\partial Y_t^N}{\partial \epsilon_t} - u'(C_t^r)\left(NX_t + \epsilon_t \frac{\partial NX_t}{\partial \epsilon_t}\right)\right] + \mu\,(R^*)^{-(t-1)} \frac{\partial NX_t}{\partial \epsilon_t} = 0$$

where $\bar{u}_t' \equiv \lambda\, u'(C_t^p) + (1-\lambda)\, u'(C_t^r)$. Solving for μ:

$$\mu = (\beta R^*)^{t-1} \cdot \frac{u'(C_t^r)\left(NX_t + \epsilon_t \dfrac{\partial NX_t}{\partial \epsilon_t}\right) - \bar{u}_t' \dfrac{\partial Y_t^N}{\partial \epsilon_t}}{\dfrac{\partial NX_t}{\partial \epsilon_t}} \tag{69}$$

Using $NX_t = \zeta \epsilon_t^{\chi} - P_t^{T*} \theta N_t$, rewrite Y_t^N as:

$$Y_t^N = A_t N_t - \zeta \epsilon_t^{1+\chi} + \epsilon_t NX_t$$

Differentiating with respect to ϵ_t and rearranging we get:

$$NX_t + \epsilon_t \frac{\partial NX_t}{\partial \epsilon_t} = \frac{\partial Y_t^N}{\partial \epsilon_t} + (1+\chi)\zeta \epsilon_t^{\chi} \tag{70}$$

Substitute (70) into the numerator of (69) and collect terms using $u'(C_t^r) - \bar{u}_t' = -\lambda[u'(C_t^p) - u'(C_t^r)]$:

$$\mu = (\beta R^*)^{t-1}\left[u'(C_t^r)\,\epsilon_t\left(1 + \frac{1}{\chi}\right) - \lambda\big[u'(C_t^p) - u'(C_t^r)\big] \frac{\dfrac{\partial Y_t^N}{\partial \epsilon_t}}{\dfrac{\partial NX_t}{\partial \epsilon_t}}\right] \tag{71}$$

Define:

$$\Omega_t \equiv \frac{\lambda\big[u'(C_t^p) - u'(C_t^r)\big]\left(-\dfrac{\partial Y_t^N/\partial \epsilon_t}{\partial NX_t/\partial \epsilon_t}\right)}{1 + \dfrac{1}{\chi}} \tag{72}$$

Setting $\tilde{\mu} \equiv \mu/(1 + 1/\chi)$:

$$(\beta R^*)^{t-1}\Big[\epsilon_t\, u'(C_t^r) + \Omega_t\Big] = \tilde{\mu} \tag{73}$$

Since $\tilde{\mu}$ is constant, equating (73) at $t = 1$ and $t = 2$ gives the intertemporal optimality condition:

$$\epsilon_1 u'(C_1^r) + \Omega_1 = \beta R^* \left[\epsilon_2 u'(C_2^r) + \Omega_2\right] \tag{74}$$

FOC with respect to N_t. Differentiating (68) with respect to N_t and substituting μ from (69):

$$v'(N_t) = \bar{u}_t' \left[\frac{\partial Y_t^N}{\partial N_t} - \frac{\partial Y_t^N}{\partial \epsilon_t} \cdot \frac{\frac{\partial NX_t}{\partial N_t}}{\frac{\partial NX_t}{\partial \epsilon_t}}\right] + u'(C_t^r)\, NX_t \cdot \frac{\frac{\partial NX_t}{\partial N_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} \tag{75}$$

Define:

$$\hat{w}_t \equiv \frac{\partial Y_t^N}{\partial N_t} + \left(NX_t - \frac{\partial Y_t^N}{\partial \epsilon_t}\right) \frac{\frac{\partial NX_t}{\partial N_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} = w_t - \tfrac{1}{\chi}\epsilon_t P_t^{T*}\theta \tag{76}$$

Define:

$$\Lambda_t \equiv \lambda\left[u'(C_t^p) - u'(C_t^r)\right] NX_t \cdot \frac{\frac{\partial NX_t}{\partial N_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} \tag{77}$$

Using $u'(C_t^r) = \bar{u}_t' - \lambda[u'(C_t^p) - u'(C_t^r)]$ we get:

$$v'(N_t) = \bar{u}_t' \hat{w}_t - \Lambda_t \tag{78}$$

The solution to the constrained planning problem, (16), $\{\epsilon_t, N_t\}_{t=1}^2$, is characterized by the intertemporal optimality condition, (74), and the intratemporal optimality conditions, (78), for $t = 1, 2$, together with the intertemporal budget constraint:

$$NX_1 + \frac{1}{R^*} NX_2 = 0 \tag{79}$$

Optimal Wedges Using the optimality conditions and the definitions of the labor wedge, (20), and the intertemporal wedge, (21), we can express the optimal wedges as:

$$\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*}\theta}{w_t} + \frac{\Lambda_t}{\bar{u}_t' w_t} \tag{80}$$

$$\tau = \frac{\Omega_1 - \beta R^* \Omega_2}{\epsilon_1 u'(C_1^r)} \tag{81}$$

A.2 Uniqueness of the Solution

The following Lemma establishes sufficient conditions for the uniqueness of the solution to the constrained planning problem, (16).

Lemma A.1. *Let $A_t > 0$ and $P_t^{T*} > 0$ for $t \in \{1, 2\}$. Let $R^* > 0$, $\beta > 0$, $\zeta > 0$, $\theta > 0$, $\chi > 0$, and $\lambda \in [0, 1)$. Let $u : \mathbb{R}_{++} \to \mathbb{R}$ be strictly increasing and strictly concave, and $v : \mathbb{R}_{++} \to \mathbb{R}$ be strictly increasing and*

strictly convex. For $\lambda = 0$, assume that there exists at least one solution, $\{N_t, \epsilon_t\}_{t=1}^2$, to equations (74), (78), and (79).

For $\lambda \in (0,1)$, let $I \subset (0,\infty)$ be an interval and assume that for each $t \in \{1,2\}$ and each $\mu \in I$, the function

$$\Phi_t(N_t, \epsilon_t; \mu) \equiv \beta^{t-1}\Big[\lambda\, u\big(C_t^p(N_t,\epsilon_t)\big) + (1-\lambda)\, u(C_t^r(N_t,\epsilon_t)) - v(N_t)\Big] + \mu\,(R^*)^{-(t-1)}\, NX_t(N_t,\epsilon_t)$$

admits a unique interior maximizer $\big(N_t^*(\mu),\ \epsilon_t^*(\mu)\big)$ on its domain. Let $h_t(\mu) \equiv NX_t\big(N_t^*(\mu), \epsilon_t^*(\mu)\big)$, and assume that there exists a common multiplier $\mu^* \in I$ such that $h_1(\mu^*) + (R^*)^{-1} h_2(\mu^*) = 0$.

Then the constrained planning problem, (16), has a unique maximizer. For $\lambda = 0$, it is the solution $\{N_t, \epsilon_t\}_{t=1}^2$ of (74), (78), and (79). For $\lambda \in (0,1)$, it is the pair of static maximizers $\big\{\big(N_t^*(\mu^*), \epsilon_t^*(\mu^*)\big)\big\}_{t=1}^2$.

Proof. Step 1: Uniqueness of the per-period maximizer.

For $\lambda \in (0,1)$ and $\mu \in I$, uniqueness of $\big(N_t^*(\mu), \epsilon_t^*(\mu)\big)$ is assumed directly.

For $\lambda = 0$, Φ_t simplifies to:

$$\Phi_t(N_t, \epsilon_t; \mu) = \beta^{t-1}\Big[u\big(A_t N_t - \zeta \epsilon_t^{1+\chi}\big) - v(N_t)\Big] + \mu\,(R^*)^{-(t-1)}\big(\zeta \epsilon_t^{\chi} - P_t^{T*}\theta N_t\big)$$

where $C_t^r = A_t N_t - \zeta \epsilon_t^{1+\chi}$ at $\lambda = 0$. Under the change of variables $z_t = \epsilon_t^{\chi}$, a bijection on $\mathbb{R}_{++}$ since $\chi > 0$, the exponent $(1+\chi)/\chi > 1$ implies that $z_t \mapsto z_t^{(1+\chi)/\chi}$ is strictly convex. Since u is strictly increasing and strictly concave, $u\Big(A_t N_t - \zeta z_t^{(1+\chi)/\chi}\Big)$ is strictly concave in (N_t, z_t). The term $-v(N_t)$ is strictly concave in N_t, and $\mu(R^*)^{-(t-1)}\big(\zeta z_t - P_t^{T*}\theta N_t\big)$ is linear. Therefore Φ_t is strictly concave in (N_t, z_t). Hence, whenever an interior maximizer exists, it is unique and maps back to a unique interior maximizer $\big(N_t^*(\mu), \epsilon_t^*(\mu)\big)$ in the original variables.

Step 2: Strict monotonicity of $h_t(\mu) \equiv NX_t\big(N_t^*(\mu), \epsilon_t^*(\mu)\big)$.

Let $\mu_2 > \mu_1$ be two multipliers at which the per-period maximizers exist, with $\mu_1, \mu_2 \in I$ when $\lambda \in (0,1)$. We first show that

$$\big(N_t^*(\mu_2), \epsilon_t^*(\mu_2)\big) \neq \big(N_t^*(\mu_1), \epsilon_t^*(\mu_1)\big)$$

Suppose for contradiction that both equal the same interior point (N_t, ϵ_t). Since the maximizer is interior, it satisfies the first-order conditions of $\Phi_t(\,\cdot\,; \mu_i)$ for $i \in \{1,2\}$:

$$\nabla_{(N_t,\epsilon_t)} U_t(N_t,\epsilon_t) + \mu_i\,(R^*)^{-(t-1)} \nabla_{(N_t,\epsilon_t)} NX_t(N_t,\epsilon_t) = 0, \qquad i \in \{1,2\}$$

where

$$U_t(N_t,\epsilon_t) \equiv \beta^{t-1}\Big[\lambda u(C_t^p(N_t,\epsilon_t)) + (1-\lambda) u(C_t^r(N_t,\epsilon_t)) - v(N_t)\Big]$$

Subtracting the two conditions gives:

$$(\mu_2 - \mu_1)(R^*)^{-(t-1)} \nabla_{(N_t,\epsilon_t)} NX_t(N_t,\epsilon_t) = 0$$

Since $\mu_2 > \mu_1$ and $(R^*)^{-(t-1)} > 0$, this requires $\nabla_{(N_t,\epsilon_t)} NX_t(N_t,\epsilon_t) = 0$. But

$$\frac{\partial NX_t}{\partial N_t} = -P_t^{T*}\theta < 0$$

at every interior point, so $\nabla_{(N_t,\epsilon_t)} NX_t \neq 0$, a contradiction. Hence the maximizers at μ_1 and μ_2 are distinct.

Denote the unique interior maximizers by (N_t^i, ϵ_t^i) for $i \in \{1,2\}$, where

$$(N_t^i, \epsilon_t^i) = \big(N_t^*(\mu_i), \epsilon_t^*(\mu_i)\big)$$

Set

$$U_t^i \equiv \beta^{t-1}\Big[\lambda u(C_t^p(N_t^i, \epsilon_t^i)) + (1-\lambda)u(C_t^r(N_t^i, \epsilon_t^i)) - v(N_t^i)\Big], \qquad H_t^i \equiv NX_t(N_t^i, \epsilon_t^i)$$

Since the maximizers are distinct, uniqueness at μ_2 and μ_1 respectively yields strict inequalities:

$$U_t^2 + \mu_2 (R^*)^{-(t-1)} H_t^2 > U_t^1 + \mu_2 (R^*)^{-(t-1)} H_t^1$$
$$U_t^1 + \mu_1 (R^*)^{-(t-1)} H_t^1 > U_t^2 + \mu_1 (R^*)^{-(t-1)} H_t^2$$

Adding and cancelling $U_t^1 + U_t^2$ yields

$$(\mu_2 - \mu_1)(R^*)^{-(t-1)}(H_t^2 - H_t^1) > 0$$

Since $\mu_2 > \mu_1$ and $(R^*)^{-(t-1)} > 0$, it follows that

$$H_t^2 > H_t^1$$

Therefore $h_t(\mu)$ is strictly increasing in μ.

Step 3: Uniqueness of μ^* and of the maximizer.

Define

$$S(\mu) \equiv h_1(\mu) + R^{*-1} h_2(\mu)$$

Since h_1 and h_2 are strictly increasing and $R^* > 0$, S is strictly increasing. Hence there exists at most one μ^* satisfying

$$S(\mu^*) = 0$$

For $\lambda = 0$, the assumed solution $\{N_t, \epsilon_t\}_{t=1}^2$ to (74), (78), and (79) defines a common multiplier μ through (69). Equation (74) implies that this multiplier is the same across the two periods, while (78) gives the corresponding first-order condition with respect to N_t. Since Φ_t is strictly concave in (N_t, z_t) by Step 1, this stationary point is the unique maximizer of $\Phi_t(\,\cdot\,;\mu)$, and the solution coincides with

$$\big\{(N_t^*(\mu), \epsilon_t^*(\mu))\big\}_{t=1}^2$$

Since (79) holds, this multiplier satisfies $S(\mu) = 0$. For $\lambda \in (0,1)$, existence of $\mu^* \in I$ with $S(\mu^*) = 0$ is assumed. In both cases, strict monotonicity of S implies that μ^* is unique, and Step 1 yields a unique pair

$$\big(N_t^*(\mu^*), \epsilon_t^*(\mu^*)\big)$$

for each $t \in \{1,2\}$.

It remains to verify that this allocation is the unique global maximizer of the constrained planning problem.

Let

$$x_t = (N_t, \epsilon_t), \qquad x_t^* = \left(N_t^*(\mu^*), \epsilon_t^*(\mu^*)\right)$$

Since x_t^* uniquely maximizes

$$\Phi_t(x_t; \mu^*) = U_t(x_t) + \mu^*(R^*)^{-(t-1)} NX_t(x_t)$$

for each $t \in \{1, 2\}$, any allocation (x_1, x_2) satisfies

$$\sum_{t=1}^{2} \left[U_t(x_t) + \mu^*(R^*)^{-(t-1)} NX_t(x_t)\right] \leqslant \sum_{t=1}^{2} \left[U_t(x_t^*) + \mu^*(R^*)^{-(t-1)} NX_t(x_t^*)\right]$$

with strict inequality unless $x_t = x_t^*$ for both $t = 1, 2$.

For any feasible allocation,

$$NX_1(x_1) + R^{*-1} NX_2(x_2) = 0$$

The allocation $\{x_t^*\}_{t=1}^2$ also satisfies this condition because $S(\mu^*) = 0$. Therefore the multiplier terms cancel, and

$$U_1(x_1) + U_2(x_2) \leqslant U_1(x_1^*) + U_2(x_2^*)$$

with strict inequality unless $(x_1, x_2) = (x_1^*, x_2^*)$. Hence $\{x_t^*\}_{t=1}^2$ is the unique global maximizer of problem (16). □

Remark. Lemma A.1 requires additional assumptions on the static optimizers for any $\lambda \in (0, 1)$ since uniqueness of the solution cannot be derived from strict concavity of u alone. This is because $C_t^p = (A_t - \epsilon_t P_t^{T*}\theta)N_t$ and C_t^r are both bilinear in (N_t, ϵ_t), and a composition with a strictly concave u does not yield a globally concave Φ_t for $\lambda \in (0, 1)$. The additional assumptions require that each per-period problem have a unique interior global maximizer for multipliers in I and that a common multiplier in I make these maximizers satisfy the external budget. Stationary points of the first-order conditions other than this pair are not optimal. In practice, these assumptions are satisfied for a large range of parameters under CRRA preferences, and verified numerically.

A.3 Proof of Lemma 1

Proof. Let $\{N_t, \epsilon_t\}_{t=1}^2$ solve (16). The intertemporal resource constraint gives

$$NX_1 + \frac{1}{R^*} NX_2 = 0$$

Define prices and quantities by

$$w_t = A_t - \epsilon_t P_t^{T*}\theta, \qquad M_t = \theta N_t, \qquad Y_t = A_t N_t, \qquad X_t = \zeta \epsilon_t^{1+\chi}$$

and define household consumption by (17) and (18). Let aggregate net foreign assets be

$$F^* = NX_1$$

and set $f^* = F^*/(1-\lambda)$. The intertemporal resource constraint implies $NX_2 = -R^*F^*$, so the two balance-of-payments equations in (13) hold.

Choose τ to satisfy the Ricardian Euler equation,

$$1+\tau = \frac{\beta R^* u'(C_2^r)\epsilon_2}{u'(C_1^r)\epsilon_1}$$

and set $T = \tau\epsilon_1 f^*$. Then the fiscal budget constraint (12) holds. Moreover, using $F^* = NX_1$,

$$C_1^r = w_1 N_1 - \epsilon_1 f^*$$

so

$$C_1^r + \epsilon_1 f^*(1+\tau) = w_1 N_1 + T$$

Similarly, using $NX_2 = -R^*F^*$,

$$C_2^r = w_2 N_2 + \epsilon_2 R^* f^*$$

Thus the Ricardian budget constraints hold. The HtM budget constraints hold by (17). Firms' optimality conditions and zero profits hold by the definitions of M_t, Y_t, and w_t above. Export demand holds by construction. Goods-market clearing follows from Walras' identity, since household budgets, firms' zero-profit conditions, the fiscal budget constraint, and the balance of payments all hold. Hence the allocation is implementable as an equilibrium under the policy $\{\tau, T\}$. □

A.4 Proof of Proposition 1

Proof. Setting $\lambda = 0$ in (72) gives $\Omega_t = 0$. Equation (74) then coincides with the undistorted Ricardian Euler equation, so $\tau = 0$. Equation (78) becomes $v'(N_t) = u'(C_t^r)\hat{w}_t$. Therefore $1-\tau_t^N = v'(N_t)/(u'(C_t^r)w_t) = \hat{w}_t/w_t$. Since $\hat{w}_t = w_t - \frac{1}{\chi}\epsilon_t P_t^{T*}\theta$, $\tau_t^N = \frac{1}{\chi}\epsilon_t P_t^{T*}\theta/w_t$. At an interior solution, $v'(N_t) > 0$ and $u'(C_t^r) > 0$ imply $\hat{w}_t > 0$, hence $w_t > 0$ and $\tau_t^N > 0$. □

A.5 Proof of Lemma 2

Proof. Let $z_t \equiv \epsilon_t^\chi$ and $q \equiv (1+\chi)/\chi > 1$, so that $C_t = AN_t - \zeta z_t^q$ and $NX_t = \zeta z_t - P_t^{T*}\theta N_t$.

Part 1. At $\lambda = 0$, the optimality system derived in the Appendix A reduces to

$$\epsilon_1 u'(C_1^r) = \beta R^* \epsilon_2 u'(C_2^r), \tag{82}$$

$$v'(N_t) = u'(C_t^r)\hat{w}_t, \qquad t \in \{1,2\}, \tag{83}$$

$$NX_1 + \frac{1}{R^*}NX_2 = 0 \tag{84}$$

Suppose $P_1^{T*} = P_2^{T*} = P^{T*}$. Let (N^0, z^0) solve the one-period system

$$\zeta z^0 - P^{T*}\theta N^0 = 0$$

and

$$v'(N^0) = u'\big(AN^0 - \zeta(z^0)^q\big)\left[A - q\zeta(z^0)^{q-1}\frac{P^{T*}\theta}{\zeta}\right]$$

This system is the first-order condition of the strictly concave one-period problem obtained after imposing $NX = 0$, and hence has at most one interior solution. Setting

$$N_1 = N_2 = N^0, \qquad z_1 = z_2 = z^0$$

satisfies (83). It also satisfies (82) because $\beta R^* = 1$, and it satisfies (84) because $NX_1 = NX_2 = 0$. Hence it is a solution to the representative-agent optimality system. By Lemma A.1, the constrained efficient allocation is unique. Therefore

$$N_1 = N_2, \qquad z_1 = z_2$$

It follows that $\epsilon_1 = \epsilon_2$ and $NX_1 = NX_2 = 0$.

Part 2. For $P > 0$ and $nx \in \mathbb{R}$, define the period value function:

$$V(P, nx) \equiv \max_{N,z} \{u(AN - \zeta z^q) - v(N)\} \quad \text{s.t.} \quad \zeta z - P\theta N = nx$$

Since the objective is concave in (N, z) and the constraint is affine, V is concave in nx. The two-period problem is equivalent to

$$\max_{nx_1} \; G(nx_1) \equiv V(P_1^{T*}, nx_1) + \beta V(P^{T*}, -R^* nx_1)$$

G is concave. Since $\beta R^* = 1$,

$$G'(0) = V_{nx}(P_1^{T*}, 0) - V_{nx}(P^{T*}, 0)$$

It suffices to show $V_{nx}(P, 0)$ is strictly decreasing in P.

At $nx = 0$ the constraint gives $\zeta z = P\theta N$. The envelope theorem yields $V_{nx}(P, 0) = -q\, C^{-\sigma} z^{q-1}$, so it suffices to show $C^{-\sigma} z^{q-1}$ is strictly increasing in P along the balanced-trade optimum.

Define $s \equiv \zeta z^q/(AN) \in (0, 1/q)$, so $C = AN(1 - s)$. The first-order condition with respect to N is:

$$A^{1-\sigma} N^{-\sigma} (1 - s)^{-\sigma} (1 - qs) = v'(N)$$

Implicit differentiation with respect to s gives:

$$\frac{d \log N}{ds} = \frac{\dfrac{\sigma}{1 - s} - \dfrac{q}{1 - qs}}{\sigma + \dfrac{N v''(N)}{v'(N)}}$$

Since v is convex, $N v''(N)/v'(N) \geqslant 0$, so

$$\frac{d \log N}{ds} < \frac{\sigma}{\sigma(1 - s)} = \frac{1}{1 - s} \tag{85}$$

From $s = (\theta^q / A\zeta^{q-1}) P^q N^{q-1}$:

$$\frac{d \log P}{ds} = \frac{1}{q} \left[\frac{1}{s} - (q - 1) \frac{d \log N}{ds} \right]$$

Since $s < 1/q$ implies $(q - 1)/(1 - s) < 1/s$, combining with (85) gives $d \log P/ds > 0$: P is strictly increasing in s.

Up to a positive constant,

$$C^{-\sigma} z^{q-1} \propto s^{\frac{q-1}{q}} (1-s)^{-\sigma} N^{\frac{q-1}{q}-\sigma}$$

Its log-derivative with respect to s is:

$$\frac{q-1}{qs} + \frac{\sigma}{1-s} + \left(\frac{q-1}{q} - \sigma\right) \frac{d \log N}{ds}$$

Since $\sigma \geqslant (q-1)/q = 1/(1+\chi)$, the last coefficient is non-positive. Using the tighter bound $d \log N/ds < \sigma/[(\sigma + N\nu''/\nu')(1-s)]$ from (85), the log-derivative is bounded below by:

$$\frac{q-1}{qs} + \frac{\sigma}{1-s}\left[1 - \frac{\sigma - (q-1)/q}{\sigma + N\nu''(N)/\nu'(N)}\right] > 0$$

Hence $C^{-\sigma} z^{q-1}$ is strictly increasing in s, and since P is strictly increasing in s, it is strictly increasing in P. Therefore $V_{nx}(P, 0) = -q\, C^{-\sigma} z^{q-1}$ is strictly decreasing in P.

Since $P_1^{T*} > P^{T*}$, we have $G'(0) < 0$. Concavity of G then gives $NX_1 = nx_1^* < 0$, and (84) gives $NX_2 = -R^* NX_1 > 0$. □

A.6 Proof of Lemma 4

Proof. With $\tau = 0$ and $\beta R^* = 1$, the laissez-faire equilibrium satisfies:

$$\epsilon_1 u'(C_1^r) = \epsilon_2 u'(C_2^r), \qquad NX_1 + \frac{1}{R^*} NX_2 = 0$$

Suppose for contradiction $NX_1 \geqslant 0$, so $NX_2 \leqslant 0$. Since $P_1^{T*} > P^{T*}$:

$$\zeta \epsilon_1^{\chi} \geqslant P_1^{T*} \theta \bar{N} > P^{T*} \theta \bar{N} \geqslant \zeta \epsilon_2^{\chi}$$

giving $\epsilon_1 > \epsilon_2$. Since $NX_1 \geqslant 0$ and $NX_2 \leqslant 0$:

$$C_1^r \leqslant Y_1^N = (A - \epsilon_1 P_1^{T*} \theta)\bar{N} < (A - \epsilon_2 P^{T*} \theta)\bar{N} = Y_2^N \leqslant C_2^r$$

Therefore $u'(C_1^r) > u'(C_2^r)$, and with $\epsilon_1 > \epsilon_2$:

$$\epsilon_1 u'(C_1^r) > \epsilon_2 u'(C_2^r)$$

contradicting the Euler equation. Hence $NX_1 < 0$ and $NX_2 = -R^* NX_1 > 0$. Since $C_t^r - C_t^p = -\frac{\epsilon_t NX_t}{1-\lambda}$, it follows that $C_1^r > C_1^p$ and $C_2^r < C_2^p$. □

A.7 Proof of Proposition 2

Proof. Strict concavity and gradient formula. Since C_t^p is affine and non-constant in ϵ_t and u is strictly concave, $W(\epsilon_1, \epsilon_2) \equiv \sum_{t=1}^{2} \beta^{t-1}[\lambda u(C_t^p) + (1-\lambda) u(C_t^r)]$ is strictly concave. Direct differentiation of W, together with the definition of Ω_t, gives

$$\frac{\partial W}{\partial \epsilon_t} = -\beta^{t-1}\left(1 + \frac{1}{\chi}\right)\left[\epsilon_t u'(C_t^r) + \Omega_t\right] \frac{\partial NX_t}{\partial \epsilon_t} \tag{86}$$

From (72), (7), and (15):

$$|\Omega_t| = \frac{\lambda\left|u'(C_t^p) - u'(C_t^r)\right| P_t^{T*}\theta\bar{N}}{(1+\chi^{-1})\,\chi\zeta\epsilon_t^{\chi-1}}$$

When $NX_t > 0$, $C_t^r < C_t^p$, so $\left|u'(C_t^p) - u'(C_t^r)\right| < u'(C_t^r)$, and $P_t^{T*}\theta\bar{N} < \zeta\epsilon_t^{\chi}$. Hence

$$|\Omega_t| < \frac{\lambda\, u'(C_t^r)\,\zeta\epsilon_t^{\chi}}{(1+\chi^{-1})\,\chi\zeta\epsilon_t^{\chi-1}} = \frac{\lambda}{1+\chi}\,\epsilon_t u'(C_t^r) < \epsilon_t u'(C_t^r) \tag{87}$$

where the last inequality uses $\lambda < 1 < 1+\chi$. Hence $\epsilon_t u'(C_t^r) + \Omega_t > 0$ whenever $NX_t > 0$.

Proof of items ii and iii. By Lemma 4, $NX_1^{LF} < 0 < NX_2^{LF}$, so $\Omega_1^{LF} > 0 > \Omega_2^{LF}$. Setting $\tau = 0$ and $\beta R^* = 1$ in (11) gives $\epsilon_1^{LF} u'(C_1^{r,LF}) = \epsilon_2^{LF} u'(C_2^{r,LF})$. Hence, using $\Omega_1^{LF} > 0$, $\Omega_2^{LF} < 0$, and (87) at $t = 2$:

$$H_1^{LF} \equiv \epsilon_1^{LF} u'(C_1^{r,LF}) + \Omega_1^{LF} \;>\; \epsilon_2^{LF} u'(C_2^{r,LF}) \;>\; \epsilon_2^{LF} u'(C_2^{r,LF}) + \Omega_2^{LF} \equiv H_2^{LF} > 0 \tag{88}$$

For any feasible $(\epsilon_1, \epsilon_2) \neq (\epsilon_1^{LF}, \epsilon_2^{LF})$, strict concavity of W, (86), and $\beta R^* = 1$ give:

$$W(\epsilon_1, \epsilon_2) < W^{LF} - \left(1+\chi^{-1}\right) \underbrace{\left[H_1^{LF}\frac{\partial NX_1^{LF}}{\partial \epsilon_1}\Delta\epsilon_1 + R^{*-1} H_2^{LF}\frac{\partial NX_2^{LF}}{\partial \epsilon_2}\Delta\epsilon_2\right]}_{\equiv \mathcal{B}}$$

where $\Delta\epsilon_t = \epsilon_t - \epsilon_t^{LF}$. Decompose $\mathcal{B}$ as:

$$\mathcal{B} = H_2^{LF}\left[\frac{\partial NX_1^{LF}}{\partial \epsilon_1}\Delta\epsilon_1 + R^{*-1}\frac{\partial NX_2^{LF}}{\partial \epsilon_2}\Delta\epsilon_2\right] + \left(H_1^{LF} - H_2^{LF}\right)\frac{\partial NX_1^{LF}}{\partial \epsilon_1}\Delta\epsilon_1$$

By concavity of $NX_1 + R^{*-1}NX_2$ and the equality constraint, the first bracket is non-negative. When $\Delta\epsilon_1 \geqslant 0$, both terms are non-negative by (88) and $\partial NX_t/\partial\epsilon_t > 0$, and at least one is strictly positive for $(\epsilon_1, \epsilon_2) \neq (\epsilon_1^{LF}, \epsilon_2^{LF})$. Hence

$$W(\epsilon_1, \epsilon_2) < W^{LF} \quad \text{for all feasible } (\epsilon_1, \epsilon_2) \text{ with } \epsilon_1 \geqslant \epsilon_1^{LF}$$

Along the constraint, $R^{*-1}\partial NX_2/\partial\epsilon_2\, d\epsilon_2 = -\partial NX_1/\partial\epsilon_1\, d\epsilon_1$, so by (86) and $\beta R^* = 1$:

$$dW^{LF} = -\left(1+\chi^{-1}\right)\left(H_1^{LF} - H_2^{LF}\right)\frac{\partial NX_1^{LF}}{\partial \epsilon_1}d\epsilon_1 > 0 \quad \text{for } d\epsilon_1 < 0$$

so LF is suboptimal and there exist feasible allocations with $W > W^{LF}$. Since every feasible allocation with $\epsilon_1 \geqslant \epsilon_1^{LF}$ satisfies $W \leqslant W^{LF}$ and LF is suboptimal, any optimum satisfies $\epsilon_1^O < \epsilon_1^{LF}$. Since both allocations satisfy the constraint and NX_t is strictly increasing in ϵ_t,

$$\epsilon_2^O > \epsilon_2^{LF}, \qquad NX_1^O < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^O$$

Proof of items iv and v. Item iv follows immediately from item ii. Since $Y_t^N = (A - \epsilon_t P_t^{T*}\theta)\bar{N}$ is strictly decreasing in $\epsilon_t P_t^{T*}$, item v follows from item iv.

Proof of item i. Since $NX_1^O < 0 < NX_2^O$, we have $\Omega_1^O > 0 > \Omega_2^O$. With $\beta R^* = 1$, (74) gives $\epsilon_1^O u'(C_1^{r,O}) + \Omega_1^O = \epsilon_2^O u'(C_2^{r,O}) + \Omega_2^O$, and (11) gives $\epsilon_1^O(1+\tau)u'(C_1^{r,O}) = \epsilon_2^O u'(C_2^{r,O})$. Substituting the latter into

the former gives

$$\tau = \frac{\Omega_1^O - \Omega_2^O}{\epsilon_1^O u'(C_1^{r,O})} > 0$$

Proof of item vi. This follows directly from Lemma A.2.

□

Lemma A.2. *Assume the conditions of Proposition 2 hold. Let* $p \equiv \frac{P_1^{T*}}{P_2^{T*}} > 1$.

i. Define the allocation that ensures equal input prices across periods, $(\epsilon_1^E, \epsilon_2^E)$*:*

$$\epsilon_1^E P_1^{T*} = \epsilon_2^E P_2^{T*}$$

and satisfies the intertemporal budget constraint, (79). If such an allocation is infeasible or it is feasible and

$$\frac{\lambda}{1+\chi} \leqslant \frac{(p+\beta)}{p(1+\beta p^\chi)}$$

then the constrained efficient allocation satisfies:

$$0 \leqslant \epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} < \epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}$$

$$0 \leqslant Y_2^{N,O} - Y_1^{N,O} < Y_2^{N,LF} - Y_1^{N,LF}$$

ii. Define the mirror allocation as one that ensures input prices are as volatile as the laissez-faire allocation, $(\epsilon_1^M, \epsilon_2^M)$*:*

$$\epsilon_1^M P_1^{T*} - \epsilon_2^M P_2^{T*} = -\left(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}\right)$$

with $\epsilon_1^M < \epsilon_1^{LF}, \epsilon_2^M > \epsilon_2^{LF}$*, and satisfies the intertemporal budget constraint, (79). If such an allocation is infeasible or it is feasible and:*

$$-NX_1^M \leqslant \left(1 - \frac{\lambda}{1+\chi}\right) P_1^{T*} \theta \bar{N}$$

then the constrained efficient allocation satisfies:

$$|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}|$$

$$\left|Y_2^{N,O} - Y_1^{N,O}\right| < \left|Y_2^{N,LF} - Y_1^{N,LF}\right|$$

Proof. By Proposition 2,

$$\epsilon_1^O < \epsilon_1^{LF}, \qquad \epsilon_2^O > \epsilon_2^{LF}$$

and

$$NX_1^O < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^O$$

Also, from the laissez-faire lemma,

$$\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*} > 0$$

Since

$$Y_2^N - Y_1^N = \theta \bar{N} \left(\epsilon_1 P_1^{T*} - \epsilon_2 P_2^{T*} \right)$$

it is enough to prove the corresponding claims for domestic input prices.

We first record an ordering property of the feasible frontier. Let $D(\epsilon_1, \epsilon_2) \equiv \epsilon_1 P_1^{T*} - \epsilon_2 P_2^{T*}$ and let (ϵ_1, ϵ_2) and $(\tilde{\epsilon}_1, \tilde{\epsilon}_2)$ both satisfy (79). If $\tilde{\epsilon}_1 > \epsilon_1$, then $NX_1(\tilde{\epsilon}_1) > NX_1(\epsilon_1)$, so (79) implies $NX_2(\tilde{\epsilon}_2) < NX_2(\epsilon_2)$. Since NX_2 is strictly increasing in ϵ_2, $\tilde{\epsilon}_2 < \epsilon_2$. Hence

$$D(\tilde{\epsilon}_1, \tilde{\epsilon}_2) > D(\epsilon_1, \epsilon_2)$$

Along the feasible frontier, D is therefore strictly increasing in ϵ_1. Since NX_2 is continuous and strictly increasing, the feasible frontier is continuous wherever it is defined.

We first prove the equal-input-price cutoff result. Suppose first that no feasible allocation satisfying (79) and

$$\epsilon_1 P_1^{T*} = \epsilon_2 P_2^{T*}$$

exists in the direction $\epsilon_1 < \epsilon_1^{LF}$. Since $D(\epsilon_1^{LF}, \epsilon_2^{LF}) > 0$ and D is strictly increasing along the feasible frontier, the constrained efficient allocation, which satisfies $\epsilon_1^O < \epsilon_1^{LF}$ by Proposition 2, must satisfy

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} \geqslant 0$$

Together with

$$\epsilon_1^O < \epsilon_1^{LF}, \qquad \epsilon_2^O > \epsilon_2^{LF}$$

this gives

$$0 \leqslant \epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} < \epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}$$

Now suppose the equal-input-price allocation $(\epsilon_1^E, \epsilon_2^E)$ is feasible. Since

$$\epsilon_1^E P_1^{T*} = \epsilon_2^E P_2^{T*}$$

and $P_1^{T*} > P_2^{T*}$, we have

$$\epsilon_1^E < \epsilon_2^E$$

Using (79), this allocation satisfies

$$NX_1^E < 0 < NX_2^E$$

Thus

$$C_1^{r,E} > C_1^{p,E} = C_2^{p,E} > C_2^{r,E}$$

Let

$$\kappa_t^E \equiv \frac{P_t^{T*} \theta \bar{N}}{(1+\chi)\zeta(\epsilon_t^E)^{\chi-1}}$$

Expanding the two Ω terms at the equal-input-price allocation gives

$$\begin{aligned}
&\epsilon_2^E u'(C_2^{r,E}) + \Omega_2^E - \epsilon_1^E u'(C_1^{r,E}) - \Omega_1^E \\
&= (\epsilon_2^E - \epsilon_1^E) u'(C_1^{p,E}) + (\epsilon_1^E - \lambda\kappa_1^E)\left[u'(C_1^{p,E}) - u'(C_1^{r,E})\right] \\
&\quad + (\epsilon_2^E - \lambda\kappa_2^E)\left[u'(C_2^{r,E}) - u'(C_2^{p,E})\right]
\end{aligned}$$

The first term is strictly positive. Since $NX_2^E > 0$,

$$P_2^{T*}\theta\bar{N} < \zeta(\epsilon_2^E)^\chi$$

so

$$\lambda\kappa_2^E < \epsilon_2^E$$

Hence the third term is strictly positive. The condition

$$\frac{\lambda}{1+\chi} \leqslant \frac{p+\beta}{p(1+\beta p^\chi)}$$

is equivalent, using (79) and $\epsilon_2^E = p\epsilon_1^E$, to

$$\lambda P_1^{T*}\theta\bar{N} \leqslant (1+\chi)\zeta(\epsilon_1^E)^\chi$$

and hence to

$$\lambda\kappa_1^E \leqslant \epsilon_1^E$$

Therefore the second term is non-negative, and so

$$\epsilon_2^E u'(C_2^{r,E}) + \Omega_2^E > \epsilon_1^E u'(C_1^{r,E}) + \Omega_1^E$$

We now use the strict concavity argument from Proposition 2. For any feasible allocation with $\epsilon_1 \leqslant \epsilon_1^E$, feasibility and concavity of the external resource constraint imply that the tangent feasibility term at $(\epsilon_1^E, \epsilon_2^E)$ is non-negative. Since the planner gradient at $(\epsilon_1^E, \epsilon_2^E)$ points strictly toward larger ϵ_1 along the feasible frontier, strict concavity implies that every feasible allocation with $\epsilon_1 \leqslant \epsilon_1^E$ is dominated. Hence the constrained efficient allocation must satisfy

$$\epsilon_1^O > \epsilon_1^E$$

Since both allocations satisfy (79) and D is strictly increasing along the feasible frontier, this implies

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} \geqslant 0$$

Together with the directional comparison in Proposition 2, this gives

$$0 \leqslant \epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} < \epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}$$

Multiplying by $\theta\bar{N}$ gives

$$0 \leqslant Y_2^{N,O} - Y_1^{N,O} < Y_2^{N,LF} - Y_1^{N,LF}$$

We next prove the mirror cutoff result. Suppose first that no feasible allocation satisfying (79) and

$$\epsilon_1 P_1^{T*} - \epsilon_2 P_2^{T*} = -\left(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}\right)$$

exists in the direction

$$\epsilon_1 < \epsilon_1^{LF}, \qquad \epsilon_2 > \epsilon_2^{LF}$$

Since $D(\epsilon_1^{LF}, \epsilon_2^{LF}) > 0$ and D is strictly increasing along the feasible frontier, the constrained efficient allocation, which satisfies $\epsilon_1^O < \epsilon_1^{LF}$ by Proposition 2, must satisfy

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} > -\left(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}\right)$$

Using again

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} < \epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}$$

we obtain

$$|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}|$$

Now suppose the mirror allocation $(\epsilon_1^M, \epsilon_2^M)$ is feasible. By construction,

$$\epsilon_1^M P_1^{T*} - \epsilon_2^M P_2^{T*} = -\left(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}\right)$$

with

$$\epsilon_1^M < \epsilon_1^{LF}, \qquad \epsilon_2^M > \epsilon_2^{LF}$$

Therefore

$$NX_1^M < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^M$$

and

$$C_1^{r,M} > C_1^{p,M} > C_2^{p,M} > C_2^{r,M}$$

Let

$$\kappa_t^M \equiv \frac{P_t^{T*}\theta\bar{N}}{(1+\chi)\zeta(\epsilon_t^M)^{\chi-1}}$$

Expanding the two Ω terms at the mirror allocation gives

$$\begin{aligned}
&\epsilon_2^M u'(C_2^{r,M}) + \Omega_2^M - \epsilon_1^M u'(C_1^{r,M}) - \Omega_1^M \\
&= (\epsilon_2^M - \epsilon_1^M) u'(C_1^{p,M}) + \epsilon_2^M \left[u'(C_2^{p,M}) - u'(C_1^{p,M})\right] \\
&\quad + (\epsilon_2^M - \lambda\kappa_2^M)\left[u'(C_2^{r,M}) - u'(C_2^{p,M})\right] \\
&\quad + (\epsilon_1^M - \lambda\kappa_1^M)\left[u'(C_1^{p,M}) - u'(C_1^{r,M})\right]
\end{aligned}$$

The first two terms are strictly positive. Since $NX_2^M > 0$,

$$\lambda\kappa_2^M < \epsilon_2^M$$

so the third term is strictly positive. The condition

$$-NX_1^M \leqslant \left(1 - \frac{\lambda}{1+\chi}\right) P_1^{T*} \theta \bar{N}$$

is equivalent to

$$\lambda P_1^{T*} \theta \bar{N} \leqslant (1+\chi)\zeta(\epsilon_1^M)^\chi$$

and hence to

$$\lambda \kappa_1^M \leqslant \epsilon_1^M$$

Therefore the fourth term is non-negative, and so

$$\epsilon_2^M u'(C_2^{r,M}) + \Omega_2^M > \epsilon_1^M u'(C_1^{r,M}) + \Omega_1^M$$

The strict-concavity argument above implies that the constrained efficient allocation lies before the mirror allocation:

$$\epsilon_1^O > \epsilon_1^M$$

Since D is strictly increasing along the feasible frontier,

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} > -\left(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}\right)$$

Together with

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} < \epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}$$

this gives

$$|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}|$$

Multiplying by $\theta\bar{N}$ gives

$$\left|Y_2^{N,O} - Y_1^{N,O}\right| < \left|Y_2^{N,LF} - Y_1^{N,LF}\right|$$

□

A.8 Proof of Proposition 3

Proof. Since $N_t = \bar{N}$ and $A_t = A$, output is $Y_t = A\bar{N}$ at all allocations and in both periods. All share comparisons therefore reduce to comparisons of levels. Proposition 2 gives $\epsilon_1^O < \epsilon_1^{LF}$, $\epsilon_2^O > \epsilon_2^{LF}$, $NX_1^O < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^O$, $Y_1^{N,O} > Y_1^{N,LF}$, and $Y_2^{N,O} < Y_2^{N,LF}$.

Export shares. Export demand satisfies $X_t = \zeta\epsilon_t^{1+\chi}$, which is strictly increasing in ϵ_t. Hence $X_1^O < X_1^{LF}$ and $X_2^O > X_2^{LF}$. The export share of output is strictly lower in period 1 and strictly higher in period 2 under the constrained efficient allocation.

Domestic consumption shares. The resource constraint gives $C_t = Y_t - X_t = A\bar{N} - X_t$. Since output is the same across allocations, the domestic share moves opposite to the export share: $C_1^O > C_1^{LF}$ and $C_2^O < C_2^{LF}$. The domestic share is strictly higher in period 1 and strictly lower in period 2 under the constrained efficient allocation.

HtM consumption shares. HtM households consume their current labor income: $C_t^p = w_t \bar{N} = Y_t^N$. Hence $C_1^{p,O} > C_1^{p,LF}$ and $C_2^{p,O} < C_2^{p,LF}$. The HtM share is strictly higher in period 1 and strictly lower in period 2 under the constrained efficient allocation.

Ricardian consumption share in period 2. From the consumption function (18), with $NX_t > 0$ in period 2, $C_2^r = Y_2^N - \epsilon_2 NX_2/(1-\lambda)$. Since $\epsilon_2^O > \epsilon_2^{LF} > 0$ and $NX_2^O > NX_2^{LF} > 0$, $\epsilon_2^O NX_2^O > \epsilon_2^{LF} NX_2^{LF}$. Therefore

$$C_2^{r,O} = Y_2^{N,O} - \frac{\epsilon_2^O NX_2^O}{1-\lambda} < Y_2^{N,LF} - \frac{\epsilon_2^{LF} NX_2^{LF}}{1-\lambda} = C_2^{r,LF}$$

The Ricardian share is strictly lower in period 2 under the constrained efficient allocation.

Ricardian consumption share in period 1. From the consumption function (18), with $NX_1 < 0$, let $B \equiv -NX_1 > 0$. Then $C_1^r = Y_1^N + \epsilon_1 B/(1-\lambda)$, so

$$C_1^{r,O} - C_1^{r,LF} = \left(Y_1^{N,O} - Y_1^{N,LF}\right) + \frac{1}{1-\lambda}\left(\epsilon_1^O B^O - \epsilon_1^{LF} B^{LF}\right)$$

The first term is strictly positive. Since $\epsilon_1^O < \epsilon_1^{LF}$ and $B^O = -NX_1^O > -NX_1^{LF} = B^{LF}$, the sign of the second term is ambiguous. Hence the sign of $C_1^{r,O} - C_1^{r,LF}$ is ambiguous. □

A.9 Proof of Proposition 4

Proof. From (78), $\hat{\tau}_t^N = 1 - v'(N_t)/(\hat{w}_t \bar{u}_t') = \Lambda_t/(\hat{w}_t \bar{u}_t')$. Since u is strictly increasing, $\bar{u}_t' > 0$, and by assumption $\hat{w}_t > 0$. Thus it suffices to show $\Lambda_t > 0$ whenever $NX_t \neq 0$.

From (15), $(\partial NX_t/\partial N_t)/(\partial NX_t/\partial \epsilon_t) < 0$. Moreover, (17) and (18) imply $C_t^r - C_t^p = -\epsilon_t NX_t/(1-\lambda)$. If $NX_t > 0$, then $C_t^r < C_t^p$, so strict concavity of u gives $u'(C_t^p) - u'(C_t^r) < 0$. If $NX_t < 0$, then $C_t^r > C_t^p$, so $u'(C_t^p) - u'(C_t^r) > 0$. Hence $\left[u'(C_t^p) - u'(C_t^r)\right] NX_t < 0$ whenever $NX_t \neq 0$. Since $\lambda \in (0,1)$, (77) gives $\Lambda_t > 0$, and therefore $\hat{\tau}_t^N > 0$. □

A.10 Proof of Proposition 5

Proof. Since $A_t = A$, $P_t^{T*} = P^{T*}$, and $\beta R^* = 1$, we have $\Phi_2(\cdot\,;\mu) = \beta\,\Phi_1(\cdot\,;\mu)$ for all μ. Since $\beta > 0$, the maximizer is unchanged, so by Lemma A.1, $N_1^*(\mu) = N_2^*(\mu)$ and $\epsilon_1^*(\mu) = \epsilon_2^*(\mu)$ for all $\mu \in I$, and in particular at the supporting multiplier μ^*. Since the two periods share the same $(N^*, \epsilon^*, A, P^{T*})$, we have $NX_1 = NX_2$, so (79) gives $NX_1 = NX_2 = 0$. Substituting into the consumption functions yields $C_t^p = C_t^r$ for $t \in \{1,2\}$, and hence $C_1^p = C_1^r = C_2^p = C_2^r$.

Since $C_t^p = C_t^r$, we have $\Omega_t = \Lambda_t = 0$. Equation (74) then reduces to the undistorted Ricardian Euler equation, giving $\tau = 0$. Equation (78) reduces to $v'(N_t) = \bar{u}_t' \hat{w}_t$, so $1 - \tau_t^N = v'(N_t)/(\bar{u}_t' w_t) = \hat{w}_t/w_t$. Hence $\tau_t^N = \frac{1}{\chi}\epsilon_t P^{T*}\theta/w_t > 0$, since $v'(N_t) > 0$ and $\bar{u}_t' > 0$ imply $\hat{w}_t > 0$ and therefore $w_t > 0$. □

A.11 Proof of Proposition 6

Proof. For a fixed import price P and $\mu \in I$, let $h(P,\mu)$ denote the trade balance chosen by the unique interior maximizer of the static problem $\Phi(N, \epsilon; \mu)$ in Lemma A.1. Step 2 in the proof of that lemma implies

that $h(P,\mu)$ is strictly increasing in μ on I.

Let $\mu^0(P) \in I$ denote the multiplier associated with the balanced-trade static optimum, $h(P,\mu^0(P)) = 0$. At balanced trade, $C^p = C^r$, so the static problem coincides with the representative-agent static problem. The first-order condition with respect to ϵ gives $\mu^0(P) = \frac{1+\chi}{\chi}\epsilon(P)C(P)^{-\sigma}$. Along the balanced-trade optimum, the proof of Lemma 2 shows that $\epsilon(P)C(P)^{-\sigma}$ is strictly increasing in P when $0 < \chi \leqslant 1$ and $\sigma > 1/(1+\chi)$. Hence $\mu^0(P_1^{T*}) > \mu^0(P_2^{T*})$.

Let μ^* denote the multiplier for the constrained efficient allocation, and write $h_t(\mu) \equiv h(P_t^{T*},\mu)$. Since the allocation satisfies the intertemporal budget constraint,

$$h_1(\mu^*) + \frac{1}{R^*}h_2(\mu^*) = 0$$

Moreover, $h_1(\mu^0(P_2^{T*})) < h_1(\mu^0(P_1^{T*})) = 0$ and $h_2(\mu^0(P_1^{T*})) > h_2(\mu^0(P_2^{T*})) = 0$. Since $h_1(\mu) + R^{*-1}h_2(\mu)$ is strictly increasing in μ, it follows that

$$\mu^0(P_2^{T*}) < \mu^* < \mu^0(P_1^{T*})$$

Therefore

$$NX_1 = h_1(\mu^*) < 0 < NX_2 = h_2(\mu^*)$$

Since $C_t^r - C_t^p = -\epsilon_t NX_t/(1-\lambda)$, we have $C_1^r > C_1^p$ and $C_2^r < C_2^p$. Strict concavity of u then gives $u'(C_1^p) > u'(C_1^r)$ and $u'(C_2^p) < u'(C_2^r)$. Since $-(\partial Y_t^N/\partial\epsilon_t)/(\partial NX_t/\partial\epsilon_t) > 0$, (72) gives $\Omega_1 > 0 > \Omega_2$. Using (81) and $\beta R^* = 1$, $\tau = (\Omega_1 - \Omega_2)/[\epsilon_1 u'(C_1^r)] > 0$.

Since $NX_t \neq 0$ for both periods and $\hat{w}_t > 0$, Proposition 4 implies $\hat{\tau}_t^N > 0$. □

A.12 Imports with a CES Production Technology and Comparative Statics

This Appendix supports the discussion in Section 3.5.1. We state the constrained planning problem, describe the calibration used in the numerical simulations, and present the results of the comparative statics exercise.

A.12.1 The Planning Problem with CES Production Technology

A competitive firm in period t chooses labor and imported inputs to solve

$$\max_{N_t,M_t} A_t\left[(1-\alpha)N_t^{\frac{\xi-1}{\xi}} + \alpha\left(\frac{M_t}{\theta}\right)^{\frac{\xi-1}{\xi}}\right]^{\frac{\xi}{\xi-1}} - w_t N_t - \epsilon_t P_t^{T*} M_t$$

For an interior allocation, the firm's first-order conditions are

$$w_t = A_t(1-\alpha)N_t^{-1/\xi}\left[(1-\alpha)N_t^{\frac{\xi-1}{\xi}} + \alpha\left(\frac{M_t}{\theta}\right)^{\frac{\xi-1}{\xi}}\right]^{\frac{1}{\xi-1}} \tag{89}$$

$$\epsilon_t P_t^{T*} = \frac{A_t\alpha}{\theta}\left(\frac{M_t}{\theta}\right)^{-1/\xi}\left[(1-\alpha)N_t^{\frac{\xi-1}{\xi}} + \alpha\left(\frac{M_t}{\theta}\right)^{\frac{\xi-1}{\xi}}\right]^{\frac{1}{\xi-1}} \tag{90}$$

The utilitarian planner chooses employment, imported inputs, and the real exchange rate in both periods subject to the firm's first-order conditions and the economy's intertemporal resource constraint:

$$\max_{\{N_t,M_t\}_{t=1}^2} \quad \sum_{t=1}^{2} \beta^{t-1}\left[\lambda\frac{(C_t^p)^{1-\sigma}}{1-\sigma}+(1-\lambda)\frac{(C_t^r)^{1-\sigma}}{1-\sigma}-\eta\frac{N_t^{1+\phi}}{1+\phi}\right] \quad \text{s.t.}$$

$$NX_1+\frac{1}{R^*}NX_2=0$$

where:

$$\epsilon_t=\frac{A_t\alpha}{P_t^{T*}\theta}\left(\frac{M_t}{\theta}\right)^{-1/\xi}\left[(1-\alpha)N_t^{\frac{\xi-1}{\xi}}+\alpha\left(\frac{M_t}{\theta}\right)^{\frac{\xi-1}{\xi}}\right]^{\frac{1}{\xi-1}}$$

$$w_t=A_t(1-\alpha)N_t^{-1/\xi}\left[(1-\alpha)N_t^{\frac{\xi-1}{\xi}}+\alpha\left(\frac{M_t}{\theta}\right)^{\frac{\xi-1}{\xi}}\right]^{\frac{1}{\xi-1}}$$

$$NX_t=\zeta\epsilon_t^{\chi}-P_t^{T*}M_t$$

$$Y_t^N=w_tN_t$$

$$C_t^p=Y_t^N$$

$$C_t^r=Y_t^N-\epsilon_t\frac{NX_t}{1-\lambda}$$

The labor and intertemporal wedges are recovered from the constrained-efficient allocation using the definitions in (20) and (21):

$$1-\tau_t^N=\frac{\eta N_t^{\phi}}{w_t\left[\lambda(C_t^p)^{-\sigma}+(1-\lambda)(C_t^r)^{-\sigma}\right]} \tag{91}$$

$$1+\tau=\frac{\beta R^*(C_2^r)^{-\sigma}\epsilon_2}{(C_1^r)^{-\sigma}\epsilon_1} \tag{92}$$

The Leontief production technology with elastic labor supply is nested in the above specification as the limit $\xi\to 0$. The perfectly inelastic labor supply specification is nested as the limit $\phi\to\infty$.

A.12.2 The Necessity Nature of Imports

The elasticity of substitution ξ and the inverse Frisch elasticity ϕ are two important parameters that together govern how essential the imports are for the domestic economy. We illustrate this point with the help of a static representative agent economy and show how the price elasticity of import demand varies with these two parameters. Consider a static representative-agent set up with a CES production technology that combines imports and labor. Let $q\equiv\epsilon P^{T*}$ denote the relative price of the imported inputs in domestic units. Firms' first order conditions satisfy equations (89) and (90). The representative household's budget constraint is given by $C=wN$ and the household's optimal employment choice is given by $C^{-\sigma}w=\eta N^{\phi}$. Let $s_M\equiv\frac{qM}{Y}, s_N\equiv\frac{wN}{Y}$ with $s_M+s_N=1$, denote the imported-input and labor shares. The firm's first-order conditions, combined with zero profits, which tie the real wage to the input price through

$s_N \, d\log w = -s_M \, d\log q$, imply that the input ratio responds to the import price according to

$$\frac{d\log(M/N)}{d\log q} = -\frac{\xi}{s_N}$$

Thus, ξ governs the extent to which firms can substitute away from imported inputs while holding the scale of production fixed. The household optimality condition implies

$$\frac{d\log N}{d\log q} = \frac{\sigma - 1}{\phi + \sigma}\frac{s_M}{s_N}$$

Thus, given σ, ϕ governs the response of employment, and hence the extent to which import demand can respond through changes in the scale of domestic activity. Combining the substitution and scale responses gives the local price response of import demand:

$$\frac{d\log M}{d\log q} = -\frac{\xi}{s_N} + \frac{\sigma - 1}{\phi + \sigma}\frac{s_M}{s_N} \tag{93}$$

where the first term captures substitution effect, and the second captures the output or scale effect. Suppose $\sigma > 1$ is given. The parameters ξ and ϕ together govern the sensitivity of import demand to its price. As $\xi \to 0$, the substitution effect disappears and production approaches the Leontief case. As $\phi \to \infty$, employment becomes less responsive and the scale effect disappears. In the joint limit,

$$\lim_{\substack{\xi \to 0 \\ \phi \to \infty}} \frac{d\log M}{d\log q} = 0$$

so that the import demand becomes locally perfectly price inelastic. Varying ξ and ϕ therefore varies the price elasticity of import demand and hence the degree to which imported inputs are essential for the domestic economy, with the Leontief, fixed-employment allocation providing the limiting case.

A.12.3 Calibration

We use an illustrative calibration to present the comparative statics. We set $A_t = 1$ and $\theta = 1$. We conjecture (and verify) that in the 'no-shock' case, i.e., when import prices are equal across periods, the constrained-efficient allocation is also symmetric across periods: $N_1 = N_2$, $M_1 = M_2$, and $\epsilon_1 = \epsilon_2 \equiv \bar{\epsilon}$. We therefore calibrate this case with $P_t^{T*} = 1$ and normalize $N_t = M_t = 1$. The firms' FOCs imply that $\bar{\epsilon} = \alpha$ and $\bar{w} = (1 - \alpha)$. α is chosen so that the domestic import bill is set equal to 4 percent of GDP:

$$\frac{\bar{\epsilon}\,\bar{P}^{T*}\bar{M}}{\bar{w}\,\bar{N}} = 0.04$$

The symmetric allocation involves $NX_1 = NX_2 = 0$. ζ is chosen so that the trade-balance condition is satisfied in the symmetric allocation:

$$\zeta \bar{\epsilon}^{\chi} = \bar{P}^{T*}\bar{M}$$

Given a value of ξ, the scale of labor disutility, η, is chosen so that the planner's optimality condition for employment is satisfied with $N_t = 1$. Since $\nu'(1) = \eta$, this calibration is independent of ϕ. Moreover, when import prices are identical across periods, the symmetric allocation satisfies $C_t^p = C_t^r$ so that heterogeneity is irrelevant and the allocation remains unchanged for any $\lambda \in [0, 1)$. The remaining parameters are set as follows: $\sigma = 2, \chi = 0.25, R^* = 1.0021$, and $\beta = \frac{1}{R^*}$. To isolate the role of heterogeneity, we compare how

optimal wedges vary between the representative-agent economy where $\lambda = 0$ and a heterogeneous-agent economy where we set $\lambda = 0.35$. The import-price 'shock' case involves setting unequal import prices across periods for which we set $P_1^{T*} = 1.5$ and $P_2^{T*} = 1$.

The comparative statics exercise involves varying ξ and ϕ and computing the optimal wedges in each of four cases: (a) $\lambda = 0$ with equal import prices, (b) $\lambda = 0$ with unequal import prices, (c) $\lambda > 0$ with equal import prices, and (d) $\lambda > 0$ with unequal import prices. This allows us to compare how the optimal wedges vary with ξ, ϕ, and λ in the presence and absence of import-price shocks.

A.12.4 Comparative Statics Results

Figures A.1, A.2, and A.3 report the optimal intertemporal wedge and the labor wedges over the (ξ, ϕ) parameter space, one sub-panel for each of the four cases described above in each of the three figures.

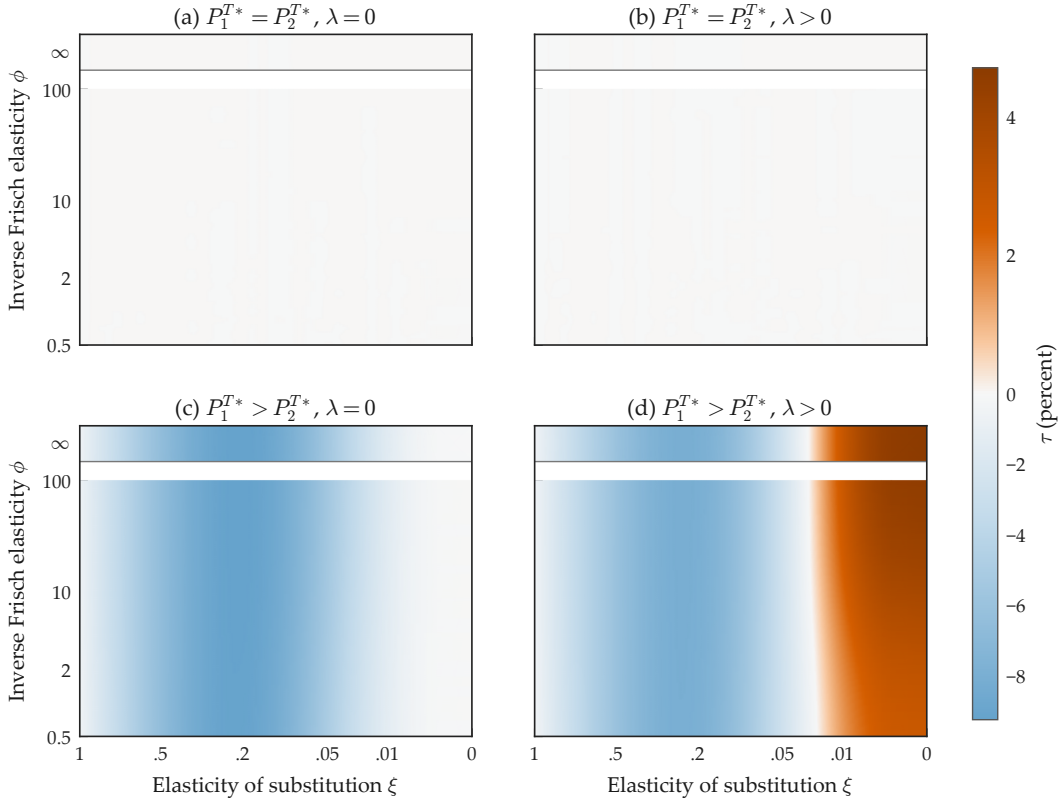


Figure A.1. Optimal intertemporal wedge τ

Note: The x-axis uses a cube-root scale.

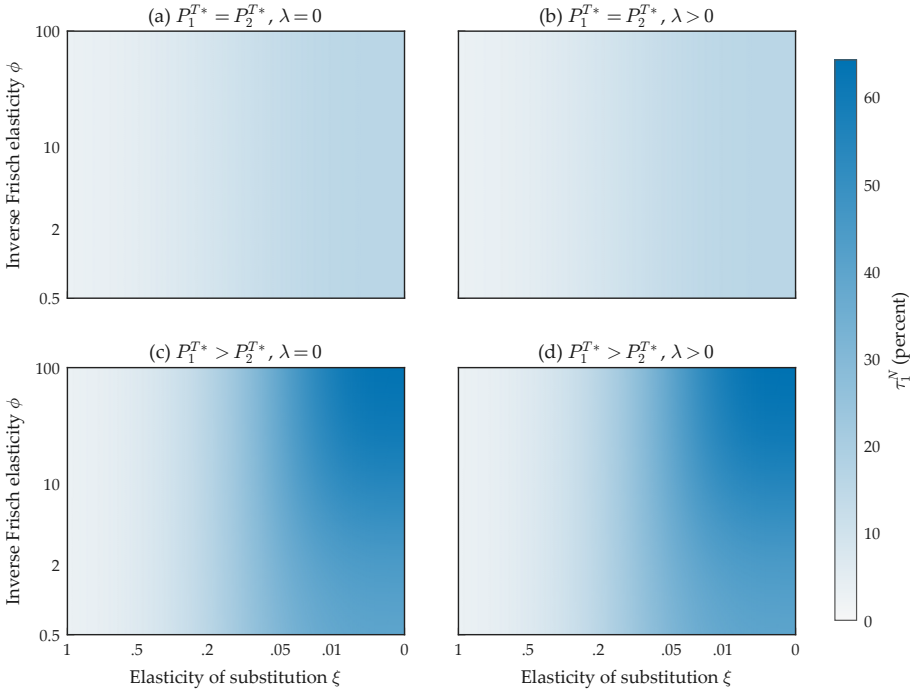


Figure A.2. Optimal period-1 labor wedge τ_1^N

Note: The x-axis uses a cube-root scale.

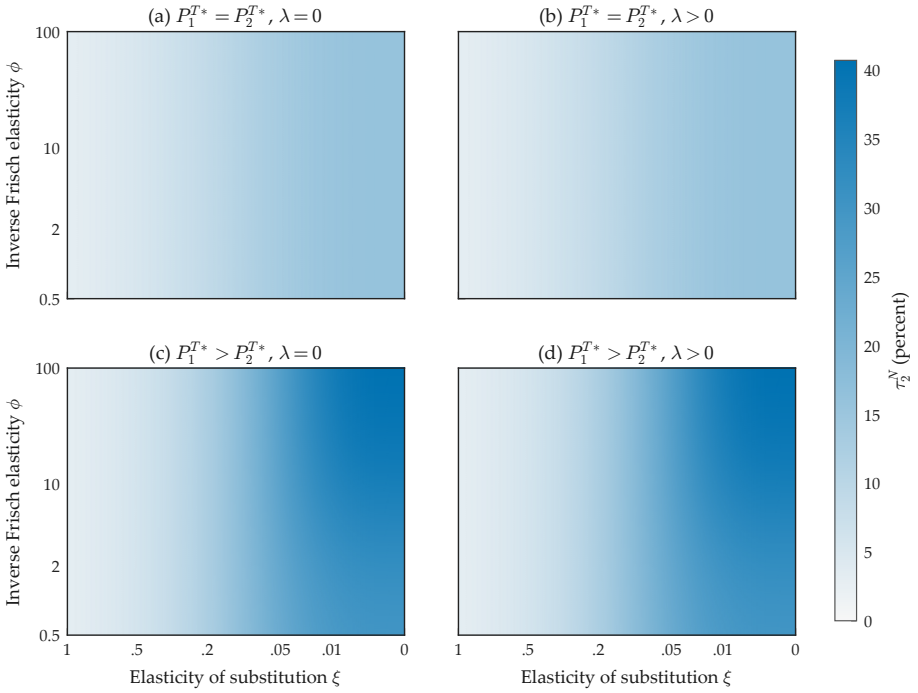


Figure A.3. Optimal period-2 labor wedge τ_2^N

Note: The x-axis uses a cube-root scale.

A.13 Imports as Final Goods and Comparative Statics

In this Appendix we consider an alternative model specification in which imports are utilized by consumers as final goods rather than being intermediate inputs in production. We continue to assume that the domestic tradeable good is produced only by the domestic economy and is either consumed domestically or exported: $Y_t = C_t + X_t$. The country faces a downward-sloping, price-elastic export demand for its own goods in the world market: $X_t = \zeta \epsilon_t^{1+\chi}$ where $P_t/e_t = 1/\epsilon_t$ is the world price of the domestic good in units of the world numeraire. The domestic economy continues to be small with respect to world asset markets where R^* is the world interest rate as well as small with respect to the market for imported goods with P_t^{T*} being the world price of imports in units of the world numeraire. By the law of one price, the domestic currency price of imports is $e_t P_t^{T*}$ and $\epsilon_t P_t^{T*}$ is the price in units of domestic output. We continue to interpret ϵ_t as the real exchange rate, as it is the endogenous relative price that determines real outcomes, although the typical price-level-based definition of the real exchange rate depends on both ϵ_t and P_t^{T*}.

Firms and Production Technology We assume that domestic tradeable output is produced by competitive firms using a linear production technology that uses just labor as the input: $Y_t = A_t N_t$. Firms' zero profit condition implies that the real wage is given by $w_t = A_t$.

Households Let C_t^j, M_t^j denote the consumption of domestic and imported goods respectively in period t by household j for $j \in \{p, r\}$. Households' period utility is given by:

$$u(C_t^j, M_t^j) - v(N_t)$$

HtM households face period-by-period static budget constraints:

$$C_t^p + \epsilon_t P_t^{T*} M_t^p = w_t N_t$$

Ricardian households are able to participate in international asset markets and are subject to capital controls:

$$C_t^r + \epsilon_t P_t^{T*} M_t^r + (1+\tau_t)\epsilon_t f_{t+1}^* = w_t N_t + \epsilon_t R^* f_t^* + T_t$$

Household optimality conditions include an intratemporal condition that equates the marginal rate of substitution between domestic and imported goods to the terms of trade:

$$\frac{u_2(C_t^j, M_t^j)}{u_1(C_t^j, M_t^j)} = \epsilon_t P_t^{T*} \quad j \in \{p, r\}$$

and an Euler equation for Ricardian households:

$$\frac{u_1(C_t^r, M_t^r)\epsilon_t}{\beta u_1(C_{t+1}^r, M_{t+1}^r)\epsilon_{t+1}} = \frac{R^*}{1+\tau_t}$$

Fiscal-Monetary Policy As in the main text, we assume that the exchange rate policy is chosen by a public agent who is able to distort the intertemporal margin by choosing capital controls with revenue

rebated as lump-sum transfers to Ricardian households:

$$\tau_t \epsilon_t F^*_{t+1} = (1-\lambda) T_t$$

Monetary policy continues to be reduced-form: the labor supply N_t is directly chosen by the public agent, reflecting the ability to distort the labor wedge. Tariffs and direct fiscal transfers to HtM households continue to remain absent.

Balance of Payments The balance of payments condition continues to be unchanged and is implied by all budget constraints and market clearing conditions:

$$NX_t = F^*_{t+1} - R^* F^*_t$$

where $NX_t = \frac{X_t}{\epsilon_t} - P^{T*}_t M_t$ with $M_t = \lambda M^p_t + (1-\lambda) M^r_t$.

Constrained Planning Problem For the purpose of numerical simulations and conducting the comparative statics exercise we consider a two-period framework. We also assume that Ricardian households begin with no initial assets. The intertemporal budget constraint is then given by $NX_1 + \frac{1}{R^*} NX_2 = 0$. The Ramsey planner's primal problem is given by:

$$\max_{\substack{\{N_t, \epsilon_t, C^p_t, C^r_t, \\ M^p_t, M^r_t\}^2_{t=1}}} \quad \sum_{t=1}^{2} \beta^{t-1} \left[\lambda u(C^p_t, M^p_t) + (1-\lambda) u(C^r_t, M^r_t) - v(N_t) \right] \quad \text{s.t.}$$

$$NX_1 + \frac{1}{R^*} NX_2 = 0$$

$$C^p_t + \epsilon_t P^{T*}_t M^p_t = A_t N_t$$

$$(1-\lambda) C^r_t + \lambda C^p_t + \zeta \epsilon_t^{1+\chi} = A_t N_t$$

$$\frac{u_2(C^j_t, M^j_t)}{u_1(C^j_t, M^j_t)} = \epsilon_t P^{T*}_t \quad j \in \{p, r\}$$

For the purpose of the comparative statics exercise, we continue to use the functional form $v(N) = \eta N^{1+\phi}/(1+\phi)$ for labor disutility and for consumption we use a Stone-Geary specification:

$$u(C, M) = \frac{\left[C^{1-\omega} (M - \underline{M})^{\omega} \right]^{1-\sigma}}{1-\sigma}, \qquad M \geqslant \underline{M} \tag{94}$$

where $\underline{M} \geqslant 0$ is a subsistence requirement for imported consumption and $\omega \in (0,1)$ is the weight on imported consumption. Absent the subsistence term, the Cobb-Douglas aggregator keeps expenditure shares constant, and domestic and imported goods are therefore neither gross substitutes nor gross complements. The Stone-Geary specification over imports allows us to vary their degree of essentiality.

A.13.1 The Necessity Nature of Imports

The parameters $\underline{M}$ and ω together govern how essential the imports are for the domestic economy. We illustrate this point with the help of a static representative agent economy and show how the price elasticity of import demand varies with these two parameters. Consider a static representative-agent set up with Stone-Geary utility over domestic and imported goods. Let $q \equiv \epsilon P^{T*}$ denote the relative price of imported goods in domestic units and let I denote the income of the household. The household's budget constraint is given by $C + qM = I$. For an interior solution, the optimality condition is

$$\frac{\omega C}{(1-\omega)(M-\underline{M})} = q$$

Combining this condition with the budget constraint gives the demand for imports:

$$M = \omega \frac{I}{q} + (1-\omega)\underline{M} \tag{95}$$

Its absolute Marshallian price elasticity, holding income fixed, is given by

$$\mathcal{E}_M \equiv -\frac{\partial \log M}{\partial \log q} = \frac{\omega I}{\omega I + (1-\omega) q \underline{M}} \tag{96}$$

When $\underline{M} = 0$, preferences reduce to Cobb-Douglas and import demand is unit elastic:

$$M = \omega \frac{I}{q}, \qquad \mathcal{E}_M = 1$$

When $\underline{M} > 0$, equation (95) separates import demand into a price-responsive component and an inelastic subsistence component. A larger subsistence component lowers the price elasticity of import demand. In the limit in which the discretionary component vanishes,

$$\omega \to 0, \qquad M \to \underline{M}, \qquad \mathcal{E}_M \to 0$$

so that import demand becomes perfectly price inelastic. The parameters ω and $\underline{M}$ therefore jointly govern the price elasticity and hence the degree of essentiality of final-good imports.

A.13.2 Calibration

The comparative statics exercise involves jointly varying the parameters ω and $\underline{M}$ that allows us to vary the price elasticity of demand, $\mathcal{E}_M$, over $[0, 1]$. We use an illustrative calibration to examine how the optimal intertemporal and labor wedges vary with the price elasticity. We set $A_t = 1$ and conjecture (and verify) that in the 'no-shock' case, i.e., when import prices are equal across periods, the constrained-efficient allocation is also symmetric across periods: $N_1 = N_2 = \bar{N}$, $\epsilon_1 = \epsilon_2 = \bar{\epsilon}$, $C_1^p = C_1^r = C_2^p = C_2^r = \bar{C}$, $M_1^p = M_1^r = M_2^p = M_2^r = \bar{M}$. We therefore calibrate this case with the normalizations $P_t^{T*} = 1$ and $\bar{N} = \bar{M} = 1$. The domestic import bill is calibrated to equal 4 percent of GDP:

$$\frac{\bar{\epsilon} \bar{P}^{T*} \bar{M}}{A \bar{N}} = 0.04 \tag{97}$$

The symmetric allocation involves $NX_1 = NX_2 = 0$. ζ is chosen so that the trade-balance condition is satisfied in the symmetric allocation: $\zeta \bar{\epsilon}^X = \bar{P}^{T*} \bar{M}$.

Let $\mathcal{E}_M \in [0,1]$ denote the absolute price elasticity of import demand evaluated at the symmetric allocation. From (96) and (97), we have:

$$\mathcal{E}_M = \frac{\omega A \bar{N}}{\bar{\epsilon}\bar{P}^{T*}\bar{M}} = \frac{\omega}{0.04} \tag{98}$$

For each value of $\mathcal{E}_M$, we therefore set

$$\omega(\mathcal{E}_M) = 0.04\mathcal{E}_M \tag{99}$$

The subsistence requirement is chosen so that import consumption remains equal to $\bar{M}$. Evaluating (95) at the symmetric allocation gives

$$\bar{M} = \omega(\mathcal{E}_M)\frac{A\bar{N}}{\bar{\epsilon}\bar{P}^{T*}} + [1-\omega(\mathcal{E}_M)]\,\underline{M}(\mathcal{E}_M)$$

It follows that

$$\underline{M}(\mathcal{E}_M) = \bar{M}\frac{1-\mathcal{E}_M}{1-0.04\mathcal{E}_M} \tag{100}$$

When $\mathcal{E}_M = 1$, $\omega = 0.04$ and $\underline{M} = 0$, so that preferences reduce to the Cobb-Douglas case and import demand is unit elastic. As $\mathcal{E}_M \to 0$, $\omega \to 0$ and $\underline{M} \to \bar{M}$, and import demand approaches the perfectly price-inelastic case.

For every value of $\mathcal{E}_M \in [0,1]$, the scale of labor disutility, η, is chosen so that the planner's optimality condition for employment is satisfied at $N_t = 1$ in the symmetric allocation. Since the symmetric allocation has $NX_t = 0$ and both household types consume the same bundle $(\bar{C}, \bar{M})$, household heterogeneity is irrelevant and the allocation remains unchanged for any $\lambda \in [0,1)$. The remaining parameters are set as follows: $\sigma = 2, \phi = 29.5, \chi = 0.25, R^* = 1.0021$ and $\beta = \frac{1}{R^*}$. To isolate the role of heterogeneity, we compare how optimal wedges vary between the representative-agent economy where $\lambda = 0$ and a heterogeneous-agent economy where we set $\lambda = 0.35$. The import-price 'shock' case involves setting unequal import prices across periods for which we set $P_1^{T*} = 1.5$ and $P_2^{T*} = 1$.

The comparative statics exercise involves varying $\mathcal{E}_M$ and computing the optimal wedges in each of four cases: (a) $\lambda = 0$ with equal import prices, (b) $\lambda = 0$ with unequal import prices, (c) $\lambda > 0$ with equal import prices, and (d) $\lambda > 0$ with unequal import prices.

A.13.3 Comparative Statics Results

Figure A.4 reports the differences $\Delta\tau$ and $\Delta\tau_1^N$ between the optimal wedges of the heterogeneous and representative-agent economies facing the unequal import price path, plotted against the price elasticity of import demand. Figures A.5, A.6, and A.7 report the optimal intertemporal and labor wedges in levels, one panel for each of the four cases.

First, with equal import prices the optimal intertemporal wedge is zero at every elasticity, for $\lambda = 0$ and $\lambda > 0$ alike, and the labor wedge alone carries the static terms-of-trade correction. This correction now runs in the opposite direction from the production case: it is largest at Cobb-Douglas and vanishes at the perfectly inelastic corner, where subsistence pins imports, balanced trade pins the terms-of-trade path,

and employment has no effect on it. With unequal prices and $\mathcal{E}_M > 0$, the planner distorts both margins against the terms-of-trade distortion absent tariffs, with $\lambda = 0$ and $\lambda > 0$ alike: the representative-agent intertemporal wedge is again the capital-control motive of Costinot, Lorenzoni and Werning (2014), and it vanishes at $\mathcal{E}_M = 0$, where the constrained-efficient allocation with $\lambda = 0$ coincides with laissez-faire. The perfectly inelastic subsistence limit is therefore the final-goods counterpart of the Leontief limit: any intertemporal distortion that arises there with $\lambda > 0$ reflects the role of heterogeneity alone.

Second, the intertemporal wedge difference rises with the degree of essentiality and is largest in the perfectly price inelastic demand limit. The real-income stabilization channel operates here directly, without any wage response: the real wage is fixed by the linear technology, and exchange rate policy reaches constrained households through the domestic price of their imported consumption.

Third, the labor wedge differences are negligible at every elasticity and approach zero as the degree of essentiality increases: heterogeneity generates no significant separate motive for monetary policy when imports are final goods. The complementarity between exchange rate policy and the monetary stance in the production economy runs through imported inputs that move with employment; with that link absent, the additional labor wedge attributable to heterogeneity is small. Exchange rate policy continues to perform the real-income stabilization task by distorting the intertemporal margin, whereas the scope and role of monetary policy are much weaker than in the production economy.

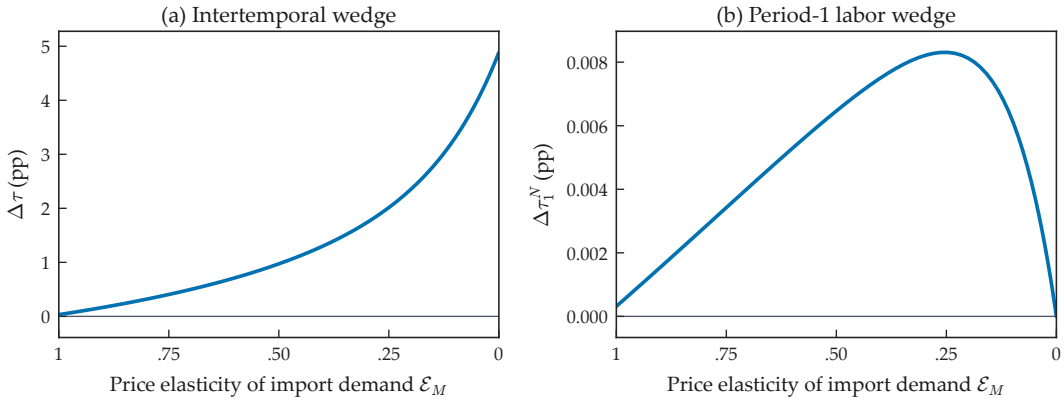


Figure A.4. Imports as final goods: optimal wedges, isolating the role of heterogeneity

Note: $\mathcal{E}_M = 1$ is the Cobb-Douglas case and $\mathcal{E}_M = 0$ the perfectly inelastic demand case.

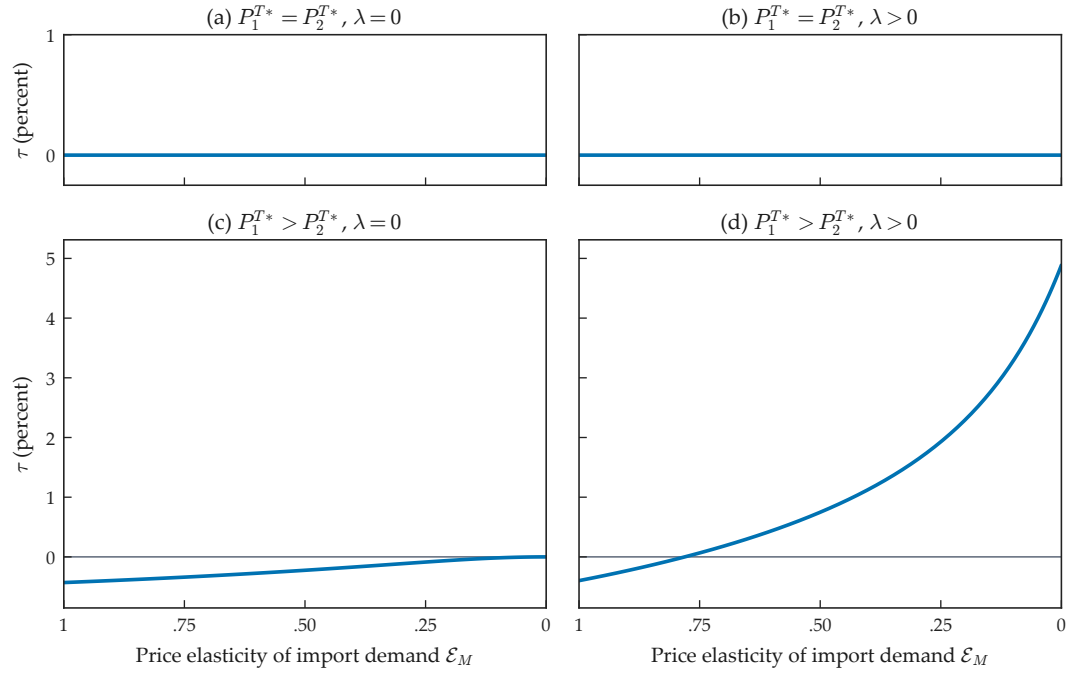


Figure A.5. Imports as final goods: optimal intertemporal wedge τ

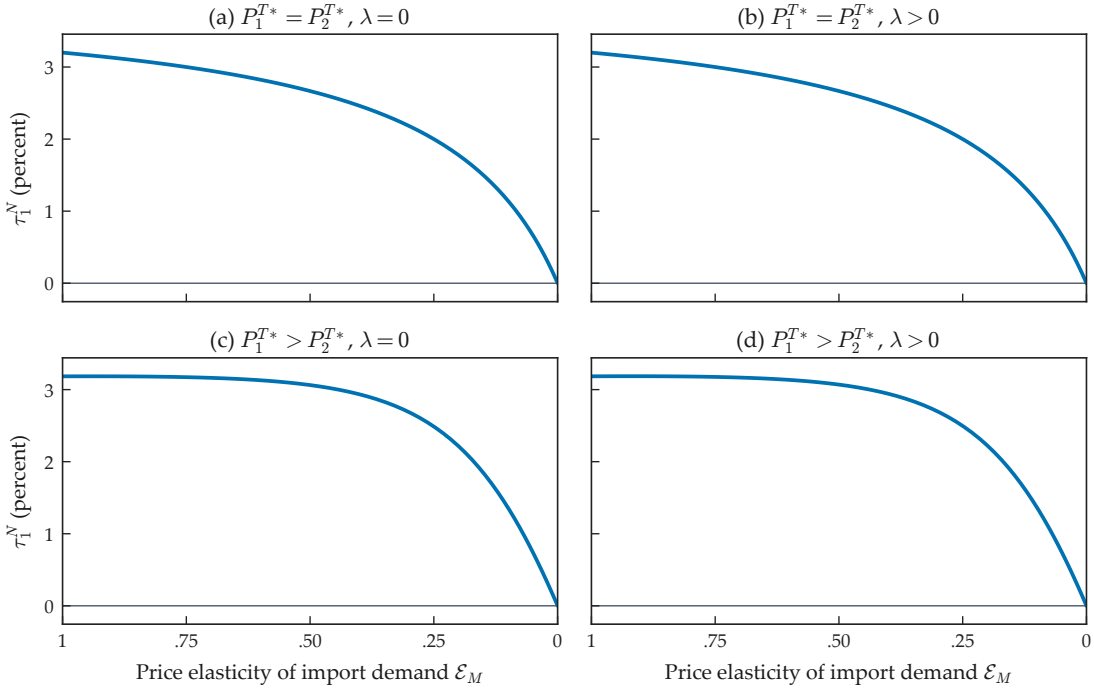


Figure A.6. Imports as final goods: optimal period-1 labor wedge τ_1^N

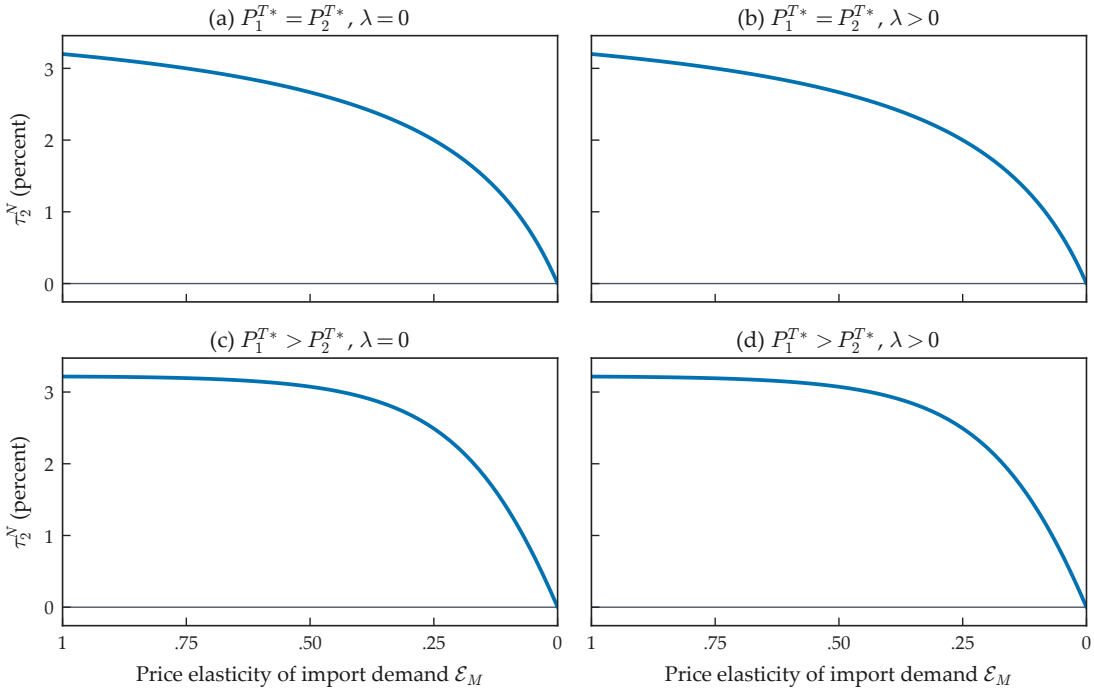


Figure A.7. Imports as final goods: optimal period-2 labor wedge τ_2^N

B Quantitative Model Appendix

B.1 Unions' Optimization Problem and the Wage Phillips Curve

At time t, union k sets the nominal wage $W_{k,t}$ to maximize (53) subject to the task demand schedule

$$N_{k,t} = \left(\frac{W_{k,t}}{W_t}\right)^{-\varepsilon_w} N_t$$

Each union is infinitesimal and takes W_t and N_t as given. Household real earnings are

$$Z_t = \frac{1}{P_t}\int_0^1 W_{k,t}\left(\frac{W_{k,t}}{W_t}\right)^{-\varepsilon_w} N_t \, dk$$

By the envelope theorem, we can evaluate how each household, j, values a marginal increase in labor incomes and hours worked. The marginal value of the earnings change is $u'(C_t^j)\frac{\partial Z_t}{\partial W_{k,t}}$, where

$$\frac{\partial Z_t}{\partial W_{k,t}} = \frac{1}{P_t} N_{k,t}(1-\varepsilon_w)$$

On the other hand, total hours worked by household j are $N_t^j \equiv \int_0^1 \left(\frac{W_{k,t}}{W_t}\right)^{-\varepsilon_w} N_t \, dk$ which falls when $W_{k,t}$ rises according to

$$\frac{\partial N_t^j}{\partial W_{k,t}} = -\varepsilon_w \frac{N_{k,t}}{W_{k,t}}$$

Therefore, the union's first order condition is given by:

$$\left(\frac{W_{k,t}}{W_{k,t-1}}-1\right)\frac{W_{k,t}}{W_{k,t-1}} = \frac{\varepsilon_w}{\psi_{nr}}\left[N_{k,t}v'(N_t)-\frac{\varepsilon_w-1}{\varepsilon_w}\frac{W_{k,t}}{P_t}N_{k,t}\bar{u}'_t\right]+\beta\left(\frac{W_{k,t+1}}{W_{k,t}}-1\right)\frac{W_{k,t+1}}{W_{k,t}}$$

where $\bar{u}'_t \equiv \lambda u'(C^p_t)+(1-\lambda)u'(C^r_t)$. In a symmetric equilibrium, $W_{k,t}=W_t$ and $N_{k,t}=N_t$ for all k. Let $\mu_w \equiv \varepsilon_w/(\varepsilon_w-1)$. Then

$$\left(\frac{W_t}{W_{t-1}}-1\right)\frac{W_t}{W_{t-1}} = \frac{\varepsilon_w}{\psi_{nr}}\left[N_t v'(N_t)-\frac{1}{\mu_w}\frac{W_t}{P_t}N_t\bar{u}'_t\right]+\beta\left(\frac{W_{t+1}}{W_t}-1\right)\frac{W_{t+1}}{W_t}$$

Using $w_t = W_t/P_t$ and $\Pi_t = P_t/P_{t-1}$, this can be written as

$$\left(\frac{w_t}{w_{t-1}}\Pi_t-1\right)\frac{w_t}{w_{t-1}}\Pi_t = \frac{\varepsilon_w}{\psi_{nr}}\left[N_t v'(N_t)-\frac{1}{\mu_w}w_t N_t\bar{u}'_t\right]+\beta\left(\frac{w_{t+1}}{w_t}\Pi_{t+1}-1\right)\frac{w_{t+1}}{w_t}\Pi_{t+1}$$

Rearranging gives $v'(N_t)=(1-\tau^N_t)w_t\bar{u}'_t$, where

$$1-\tau^N_t = \frac{1}{\mu_w}+\frac{\left(\frac{w_t}{w_{t-1}}\Pi_t-1\right)\frac{w_t}{w_{t-1}}\Pi_t-\beta\left(\frac{w_{t+1}}{w_t}\Pi_{t+1}-1\right)\frac{w_{t+1}}{w_t}\Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{nr}}w_t N_t\bar{u}'_t}$$

B.2 Balance of Payments and Intertemporal Budget Constraint

Aggregating the Ricardian and HtM household budget constraints gives the aggregate household budget constraint:

$$C_t + B_{t+1} = w_t N_t + R_t B_t + (1-\lambda)\frac{T_t}{P_t} \tag{101}$$

Adding (101) to the government budget constraint (62) eliminates transfers and gives

$$C_t + B_{t+1} + B^G_{t+1} + \epsilon_t B^{G*}_{t+1} = w_t N_t + R_t(B_t + B^G_t) + \epsilon_t R^* B^{G*}_t \tag{102}$$

Firm zero profits and goods market clearing are given by

$$w_t N_t = A_t N_t - \epsilon_t P^{T*}_t \theta N_t \qquad C_t + \zeta\epsilon_t^{1+\chi} = A_t N_t \tag{103}$$

Substituting (103) into (102) yields

$$B_{t+1} + B^G_{t+1} + \epsilon_t B^{G*}_{t+1} = \zeta\epsilon_t^{1+\chi} - \epsilon_t P^{T*}_t \theta N_t + R_t(B_t + B^G_t) + \epsilon_t R^* B^{G*}_t \tag{104}$$

Finally, using domestic bond market clearing (65), and dividing by ϵ_t, gives the period-by-period balance of payments condition (66).

To obtain the intertemporal budget constraint, use the intermediary asset demand rule (59) for the inherited position F^*_t when $t \geqslant 2$:

$$R_t\frac{\epsilon_{t-1}}{\epsilon_t}F^*_t = R^*F^*_t - \Gamma R^*(F^*_t)^2 \tag{105}$$

Substituting (105) into (66) gives

$$P^{T*}_t\theta N_t + \Gamma R^*(F^*_t)^2 + F^*_{t+1} + B^{G*}_{t+1} = \zeta\epsilon^\chi_t + R^*(F^*_t + B^{G*}_t) \tag{106}$$

for all $t \geqslant 2$. The period-1 condition stays unchanged:

$$P_1^{T*}\theta N_1 + F_2^* + B_2^{G*} = \zeta\epsilon_1^{\chi} + R_1\frac{\epsilon_0}{\epsilon_1}F_1^* + R^*B_1^{G*} \tag{107}$$

The no-Ponzi condition on households, (50), and the government's domestic asset position, (63), together with the domestic bonds market clearing, (65), imply:

$$\lim_{T\to\infty}\frac{\epsilon_T F_{T+1}^*}{\prod_{s=2}^{T} R_s} = 0 \tag{108}$$

Since $\prod_{s=2}^{T}\tilde{R}_s = (\prod_{s=2}^{T} R_s)\epsilon_1/\epsilon_T$, (108) implies $F_{T+1}^*/\prod_{s=2}^{T}\tilde{R}_s \to 0$. The interest-rate consistency condition, (64), then gives

$$\lim_{T\to\infty}\frac{F_{T+1}^*}{R^{*T-1}} = 0 \tag{109}$$

Together with the government no-Ponzi condition on B_{T+1}^{G*}, (63), we have:

$$\lim_{T\to\infty}\frac{F_{T+1}^* + B_{T+1}^{G*}}{R^{*T-1}} = 0 \tag{110}$$

Combining (107) with (106), discounting by R^{*t-1}, summing from $t = 1$ to T, gives

$$\sum_{t=1}^{T}\frac{P_t^{T*}\theta N_t}{R^{*t-1}} + \sum_{t=1}^{T-1}\frac{\Gamma(F_{t+1}^*)^2}{R^{*t-1}} + \frac{F_{T+1}^* + B_{T+1}^{G*}}{R^{*T-1}} = \sum_{t=1}^{T}\frac{\zeta\epsilon_t^{\chi}}{R^{*t-1}} + R^*B_1^{G*} + R_1\frac{\epsilon_0}{\epsilon_1}F_1^* \tag{111}$$

Taking $T \to \infty$, using (110) and substituting $\Gamma(F_{t+1}^*)^2 = \Gamma^{-1}(1 - \tilde{R}_{t+1}/R^*)^2$ from (59) yields the intertemporal budget constraint, (67).

B.3 Decentralized Equilibrium Definition

Given:

- Initial states: $\{R_1^M, B_1, B_1^{G*}, \epsilon_0, F_1^*, w_0\}$,
- Exogenous sequence of import prices and productivity, $\{P_t^{T*}, A_t\}_{t=1}^{\infty}$,
- Fiscal-monetary policy: $\{B_{t+1}^G, B_{t+1}^{G*}, R_{t+1}^M, \frac{T_t}{P_t}\}_{t=1}^{\infty}, R_1$,

a decentralized equilibrium consists of an allocation: $\{C_t^r, C_t^p, B_{t+1}, N_t, M_t, Y_t, F_{t+1}^*, X_t\}_{t=1}^{\infty}$ and prices $\{\epsilon_t, w_t, \Pi_t, R_{t+1}\}_{t=1}^{\infty}$ such that:

- Given prices, Ricardian households solve (48) and HtM households satisfy their period-by-period budget constraints:

$$\frac{u'(C_t^r)}{\beta u'(C_{t+1}^r)} = R_{t+1}$$

$$C_t^r + \frac{B_{t+1}}{1-\lambda} = w_t N_t + R_t\frac{B_t}{1-\lambda} + \frac{T_t}{P_t}$$

$$\lim_{T\to\infty}\frac{B_{T+1}}{\prod_{s=2}^{T} R_s} = 0$$

$$C_t^p = w_t N_t$$

- Given prices, firms solve (3):

$$M_t = \theta N_t, \quad Y_t = A_t N_t$$

Zero profits imply:

$$w_t = A_t - \epsilon_t P_t^{T*} \theta$$

- Labor aggregator and unions' optimization gives the wage Phillips curve:

$$\left(\frac{w_t}{w_{t-1}}\Pi_t - 1\right)\frac{w_t}{w_{t-1}}\Pi_t = \frac{\varepsilon_w}{\psi_{nr}}\left[N_t \nu'(N_t) - \frac{1}{\mu_w} w_t N_t \bar{u}_t'\right] + \beta\left[\left(\frac{w_{t+1}}{w_t}\Pi_{t+1} - 1\right)\frac{w_{t+1}}{w_t}\Pi_{t+1}\right]$$

- Given prices, international financial intermediaries optimize:

$$F_{t+1}^* = \frac{1}{\Gamma}\left[1 - \frac{R_{t+1}}{R^*}\left(\frac{\epsilon_t}{\epsilon_{t+1}}\right)\right]$$

- World demand for domestic output is given by:

$$X_t = \zeta \epsilon_t^{1+\chi}$$

- Fiscal policy satisfies the period-by-period budget constraints:

$$B_{t+1}^G + \epsilon_t B_{t+1}^{G*} + (1-\lambda)\frac{T_t}{P_t} = R_t B_t^G + \epsilon_t R^* B_t^{G*}$$

where $B_1^G = \epsilon_0 F_1^* - B_1$; satisfies no-Ponzi conditions:

$$\lim_{T\to\infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = 0, \quad \lim_{T\to\infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0$$

together with the interest rate consistency condition:

$$0 < \inf_{T\geqslant 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} \leqslant \sup_{T\geqslant 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} < \infty$$

- Inflation is implied by monetary policy:

$$R_t^M = R_t \Pi_t, \qquad t \geqslant 1$$

- Final goods market clears:

$$\lambda C_t^p + (1-\lambda) C_t^r + X_t = A_t N_t$$

- Labor market clears.
- Domestic bonds market clears:

$$B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^*$$

- Balance of payments is implied by all other budget constraints and market clearing conditions holding:

$$P_t^{T*}\theta N_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^{\chi} + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* + R^* B_t^{G*}$$

B.4 Optimal Policy Problem

Given B_1^{G*} and under the assumption that $F_1^* = 0$, consider the constrained planning problem:

$$\max_{\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} \left[\lambda u\left(C_t^p\right) + (1-\lambda) u\left(C_t^r\right) - v(N_t)\right] \qquad \text{s.t.}$$

$$\sum_{t=1}^{\infty} \frac{\frac{1}{\Gamma}\left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}} + \sum_{t=1}^{\infty} \frac{P_t^{T*}\theta N_t}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^{\chi}}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1}$$

$$\tilde{R}_{t+1} = \frac{u'\left(C_t^r\right)\epsilon_t}{\beta u'\left(C_{t+1}^r\right)\epsilon_{t+1}} \tag{112}$$

$$C_t^r = A_t N_t - \frac{\zeta \epsilon_t^{1+\chi}}{1-\lambda} + \frac{\lambda}{1-\lambda} \epsilon_t P_t^{T*} \theta N_t$$

$$C_t^p = (A_t - \epsilon_t P_t^{T*}\theta) N_t$$

$$N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p \geqslant 0$$

The following Lemmas establish conditions under which a solution to (112) is implementable as a decentralized equilibrium defined in Section B.3.

Lemma B.1. *Given $F_1^* = 0$, B_1^{G*}, $\beta R^* = 1$, and positive sequences $\{A_t, P_t^{T*}\}_{t\geqslant 1}$ satisfying $A_t \leqslant \bar{A} < \infty$ and $P_t^{T*} \geqslant \underline{P} > 0$ for all $t \geqslant 1$, let $\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t\geqslant 1}$ solve (112). Suppose further that the solution satisfies for all $t \geqslant 1$:*

$$\epsilon_t \geqslant \underline{\epsilon}, \qquad \underline{C}^r \leqslant C_t^r \leqslant \bar{C}^r$$

where $\underline{\epsilon}, \underline{C}^r, \bar{C}^r > 0$. In addition, assume the conditions of Lemma B.2 hold for $w_t = A_t - \epsilon_t P_t^{T*}\theta$ and $\tilde{u}_t' = \lambda u'(C_t^p) + (1-\lambda)u'(C_t^r)$. Then there exists a non-unique sequence of transfers $\{T_t/P_t\}_{t\geqslant 1}$ such that the solution to (112) is implementable as a decentralized equilibrium defined in Section B.3.

Proof. Construct w_t, M_t, Y_t, X_t, F_{t+1}^*, and R_{t+1} directly:

$$w_t = A_t - \epsilon_t P_t^{T*}\theta, \qquad M_t = \theta N_t, \qquad Y_t = A_t N_t, \qquad X_t = \zeta \epsilon_t^{1+\chi}$$

$$F_{t+1}^* = \frac{1}{\Gamma}\left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)$$

$$R_{t+1} = \tilde{R}_{t+1} \frac{\epsilon_{t+1}}{\epsilon_t}$$

Since Ricardian Euler equation and HtM budget constraint are implementability conditions of (112), they are already satisfied. Using the wage determination equation, firms' zero profit condition: $A_t N_t = w_t N_t + \epsilon_t P_t^{T*}\theta N_t$ also holds. The condition $C_t^r = A_t N_t - \frac{\zeta \epsilon_t^{1+\chi}}{1-\lambda} + \frac{\lambda}{1-\lambda}\epsilon_t P_t^{T*}\theta N_t$ then implies that the goods market clearing condition holds: $(1-\lambda)C_t^r + \lambda C_t^p + X_t = A_t N_t$.

The Ricardian Euler equation and $\beta R^* = 1$ imply

$$\prod_{s=2}^{T} \frac{\tilde{R}_s}{R^*} = \frac{\epsilon_1 u'(C_1^r)}{\epsilon_T u'(C_T^r)}$$

Since $C_t^p = (A_t - \epsilon_t P_t^{T*}\theta)N_t > 0$ and $N_t > 0$,

$$\epsilon_t < \frac{A_t}{P_t^{T*}\theta} \leqslant \frac{\bar{A}}{\underline{P}\theta}$$

The lower bound on ϵ_t and the bounds assumptions on C_t^r therefore imply that there exist constants $0 < \underline{m} < \overline{m} < \infty$ such that $\underline{m} \leqslant \epsilon_t u'(C_t^r) \leqslant \overline{m}$ for all $t \geqslant 1$. Hence the interest rate consistency condition holds:

$$0 < \inf_{T \geqslant 2} \prod_{s=2}^{T} \frac{\tilde{R}_s}{R^*} \leqslant \sup_{T \geqslant 2} \prod_{s=2}^{T} \frac{\tilde{R}_s}{R^*} < \infty$$

Choose any sequence of transfers $\{T_t/P_t\}_{t \geqslant 1}$ satisfying

$$\sum_{t=1}^{\infty} \frac{(1-\lambda)T_t/P_t}{\prod_{s=2}^{t} R_s} = \sum_{t=1}^{\infty} \frac{C_t - w_t N_t}{\prod_{s=2}^{t} R_s} - R_1 B_1 \tag{113}$$

where $C_t = (1-\lambda)C_t^r + \lambda C_t^p$. This choice is non-unique. One admissible choice sets $(1-\lambda)T_1/P_1$ equal to the right-hand side of (113) and sets $T_t/P_t = 0$ for all $t \geqslant 2$. Using goods market clearing and zero profits, we have $C_t - w_t N_t = \epsilon_t \left(P_t^{T*}\theta N_t - \zeta \epsilon_t^{\chi}\right)$. Then

$$\frac{C_t - w_t N_t}{\prod_{s=2}^{t} R_s} = \epsilon_1 \frac{P_t^{T*}\theta N_t - \zeta \epsilon_t^{\chi}}{\prod_{s=2}^{t} \tilde{R}_s}$$

By the interest rate consistency condition, there exists $K < \infty$ such that $\frac{1}{\prod_{s=2}^{t} \tilde{R}_s} \leqslant \frac{K}{R^{*t-1}}$ for all $t \geqslant 1$. The intertemporal budget constraint in (112) implies

$$\sum_{t=1}^{\infty} \left| \frac{C_t - w_t N_t}{\prod_{s=2}^{t} R_s} \right| \leqslant K\epsilon_1 \sum_{t=1}^{\infty} \frac{P_t^{T*}\theta N_t + \zeta \epsilon_t^{\chi}}{R^{*t-1}} < \infty$$

Define $\{B_{t+1}\}_{t \geqslant 1}$ recursively by the consolidated household budget

$$B_{t+1} = w_t N_t + R_t B_t + (1-\lambda)\frac{T_t}{P_t} - C_t, \qquad B_1 \text{ given} \tag{114}$$

Since $C_t^p = w_t N_t$ holds, subtracting λ times this from (114) and dividing by $(1-\lambda)$ yields the Ricardian budget:

$$C_t^r + \frac{B_{t+1}}{1-\lambda} = w_t N_t + R_t \frac{B_t}{1-\lambda} + \frac{T_t}{P_t}$$

Iterating (114) from $t = 1$ to T gives

$$\frac{B_{T+1}}{\prod_{s=2}^{T} R_s} = R_1 B_1 + \sum_{t=1}^{T} \frac{w_t N_t + (1-\lambda)T_t/P_t - C_t}{\prod_{s=2}^{t} R_s}$$

By (113), the right-hand side converges to zero as $T \to \infty$, so $\lim_{T\to\infty} \frac{B_{T+1}}{\prod_{s=2}^{T} R_s} = 0$.

Construct B^G_{t+1} from the bonds market clearing condition:

$$B^G_{t+1} = \epsilon_t F^*_{t+1} - B_{t+1}$$

Since $F^*_1 = 0$, the initial condition $B^G_1 = \epsilon_0 F^*_1 - B_1 = -B_1$ holds.

Construct B^{G*}_{t+1} from the period-by-period BoP conditions:

$$B^{G*}_{t+1} = \zeta \epsilon^{\chi}_t + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F^*_t + R^* B^{G*}_t - P^{T*}_t \theta N_t - F^*_{t+1}$$

with $F^*_1 = 0, B^{G*}_1$ given. Since the solution satisfies the intertemporal budget constraint, it satisfies the no-Ponzi condition on the net foreign asset position:

$$\lim_{T\to\infty} \frac{B^{G*}_{T+1} + F^*_{T+1}}{R^{*\,T-1}} = 0$$

Moreover, since the solution satisfies:

$$\sum_{t=1}^{\infty} \frac{\frac{1}{\Gamma}\left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*\,t-1}} = \sum_{t=1}^{\infty} \frac{\Gamma\left(F^*_{t+1}\right)^2}{R^{*\,t-1}} < \infty$$

therefore $\left(F^*_{T+1}\right)^2 / R^{*\,T-1} \to 0$. Hence, the following hold separately

$$\lim_{T\to\infty} \frac{F^*_{T+1}}{R^{*\,T-1}} = 0, \quad \lim_{T\to\infty} \frac{B^{G*}_{T+1}}{R^{*\,T-1}} = 0$$

Using the bonds market clearing condition we have:

$$\frac{B^G_{T+1}}{\prod_{s=2}^{T} R_s} = \frac{\epsilon_1 F^*_{T+1}}{\prod_{s=2}^{T} \tilde{R}_s} - \frac{B_{T+1}}{\prod_{s=2}^{T} R_s}$$

Taking limits: $T \to \infty$, the second term on the RHS vanishes by Household no-Ponzi condition holding. For the first: since $F^*_{T+1}/R^{*\,T-1} \to 0$ and $\sup_T \prod_{s=2}^{T}(R^*/\tilde{R}_s) < \infty$, then $\frac{F^*_{T+1}}{\prod_{s=2}^{T}\tilde{R}_s} = \frac{F^*_{T+1}}{R^{*\,T-1}} \cdot \prod_{s=2}^{T} \frac{R^*}{\tilde{R}_s} \;\Rightarrow\; 0$. Hence $\lim_{T\to\infty} \frac{B^G_{T+1}}{\prod_{s=2}^{T} R_s} = 0$.

The period-by-period government budget constraints hold by Walras' identity: multiplying period-by-period BoP conditions by ϵ_t and substituting $C_t + X_t = A_t N_t$ and $w_t N_t = A_t N_t - \epsilon_t P^{T*}_t \theta N_t$ gives

$$\epsilon_t F^*_{t+1} + \epsilon_t B^{G*}_{t+1} = w_t N_t - C_t + R_t \epsilon_{t-1} F^*_t + \epsilon_t R^* B^{G*}_t$$

Substituting $w_t N_t - C_t = B_{t+1} - R_t B_t - (1-\lambda)T_t/P_t$ and $B_{t+1} + B^G_{t+1} = \epsilon_t F^*_{t+1}$ gives

$$B^G_{t+1} + \epsilon_t B^{G*}_{t+1} + (1-\lambda)\frac{T_t}{P_t} = R_t B^G_t + \epsilon_t R^* B^{G*}_t$$

Finally, to construct the inflation path and the nominal interest rates, construct the labor wedge as:

$$\tau^N_t = 1 - \frac{\nu'(N_t)}{\tilde{u}'_t w_t}$$

Then construct Π_t as a solution to:

$$1-\tau_t^N=\frac{1}{\mu_w}+\left[\frac{\left(\frac{w_t}{w_{t-1}}\Pi_t-1\right)\frac{w_t}{w_{t-1}}\Pi_t-\beta\left[\left(\frac{w_{t+1}}{w_t}\Pi_{t+1}-1\right)\frac{w_{t+1}}{w_t}\Pi_{t+1}\right]}{\frac{\varepsilon_w}{\psi_{nr}}w_tN_t\bar{u}_t'}\right]$$

By Lemma B.2, there exists $\{\Pi_t\}_{t\geqslant 1}\subset(0,\infty)$ satisfying the above equation.

Construct R_1, R_{t+1}^M using:

$$R_1=\frac{R_1^M}{\Pi_1},\qquad R_{t+1}^M=R_{t+1}\Pi_{t+1},\quad t\geqslant 1$$

All decentralized equilibrium conditions in Section B.3 are therefore satisfied. Non-uniqueness follows from the non-uniqueness of any transfer sequence satisfying (113). □

Lemma B.2. *Let $\{N_t, w_t, \bar{u}_t'\}_{t\geqslant 1}$ be strictly positive sequences, with $w_0>0$ given. Suppose that, for every $t\geqslant 1$,*

$$\mathcal{X}_t\equiv\sum_{j=t}^{\infty}\beta^{j-t}\frac{\varepsilon_w}{\psi_{nr}}\left[N_jv'(N_j)-\frac{1}{\mu_w}w_jN_j\bar{u}_j'\right]\ \textit{is finite and lies in}\ \left(-\frac{1}{4},\infty\right)\tag{115}$$

Then there exists a sequence $\{\Pi_t\}_{t\geqslant 1}\subset(0,\infty)$ such that the wage Phillips curve (54) holds for all $t\geqslant 1$.

Proof. Define

$$\Pi_t=\frac{w_{t-1}}{w_t}\frac{1+\sqrt{1+4\mathcal{X}_t}}{2}$$

By (115), $\Pi_t>0$ for all $t\geqslant 1$ and

$$\mathcal{X}_t=\left(\frac{w_t}{w_{t-1}}\Pi_t-1\right)\frac{w_t}{w_{t-1}}\Pi_t$$

Moreover, the definition of $\mathcal{X}_t$ implies

$$\mathcal{X}_t=\frac{\varepsilon_w}{\psi_{nr}}\left[N_tv'(N_t)-\frac{1}{\mu_w}w_tN_t\bar{u}_t'\right]+\beta\mathcal{X}_{t+1}$$

Therefore the wage Phillips curve holds for all $t\geqslant 1$. □

Lemma B.3. *Given $F_1^*=0$, B_1^{G*}, $\beta R^*=1$, $w_0>0$, and positive sequences $\{A_t, P_t^{T*}\}_{t\geqslant 1}$ satisfying $A_t\to A_{ss}>0$, and $P_t^{T*}\to P_{ss}^{T*}>0$, let $\{N_t,\epsilon_t,\tilde{R}_{t+1},C_t^r,C_t^p\}_{t\geqslant 1}$ solve (112) with*

$$N_t\to N_{ss}>0,\qquad \epsilon_t\to\epsilon_{ss}>0,\qquad C_t^r\to C_{ss}^r>0,\qquad C_t^p\to C_{ss}^p>0,\qquad \tilde{R}_{t+1}\to R^*>0$$

Suppose also that condition (115) holds for every $t\geqslant 1$. Then this solution is implementable as a decentralized equilibrium. Moreover, under the larger-root convention, the inflation path $\{\Pi_t\}_{t\geqslant 1}\subset(0,\infty)$ satisfying the wage Phillips curve and converging to a positive finite limit is unique and converges to

$$\Pi_{ss}=\frac{1}{2}+\sqrt{\frac{1}{4}+\frac{1}{1-\beta}\frac{\varepsilon_w}{\psi_{nr}}\left[N_{ss}v'(N_{ss})-\frac{1}{\mu_w}w_{ss}N_{ss}\bar{u}_{ss}'\right]}$$

where $w_{ss}=A_{ss}-\epsilon_{ss}P_{ss}^{T*}\theta$.

Proof. Since $A_t \to A_{ss} > 0$ and $P_t^{T*} \to P_{ss}^{T*} > 0$, there exist $\bar{A} < \infty$ and $\underline{P} > 0$ such that $A_t \leqslant \bar{A}$ and $P_t^{T*} \geqslant \underline{P}$ for all $t \geqslant 1$. Since $\epsilon_t \to \epsilon_{ss} > 0$ and $C_t^r \to C_{ss}^r > 0$, there exist $\underline{\epsilon}, \underline{C}^r, \bar{C}^r > 0$ such that $\epsilon_t \geqslant \underline{\epsilon}$ and $\underline{C}^r \leqslant C_t^r \leqslant \bar{C}^r$ for all $t \geqslant 1$. Hence all conditions of Lemma B.1 are satisfied. Implementability follows as a corollary. Moreover by Lemma B.2, there exists an inflation path $\{\Pi_t\}_{t\geqslant 1} \subset (0, \infty)$ satisfying the wage Phillips curve. It remains to show that the inflation path is unique. Let

$$\mathcal{A}_t = \frac{\varepsilon_w}{\psi_{nr}} \left[N_t v'(N_t) - \frac{1}{\mu_w} w_t N_t \bar{u}_t' \right]$$

Since $w_t = A_t - \epsilon_t P_t^{T*} \theta$, convergence gives $w_t \to w_{ss} > 0$ then

$$\mathcal{A}_t \to \mathcal{A}_{ss} = \frac{\varepsilon_w}{\psi_{nr}} \left[N_{ss} v'(N_{ss}) - \frac{1}{\mu_w} w_{ss} N_{ss} \bar{u}_{ss}' \right]$$

For any inflation path satisfying the wage Phillips curve, the associated $\mathcal{X}_t$ satisfies

$$\mathcal{X}_t = \mathcal{A}_t + \beta \mathcal{X}_{t+1}$$

Thus

$$\mathcal{X}_t = \sum_{j=t}^{\infty} \beta^{j-t} \mathcal{A}_j + \kappa \beta^{-t}, \qquad \kappa \in \mathbb{R}$$

Since $\mathcal{A}_t \to \mathcal{A}_{ss}$,

$$\sum_{j=t}^{\infty} \beta^{j-t} \mathcal{A}_j \to \frac{\mathcal{A}_{ss}}{1-\beta}$$

If $\kappa < 0$, then $\mathcal{X}_t \to -\infty$, so $\mathcal{X}_t < -1/4$ for all sufficiently large t, which is incompatible with $\Pi_t > 0$. If $\kappa > 0$, then $\mathcal{X}_t \to +\infty$, so the larger-root inflation path cannot converge to a finite limit. Hence any positive larger-root inflation path with a finite limit must have $\kappa = 0$. In that case,

$$\mathcal{X}_t = \sum_{j=t}^{\infty} \beta^{j-t} \mathcal{A}_j \to \frac{\mathcal{A}_{ss}}{1-\beta}$$

Under the larger-root convention,

$$\Pi_t = \frac{w_{t-1}}{w_t} \frac{1 + \sqrt{1 + 4\mathcal{X}_t}}{2}$$

Since $w_{t-1}/w_t \to 1$, it follows that

$$\Pi_t \to \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{1}{1-\beta} \frac{\varepsilon_w}{\psi_{nr}} \left[N_{ss} v'(N_{ss}) - \frac{1}{\mu_w} w_{ss} N_{ss} \bar{u}_{ss}' \right]} \equiv \Pi_{ss} \in (0, \infty)$$

The case $\kappa = 0$ is the only case admitting a positive larger-root inflation path with a finite positive limit. Hence the convergent inflation path is unique. □

B.5 Calibration

We calibrate the quantitative model to Japan and interpret the imported input as fossil fuel imports. A period is one quarter. The calibration is designed for unanticipated perfect-foresight paths of the imported-input price, P_t^{T*}, and chooses the no-shock steady state as the initial state for all policy experiments. Since

$\beta R^* = 1$, the net foreign asset position is non-stationary: a transitory import-price shock changes the external position permanently, and each experiment converges to a terminal steady state that differs from the initial one. The economy repays the foreign purchasing power that finances the high-price quarters through a permanently higher trade balance, which is the infinite-horizon counterpart of the second period of the two-period model. Table B.1 reports the parameter values used in the quantitative exercises.

The normalizations are $A_{ss} = 1$, $P^{T*}_{ss} = 1$, and $\theta = 1$. The scale parameter ζ is chosen jointly with the steady real exchange rate to set the fossil-fuel import bill at 4 percent of GDP and net exports at -0.5 percent of GDP in the no-shock steady state. Since the no-shock steady state has $F^* = 0$, the public foreign asset position B^{G*} is calibrated to make the domestic value of net foreign assets equal to 60 percent of annual GDP. The world gross interest rate R^* is then chosen to satisfy the steady-state balance of payments at this net-export target.

Preferences use $u(C) = C^{1-\sigma}/(1-\sigma)$ and $v(N) = \eta N^{1+\phi}/(1+\phi)$. We choose the labor-disutility scale η to normalize employment in the no-shock steady state to one, $N_{ss} = 1$. The HtM household share λ is set to 0.35 for Japan, close to the estimate in Kaplan, Violante and Weidner (2014) that about one third of US households are hand-to-mouth. We set $\sigma = 4$, i.e., an elasticity of intertemporal substitution of 0.25 (Hall 1988; Yogo 2004). In this model, χ is the Marshall-Lerner export revenue elasticity: the elasticity of export revenue in foreign units, $X_t/\epsilon_t = \zeta \epsilon_t^{\chi}$, with respect to the real exchange rate. We set $\chi = 0.25$: a one percent real depreciation raises the foreign purchasing power supplied by exports by 0.25 percent, and the export-demand price elasticity is $1 + \chi = 1.25$. This imposes a positive but modest Marshall-Lerner response, consistent with empirical work that finds limited short-run trade-balance responses to exchange-rate movements; see Rose (1991) and the survey in Bahmani-Oskooee and Ratha (2004). The financial-intermediation parameter is set to $\Gamma = 9$, following Fanelli and Straub (2021).

The labor-disutility curvature parameter ϕ disciplines the real response to the benchmark import-price shock. We set $\phi = 29.36$, at which a doubling of import prices lowers gross output, $A_t N_t$, by 1 percent on impact in the representative-agent economy of Appendix B.6. Scaled to a doubling, the IMF rule of thumb that every persistent 10 percent increase in oil prices lowers global output by 0.1 to 0.2 percent (Georgieva 2026) implies a decline of 1 to 2 percent, and our target lies at the lower end of this range. The persistence of the imported-input price process is $\rho_{P^{T*}} = 0.945$, estimated from an AR(1) regression on detrended quarterly observations of the IMF's Global Price of Energy Index (IMF 2026).

We calibrate the nominal block to the Statistics Bureau of Japan's seasonally adjusted core CPI, defined as all items less fresh food and including energy. Its geometric quarterly growth rate is 0.097 percent from 2014Q4 to 2019Q4 and 0.411 percent from 2020Q4 through 2022Q3, or 0.314 percentage points above the pre-shock rate. We choose $\varepsilon_w = 6.129$ and $\psi_{nr} = 10598.1$ in the representative-agent economy: ε_w matches the pre-shock rate as steady-state inflation, and ψ_{nr} matches the average rate along the realized path of world energy prices through its peak in 2022Q3.

In the representative-agent economy, steady-state inflation depends on how the wage markup compares with the labor wedge that optimal policy requires. Monopolistic wage setting restricts labor supply: in a steady state with zero inflation, the labor wedge in (55) equals the markup wedge, $1/\varepsilon_w$. Optimal policy instead

Table B.1: Calibration

Parameter	Description	Value	Target or source
R^*	Gross world interest rate	1.0021	Net exports of −0.5 percent of GDP
β	Discount factor	0.9979	$\beta R^* = 1$
σ	Inverse intertemporal elasticity	4.0	Intertemporal elasticity of 0.25; Hall (1988), Yogo (2004)
η	Labor-disutility scale	0.9261	$N_{ss} = 1$ and steady labor FOC
ϕ	Inverse Frisch curvature	29.36	1 percent output decline on impact after doubling P_t^{T*} in the representative-agent economy; Georgieva (2026)
λ	HtM household share	0.35	Kaplan, Violante and Weidner (2014)
ζ	Export demand scale	1.9758	Fossil-fuel import bill of 4 percent of GDP and net exports of −0.5 percent of GDP
χ	Marshall-Lerner export revenue elasticity	0.25	Modest short-run response; Rose (1991), Bahmani-Oskooee and Ratha (2004)
$\rho_{P^{T*}}$	Import-price persistence	0.945	AR(1) on detrended quarterly IMF energy-price index from FRED
Γ	Financial-intermediation friction	9.0	Fanelli and Straub (2021)
ε_w	Wage substitution elasticity	6.129	Pre-shock quarterly core CPI inflation of 0.097 percent in the representative-agent steady state
ψ_{nr}	Nominal wage rigidity	10598.1	Average quarterly core CPI inflation of 0.411 percent through 2022Q3 in the representative-agent economy
B_{ss}^{G*}	Public foreign asset position	60.0	Net foreign assets of 60 percent of annual GDP

sets the steady-state labor wedge at the level that implements the optimal terms of trade in Proposition 1, $\chi^{-1}\epsilon_{ss}P_{ss}^{T*}\theta/w_{ss}$, i.e., the import bill as a share of GDP divided by the Marshall-Lerner export revenue elasticity, and this level does not depend on the nominal parameters. With no labor tax available, wage inflation closes the gap between the two wedges at the rate given in Lemma B.3: since unions discount the future utility cost of changing wages at $\beta < 1$, a constant rate of wage inflation lowers the effective markup when it is positive and raises it when it is negative. Steady-state inflation is therefore positive when $\varepsilon_w < \chi w_{ss}/(\epsilon_{ss}P_{ss}^{T*}\theta)$, i.e., when monopolistic wage setting restricts labor supply by more than optimal policy requires, negative when the inequality is reversed, and zero only in the knife-edge case of equality.

Hence the pre-shock rate disciplines ε_w through the markup wedge, while the slope of the wage Phillips curve, ε_w/ψ_{nr}, governs the inflation response along the shock path and disciplines ψ_{nr}. Neither parameter enters the Ramsey problem (112); both appear only in the wage Phillips curve, which pins down the inflation path that implements the Ramsey allocation. The resulting wage elasticity lies close to the conventional value of 6 in the staggered-wage formulation of Erceg, Henderson and Levin (2000). In the heterogeneous-agent economy, steady-state inflation follows from the same lemma, and the inflation paths are outcomes of the policy experiments.

B.6 Representative-Agent Benchmark

This appendix reports the responses of the representative-agent economy, in which no household is borrowing constrained and every household remains on its Euler equation, to the same two imported-energy price paths studied in the main text. By Proposition 1, optimal policy leaves the intertemporal margin undistorted and sets the terms-of-trade-adjusted labor wedge to zero; the paths below therefore coincide with the RA-equivalent policy applied in this economy. The policy figures report the labor wedge τ_t^N, which also contains the static terms-of-trade distortion; its terms-of-trade-adjusted counterpart is zero. All deviations are relative to the representative-agent economy's own no-shock steady state, which differs from the heterogeneous economy's steady state; comparisons with the main-text figures should therefore be made in proportionate terms only.

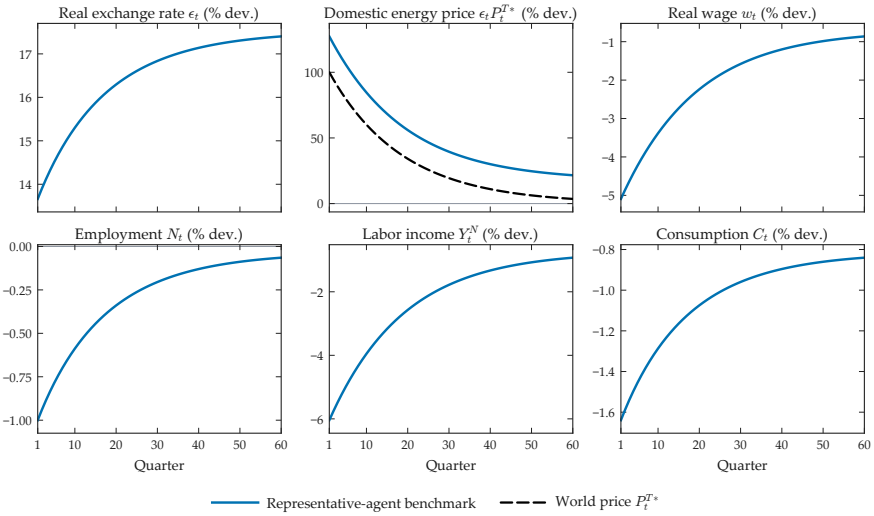


Figure B.1. Representative-agent benchmark: MIT-shock allocations

Note: The solid black line is the world energy price.

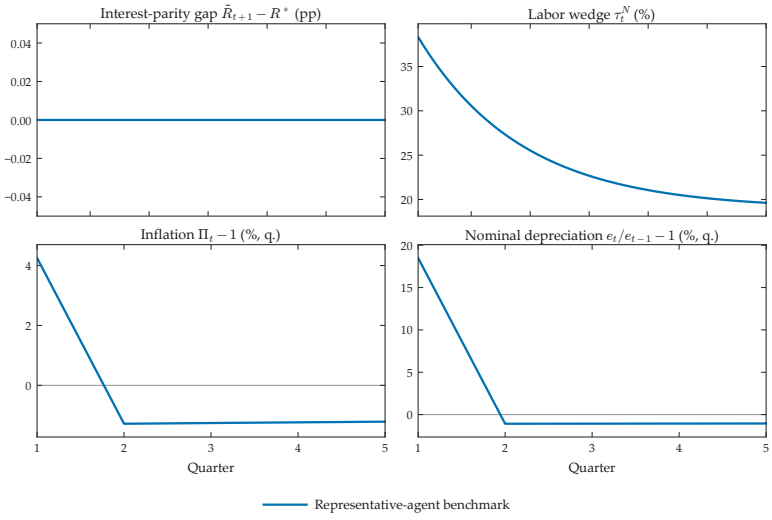


Figure B.2. Representative-agent benchmark: MIT-shock policy

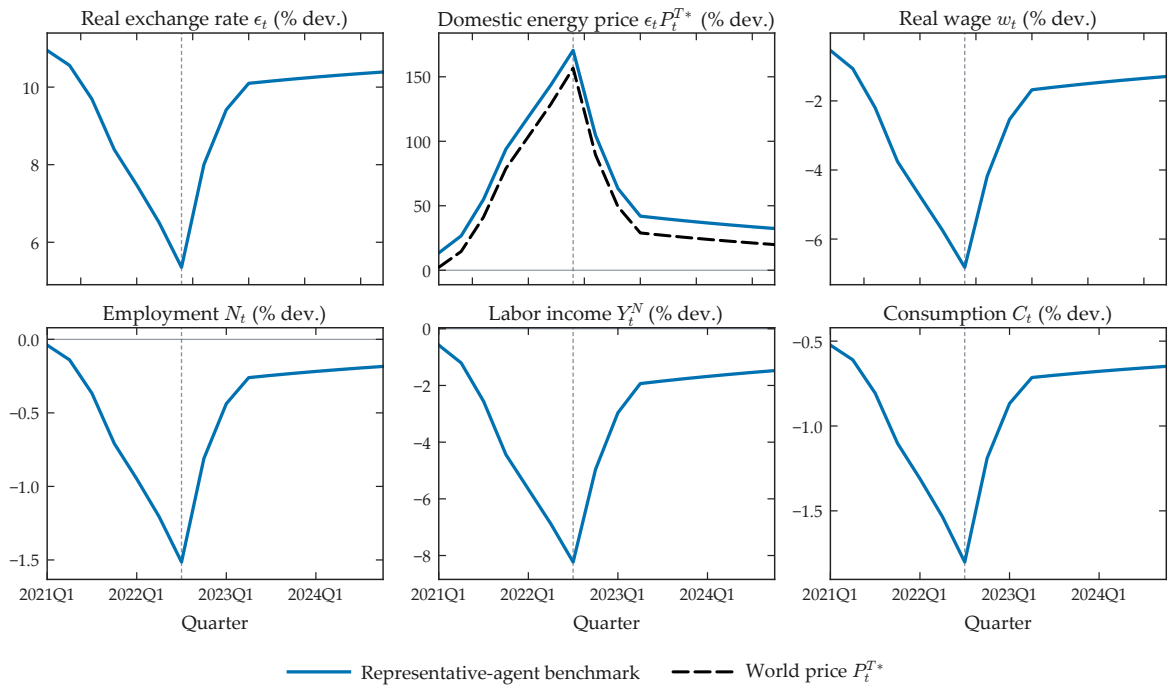


Figure B.3. Representative-agent benchmark: 2021-2023 allocations

Note: The solid black line is the world energy price; the vertical line marks its peak.

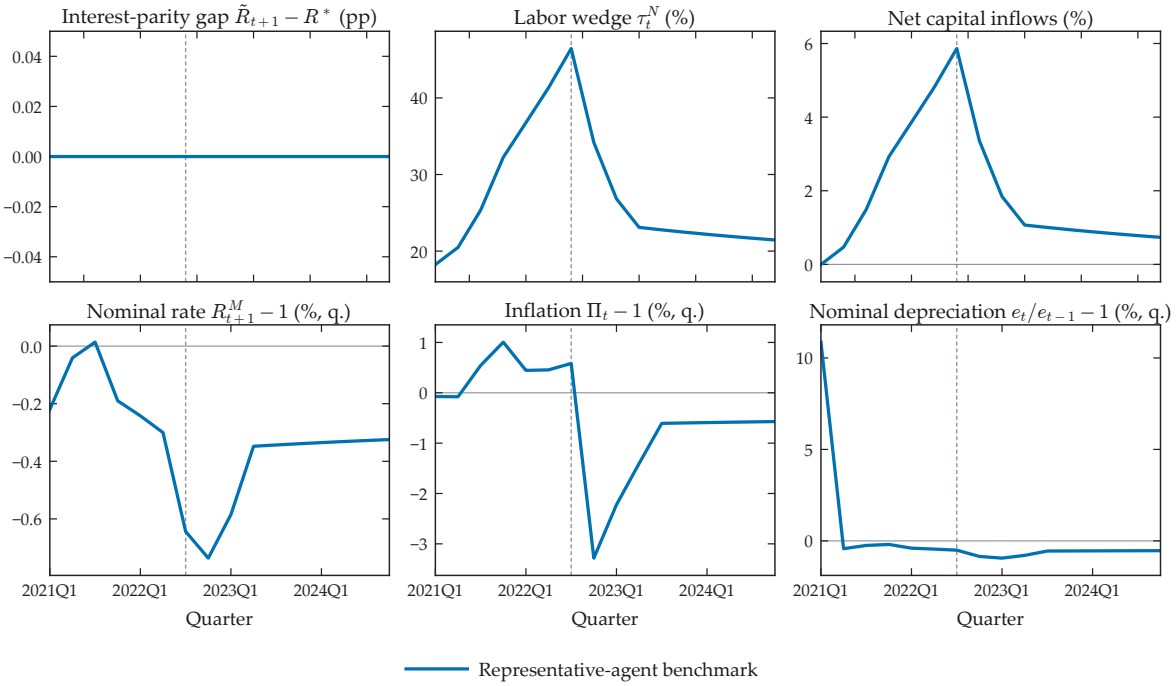


Figure B.4. Representative-agent benchmark: 2021-2023 policy

Note: Net capital inflows are shown as a percent of initial steady-state gross output. The vertical line marks the energy-price peak.

B.7 Resource Shares

The figures below report the domestic output exported and the consumption of the two household types as shares of gross output, together with domestic expenditure on imported inputs relative to gross output. The main text reads the reallocation off these shares in Section 4.

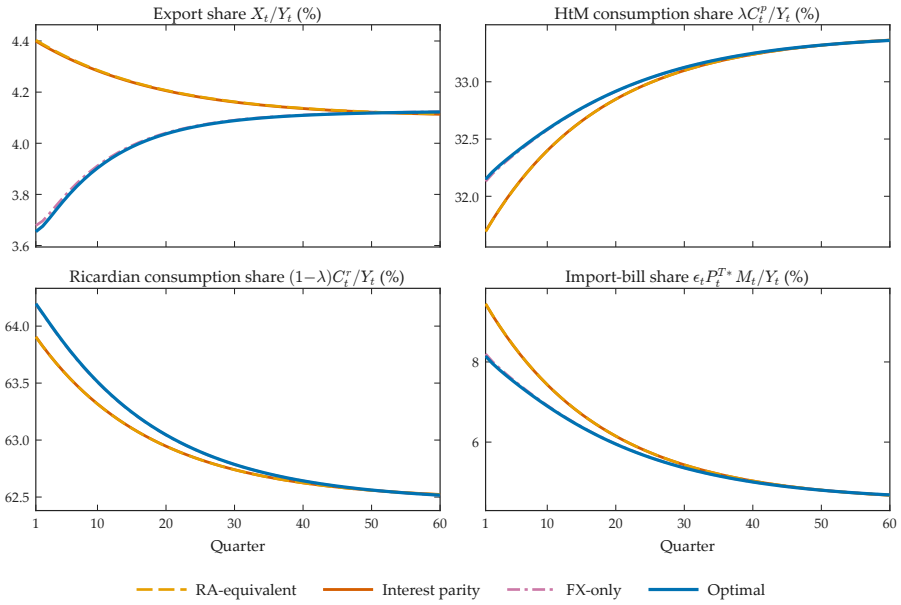


Figure B.5. MIT shock: resource shares

Note: The import bill is expressed in domestic units, $\epsilon_t P_t^{T*} M_t$.

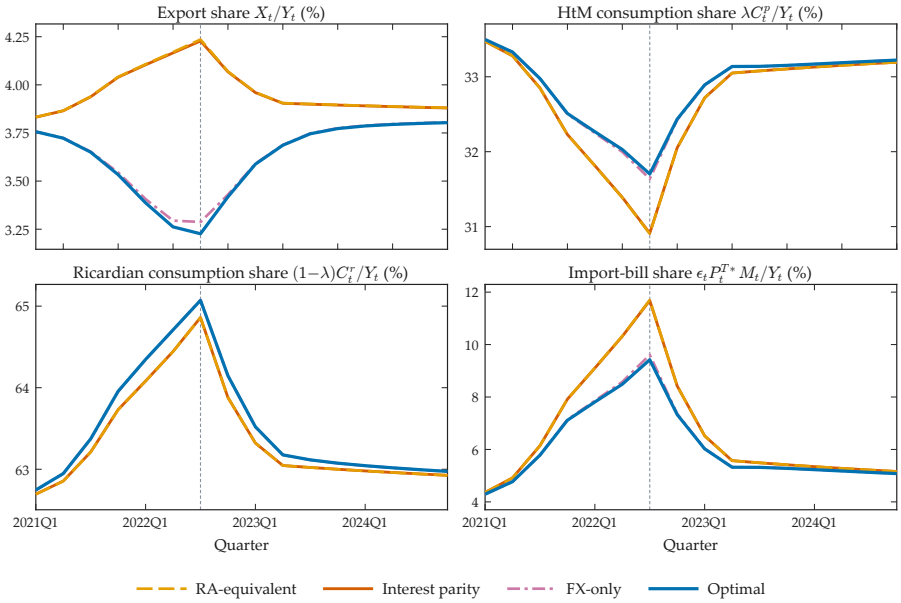


Figure B.6. 2021-2023 shock: resource shares

Note: The import bill is expressed in domestic units, $\epsilon_t P_t^{T*} M_t$. The vertical line marks the energy-price peak.

B.8 Production Substitutability and Policy Wedges

The benchmark Leontief technology excludes short-run substitution between labor and the imported input. We repeat the MIT shock for a Cobb-Douglas technology, interior CES technologies with $\xi = 0.5$ and $\xi = 0.05$, and the Leontief limit. For each value of ξ, the production share and labor-disutility scale are recalibrated so that employment, gross output, the import-bill share, and the external steady-state targets remain unchanged. All other parameters retain their benchmark values.

For any λ admitting the no-shock Ramsey steady state, constant fundamentals imply $\tilde{R} = R^*$ and a zero intertemporal wedge, as in the two-period model. Household consumption levels and marginal utilities can remain unequal in this steady state.

We first remove household heterogeneity in Figure B.7. Away from Leontief, import demand responds to its domestic price, and the planner internalizes the economy's market power over its exports. With tariffs unavailable, the labor and intertemporal wedges jointly alter the real exchange rate path. This is the dynamic terms-of-trade manipulation motive discussed in Appendix A.12 and in Costinot, Lorenzoni and Werning (2014). The intertemporal response balances this motive against the carry cost of departing from interest parity. Both wedges are therefore distorted under Cobb-Douglas and the two interior CES technologies. In the Leontief limit, the fixed input ratio permits the contemporaneous terms-of-trade distortion to be assigned to the labor wedge, while the intertemporal wedge remains zero.

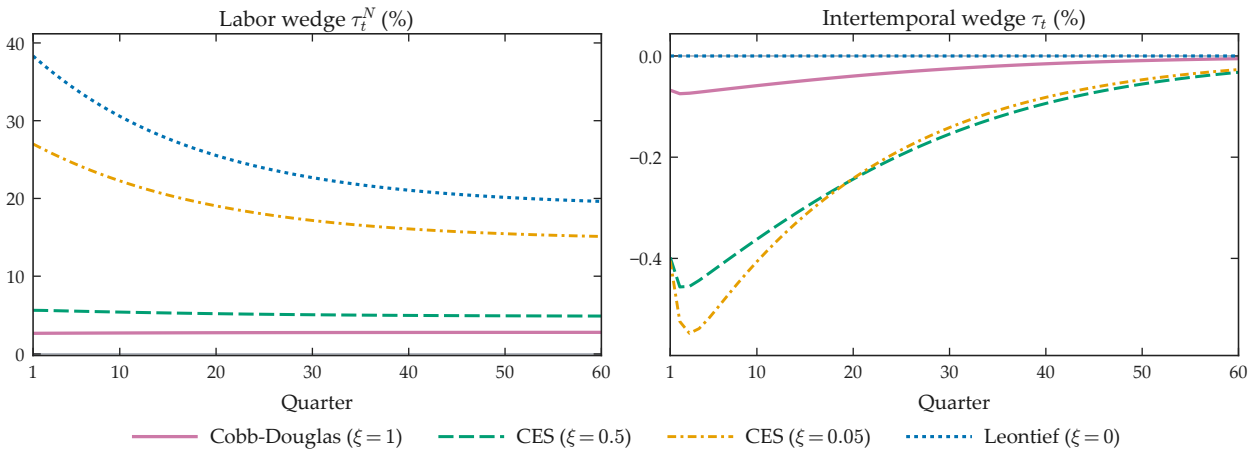


Figure B.7. Production substitutability and optimal wedges without household heterogeneity

Note: Responses to the benchmark MIT shock. The intertemporal wedge is $\tau_t = 1 - \tilde{R}_{t+1}/R^*$.

We then restore the benchmark share of constrained households and compare all four regimes in Figure B.8. For non-Leontief technologies, the RA-equivalent intertemporal wedge need not be zero because it preserves the representative-agent terms-of-trade motive while holding the heterogeneity-specific terms at their steady-state values. Interest parity sets that wedge to zero by construction. Allowing the intertemporal heterogeneity term to respond, as under FX-only, shifts the wedge upward for every technology. The shift weakens the representative-agent distortion under Cobb-Douglas and $\xi = 0.5$, reverses its sign at $\xi = 0.05$, and generates the positive wedge in the Leontief case. This movement is the incentive to transfer foreign

purchasing power toward the quarters in which constrained households bear the largest real-income loss.

The comparison between optimal and FX-only holds the intertemporal rule fixed and isolates the employment response to heterogeneity. The optimal labor wedge is higher on impact for every displayed elasticity, with the difference increasing as labor and imports become more complementary. This direction agrees with the main-text result that, through the redistributive price effect, employment-induced import demand and depreciation shift purchasing power away from constrained households. Comparisons involving interest parity are not a pure decomposition of heterogeneity away from Leontief, since imposing interest parity also removes the representative-agent dynamic terms-of-trade motive. The Leontief specification is therefore useful analytically and quantitatively: it leaves the representative-agent intertemporal wedge undistorted, and the intertemporal response can therefore be attributed to constrained-household consumption smoothing without the dynamic terms-of-trade motive.

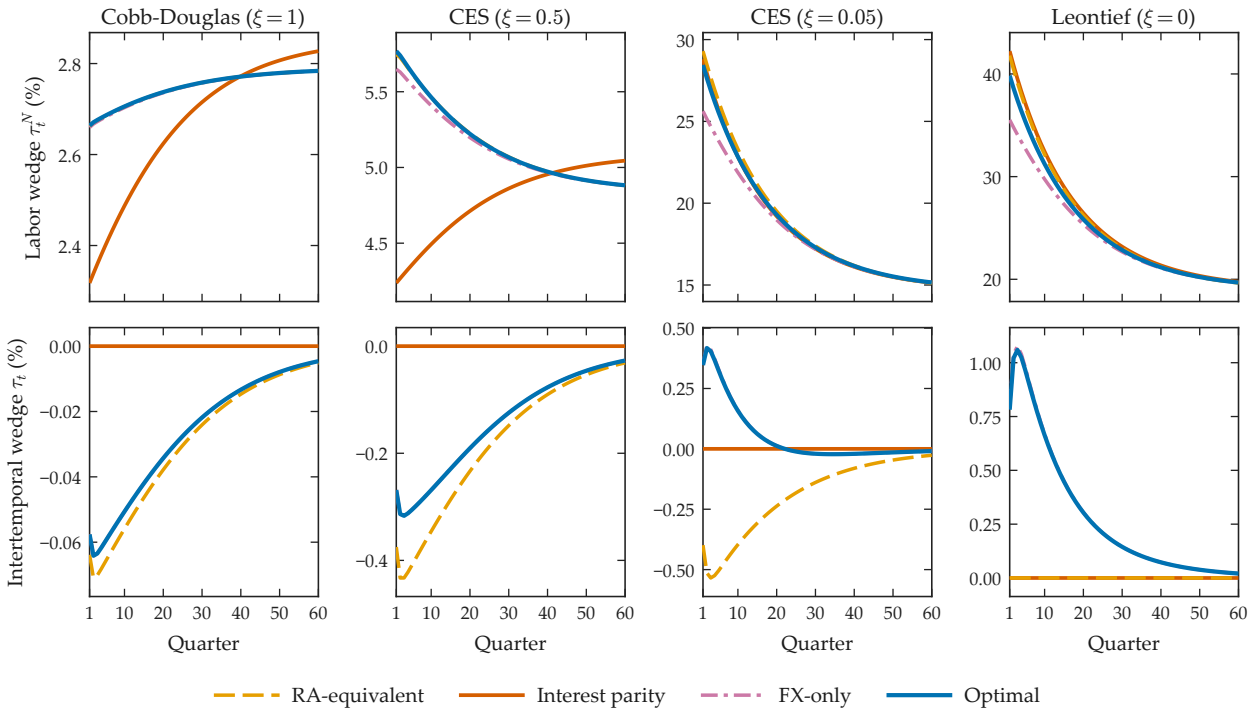


Figure B.8. Production substitutability and policy wedges with household heterogeneity

Note: Responses to the benchmark MIT shock with $\lambda = 0.35$. The intertemporal wedge is $\tau_t = 1 - \tilde{R}_{t+1}/R^*$.

C Beyond the Two-Agent Model

C.1 A Two-Agent Model without Lump-Sum Transfers

The main text's quantitative model assumes that the public agent can selectively make lump-sum transfers or impose lump-sum taxes on Ricardian households to finance carry trade losses or to distribute carry trade profits. In this appendix we withdraw this instrument entirely: lump-sum taxes and transfers are unavailable to the public agent. This restriction must be paired with a second one, that returns in domestic

asset markets are inflation indexed, i.e., denominated in real domestic units. Without real indexation, the public agent would have an incentive to use inflation ex post to manipulate the real value of Ricardian asset holdings, a time-inconsistency that was absent in the framework with lump-sum transfers and taxes for Ricardian households.

The Ricardian households' budget constraint is now given by:

$$C_t^r + B_{t+1}^r = w_t N_t + R_t B_t^r \tag{116}$$

The Ricardian Euler equation and the no-Ponzi-transversality condition on the household budget,

$$\lim_{T\to\infty} \frac{B_{T+1}^r}{\prod_{s=2}^{T} R_s} = 0$$

are unchanged. The period-by-period fiscal budget constraint is now given by:

$$B_{t+1}^G + \epsilon_t B_{t+1}^{G*} = R_t B_t^G + \epsilon_t R^* B_t^{G*} \tag{117}$$

together with unchanged no-Ponzi conditions:

$$\lim_{T\to\infty} \frac{B_{T+1}^G}{\prod_{s=2}^{T} R_s} = 0, \qquad \lim_{T\to\infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0$$

and an unchanged consistency condition on interest rates which ensures that government asset positions satisfy no-Ponzi conditions in domestic as well as foreign units:

$$0 < \inf_{T\geqslant 2} \prod_{s=2}^{T} \frac{\tilde{R}_s}{R^*} \leqslant \sup_{T\geqslant 2} \prod_{s=2}^{T} \frac{\tilde{R}_s}{R^*} < \infty$$

The choice of monetary policy is now represented directly by a money-growth rule: money supply satisfies the standard exchange equation $\mathcal{M}_t = P_t Y_t$, and the monetary agent sets

$$g_t^{\mathcal{M}} = \Pi_t \frac{Y_t}{Y_{t-1}} \tag{118}$$

This rule was already implicitly present in the main text's quantitative model, but the choice of monetary policy was represented through the monetary agent choosing the ex-post real rate, R_1, and the forward-looking nominal interest rates, R_{t+1}^M. Because bonds are denominated in real domestic units, the ex-post real rate, R_1, is a state variable independent of the ex-post choice of inflation, while inflation itself remains under the monetary agent's control through the money-growth rule.

The balance of payments condition and the remaining equilibrium conditions of Appendix B.3 are unchanged. Given

- initial states $\{R_1, B_1, B_1^{G*}, \epsilon_0, F_1^*, w_0, Y_0\}$,
- an exogenous sequence of import prices and productivity $\{P_t^{T*}, A_t\}_{t=1}^{\infty}$, and
- fiscal-monetary policy $\{B_{t+1}^G, B_{t+1}^{G*}, g_t^{\mathcal{M}}\}_{t=1}^{\infty}$,

a decentralized equilibrium is a sequence of allocations $\{C_t^r, C_t^p, B_{t+1}, N_t, M_t, Y_t, F_{t+1}^*, X_t\}_{t=1}^{\infty}$ and prices $\{\epsilon_t, w_t, \Pi_t, R_{t+1}\}_{t=1}^{\infty}$ that satisfy the conditions of Appendix B.3, with equations (116), (117), and (118) replacing the corresponding conditions.

C.1.1 Optimal Policy Problem

As in the main text, we restrict optimal policy to the case where $F_1^* = 0$ to avoid any time-inconsistency arising from the public agent's incentive to ex-post manipulate the foreign-unit value of the claims on financial intermediaries. With transfers absent, the Ricardian households' present-value budget constraint becomes an additional implementability constraint in the optimal policy problem, and the initial claims on Ricardian households, $R_1 B_1/(1-\lambda)$, become a state variable of the problem. Given B_1^{G*}, $R_1 B_1$, and $F_1^* = 0$, the Ramsey planner's optimization problem is to solve:

$$\begin{aligned}
&\max_{\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} \left[\lambda u\left(C_t^p\right) + (1-\lambda) u\left(C_t^r\right) - v(N_t)\right] \qquad \text{s.t.} \\
&\sum_{t=1}^{\infty} \frac{\frac{1}{\Gamma}\left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}} + \sum_{t=1}^{\infty} \frac{P_t^{T*}\theta N_t}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^{\chi}}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1} \\
&\sum_{t=1}^{\infty} \beta^{t-1} u'(C_t^r) \left[C_t^r - (A_t - \epsilon_t P_t^{T*}\theta) N_t\right] = \frac{R_1 B_1}{1-\lambda} u'(C_1^r) \\
&\tilde{R}_{t+1} = \frac{u'\left(C_t^r\right) \epsilon_t}{\beta u'\left(C_{t+1}^r\right) \epsilon_{t+1}} \\
&C_t^r = A_t N_t - \frac{\zeta \epsilon_t^{1+\chi}}{1-\lambda} + \frac{\lambda}{1-\lambda} \epsilon_t P_t^{T*}\theta N_t \\
&C_t^p = (A_t - \epsilon_t P_t^{T*}\theta) N_t \\
&N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p \geqslant 0
\end{aligned} \tag{119}$$

The following lemma establishes conditions under which a solution to (119) is implementable as a decentralized equilibrium.

Lemma C.1. *Given $F_1^* = 0$, B_1, B_1^{G*}, $R_1 > 0$, $\beta R^* = 1$, and positive sequences $\{A_t, P_t^{T*}\}_{t \geqslant 1}$ satisfying $A_t \leqslant \bar{A} < \infty$ and $P_t^{T*} \geqslant \underline{P} > 0$ for all $t \geqslant 1$, let $\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t \geqslant 1}$ solve (119). Suppose further that the solution satisfies, for all $t \geqslant 1$,*

$$\epsilon_t \geqslant \underline{\epsilon}, \qquad \underline{C}^r \leqslant C_t^r \leqslant \bar{C}^r$$

where $\underline{\epsilon}, \underline{C}^r, \bar{C}^r > 0$, and that the conditions of Lemma B.2 hold for $w_t = A_t - \epsilon_t P_t^{T*}\theta$ and $\bar{u}_t' = \lambda u'(C_t^p) + (1-\lambda)u'(C_t^r)$. Then the Ramsey allocation is implementable as a decentralized equilibrium of the economy without transfers. Conditional on the Ramsey allocation and the initial positions, the implementing paths $\{B_{t+1}, B_{t+1}^G, B_{t+1}^{G*}\}_{t \geqslant 1}$ are unique.

Proof. Construct w_t, M_t, Y_t, X_t, F_{t+1}^*, and R_{t+1} as in the proof of Lemma B.1:

$$w_t = A_t - \epsilon_t P_t^{T*}\theta, \qquad M_t = \theta N_t, \qquad Y_t = A_t N_t, \qquad X_t = \zeta \epsilon_t^{1+\chi}$$

$$F^*_{t+1} = \frac{1}{\Gamma}\left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right), \qquad R_{t+1} = \tilde{R}_{t+1}\frac{\epsilon_{t+1}}{\epsilon_t}$$

The constraint $\tilde{R}_{t+1} = u'(C^r_t)\epsilon_t/[\beta u'(C^r_{t+1})\epsilon_{t+1}]$ and the construction of R_{t+1} imply the Ricardian Euler equation for real-indexed bonds, $u'(C^r_t) = \beta R_{t+1}u'(C^r_{t+1})$. The HtM budget constraint, firms' zero-profit condition, and goods market clearing hold as in the proof of Lemma B.1, and the interest-rate consistency condition follows by the same argument, which uses only the bounds on $\{A_t, P^{T*}_t, \epsilon_t, C^r_t\}$ assumed here.

The Euler equation and $\beta R^* = 1$ imply

$$\frac{1}{\prod_{s=2}^t R_s} = \beta^{t-1}\frac{u'(C^r_t)}{u'(C^r_1)}$$

Let $C_t = (1-\lambda)C^r_t + \lambda C^p_t$. Since $C^p_t = w_t N_t$, we have $C_t - w_t N_t = (1-\lambda)\left[C^r_t - w_t N_t\right]$. Dividing the household implementability constraint in (119) by $u'(C^r_1)$, multiplying by $(1-\lambda)$, and substituting the two displays above shows that the constraint is equivalent to

$$\sum_{t=1}^{\infty}\frac{C_t - w_t N_t}{\prod_{s=2}^t R_s} = R_1 B_1 \tag{120}$$

The series converges absolutely: goods market clearing and zero profits give $C_t - w_t N_t = \epsilon_t(P^{T*}_t \theta N_t - \zeta\epsilon^X_t)$, so

$$\frac{C_t - w_t N_t}{\prod_{s=2}^t R_s} = \epsilon_1\frac{P^{T*}_t\theta N_t - \zeta\epsilon^X_t}{\prod_{s=2}^t \tilde{R}_s}$$

and, by the interest-rate consistency condition, there exists $K < \infty$ with $1/\prod_{s=2}^t \tilde{R}_s \leqslant K/R^{*t-1}$, so the external intertemporal budget constraint in (119) bounds the series by $K\epsilon_1\sum_t\left(P^{T*}_t\theta N_t + \zeta\epsilon^X_t\right)/R^{*t-1} < \infty$.

Define $\{B_{t+1}\}_{t\geqslant 1}$ recursively by the consolidated household budget

$$B_{t+1} = w_t N_t + R_t B_t - C_t, \qquad B_1 \text{ given} \tag{121}$$

Since $C^p_t = w_t N_t$, subtracting λ times the HtM budget from (121) and dividing by $(1-\lambda)$ yields the Ricardian budget constraint (116) with $B^r_t = B_t/(1-\lambda)$. Iterating (121) from $t = 1$ to T gives

$$\frac{B_{T+1}}{\prod_{s=2}^T R_s} = R_1 B_1 + \sum_{t=1}^{T}\frac{w_t N_t - C_t}{\prod_{s=2}^t R_s}$$

By (120) and absolute convergence, the right-hand side converges to zero as $T \to \infty$, so the household no-Ponzi condition $\lim_{T\to\infty} B_{T+1}/\prod_{s=2}^T R_s = 0$ holds.

Construct B^G_{t+1} from the bonds market clearing condition, $B^G_{t+1} = \epsilon_t F^*_{t+1} - B_{t+1}$; since $F^*_1 = 0$, the initial condition $B^G_1 = -B_1$ holds. Construct B^{G*}_{t+1} from the period-by-period balance of payments,

$$B^{G*}_{t+1} = \zeta\epsilon^X_t + R_t\frac{\epsilon_{t-1}}{\epsilon_t}F^*_t + R^*B^{G*}_t - P^{T*}_t\theta N_t - F^*_{t+1}, \qquad B^{G*}_1 \text{ given}$$

Since the solution satisfies the external intertemporal budget constraint, the no-Ponzi condition on the net foreign asset position, $\lim_{T\to\infty}(B^{G*}_{T+1} + F^*_{T+1})/R^{*T-1} = 0$, holds; the carry-cost summability in (119)

gives $(F^*_{T+1})^2/R^{*T-1} \to 0$, so the two limits hold separately; and the bonds market clearing condition, the household no-Ponzi condition, and the interest-rate consistency condition give $\lim_{T\to\infty} B^G_{T+1}/\prod_{s=2}^{T} R_s = 0$, as in the proof of Lemma B.1.

The fiscal budget constraint (117) holds period by period by Walras' identity: multiplying the balance of payments by ϵ_t and substituting $C_t + X_t = A_t N_t$ and $w_t N_t = A_t N_t - \epsilon_t P^{T*}_t \theta N_t$ gives

$$\epsilon_t F^*_{t+1} + \epsilon_t B^{G*}_{t+1} = w_t N_t - C_t + R_t \epsilon_{t-1} F^*_t + \epsilon_t R^* B^{G*}_t$$

and substituting $w_t N_t - C_t = B_{t+1} - R_t B_t$ from (121) and $B_{t+1} + B^G_{t+1} = \epsilon_t F^*_{t+1}$ yields (117).

Finally, construct the labor wedge $\tau^N_t = 1 - v'(N_t)/(\bar{u}'_t w_t)$ and, by Lemma B.2, a sequence $\{\Pi_t\}_{t\geqslant 1} \subset (0,\infty)$ satisfying the wage Phillips curve. Set

$$g^{\mathcal{M}}_t = \Pi_t \frac{Y_t}{Y_{t-1}}$$

which implements this inflation path under $\mathcal{M}_t = P_t Y_t$. All decentralized-equilibrium conditions of the economy without transfers therefore hold.

For uniqueness, fix the Ramsey allocation and the initial positions (B_1, B^{G*}_1, R_1). The recursion (121) determines $\{B_{t+1}\}_{t\geqslant 1}$ uniquely, the bonds market clearing condition then determines $\{B^G_{t+1}\}_{t\geqslant 1}$ uniquely, and the balance-of-payments recursion determines $\{B^{G*}_{t+1}\}_{t\geqslant 1}$ uniquely. Unlike Lemma B.1, no transfer sequence is available to reallocate claims across dates, and the implementing asset paths therefore admit no other solution. □

C.1.2 Optimal Policy at the Quantitative Calibration

We solve the Ramsey problem (119) at the calibration of Appendix B.5, for the heterogeneous economy ($\lambda = 0.35$) and the representative-agent economy ($\lambda = 0$), under the MIT shock of Section 4. Under constant import prices the computed solution reproduces the no-shock steady state at both values of λ: the multiplier on the household present-value constraint is zero, and the inherited claims $R_1 B_1$ create no incentive to distort any margin at date one. Real indexation of domestic bonds removes the incentive to revalue these claims through surprise inflation by construction. The constant-price solution shows that the inherited claims support no further distortion at this Ramsey steady state, and the removal of transfers therefore leaves steady-state policy unchanged.

Under the shock, however, this coincidence with the transfer economy ends, and it ends already in the representative-agent case: the planner opens interest parity gaps and moves the labor wedge (Figure C.1). The source is the denomination of the claims that the shock leaves mismatched. The public agent holds the country's foreign assets, B^{G*}_1, in foreign units, and owes Ricardian households the claims $R_1 B_1$ in domestic goods; at the steady-state real exchange rate the two match in value. A shock that raises the present value of the import bill requires a more depreciated real exchange rate on impact, which lowers the foreign value $R_1 B_1/\epsilon_1$ of households' claims below that of the reserves and leaves the public agent with a surplus. With transfers, the surplus is rebated lump sum and the representative-agent economy keeps interest parity, as Proposition 1 and the comparative statics of Section 3.5 establish under Leontief production. Without

transfers, the household present-value constraint binds, and the surplus can be paid out only through the return on households' claims: the planner raises that return during the first quarters of the shock by opening interest parity gaps, which depreciates the real exchange rate on impact, appreciates it afterward, and raises the value of households' claims in terms of future foreign resources. The labor wedge continues to correct the static terms-of-trade distortion period by period, as in the economy with transfers; the interest parity gaps of the $\lambda = 0$ economy come from the revaluation of households' claims.

Heterogeneity enters on top of this revaluation, and its contribution is the difference between the two paths in Figure C.1. Relative to the $\lambda = 0$ economy, the planner holds the parity gap lower throughout: smaller at impact, and negative while the import price remains high. The real exchange rate barely depreciates on impact and approaches its long-run level only as the import price recedes. The domestic price of the import is therefore lower, and the real wage higher, in the quarters in which the marginal utility of hand-to-mouth households is highest. The nominal depreciation is deferred in the same way, while the foreign asset drawdown that finances the import bill is nearly identical across the two economies. Monetary policy retains its complementary role: relative to the $\lambda = 0$ path, impact inflation is lower and the labor wedge larger. The appreciation lowers the imported-input cost, the wedge keeps employment and imported-input demand from expanding in response, and the cost reduction passes into the real wage. Exchange rate policy sets the intertemporal distortion. The real-income stabilization channel of the main text therefore holds without lump-sum transfers: both wedges combine motives present at either value of λ, the terms-of-trade correction and the revaluation of households' claims, with the heterogeneity motive, and differencing the $\lambda = 0.35$ and $\lambda = 0$ paths isolates the part contributed by restricted asset market access.

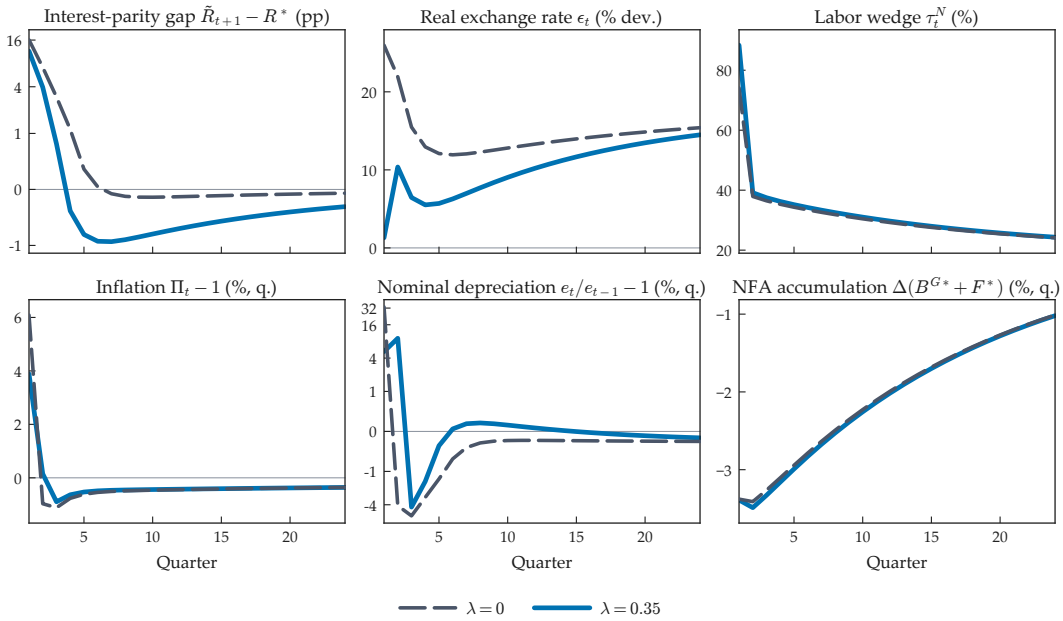


Figure C.1. Economy without lump-sum transfers: MIT-shock policy

Note: Deviations are relative to each economy's initial steady state; NFA accumulation is measured relative to steady-state output. The interest-parity-gap and nominal-depreciation panels use a symmetric log scale, linear within one percentage point of zero.